

M.A. Hayat
Editor

Tumors of the Central Nervous System

Volume 14

Glioma, Meningioma,
Neuroblastoma, and Spinal
Tumors

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Volume 14

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Glioma, Meningioma, Neuroblastoma,
and Spinal Tumors

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Although touched by technology, surgical pathology always has been, and remains, an art. Surgical pathologists, like all artists, depict in their artwork (surgical pathology reports) their interactions with nature: emotions, observations, and knowledge are all integrated. The resulting artwork is a poor record of complex phenomena.

Richard J. Reed, MD

Preface

It is recognized that scientific journals and books not only provide current information but also facilitate exchange of information, resulting in rapid progress in the medical field. In this endeavor, the main role of scientific books is to present current information in more detail after careful additional evaluation of the investigational results, especially those of new or relatively new therapeutic methods and their potential toxic side effects.

Although subjects of diagnosis, drug development, therapy and its assessment, prognosis of tumors of the central nervous system, cancer recurrence, and resistance to chemotherapy are scattered in a vast number of journals and books, there is a need to combine these subjects in single volumes. An attempt will be made to accomplish this goal in the projected 14-volume series of handbooks.

In the era of cost-effectiveness, my opinion may be minority perspective, but it needs to be recognized that the potential for false-positive or false-negative interpretation on the basis of a single laboratory test in clinical pathology does exist. Interobserver or intraobserver variability in the interpretation of results in pathology is not uncommon. Interpretative differences often are related to the relative importance of the criteria being used.

Generally, no test always performs perfectly. Although there is no perfect remedy to this problem, standardized classifications with written definitions and guidelines will help. Standardization of methods to achieve objectivity is imperative in this effort. The validity of a test should be based on the careful, objective interpretation of the tomographic images, photomicrographs, and other tests. The interpretation of the results should be explicit rather than implicit. To achieve accurate diagnosis and correct prognosis, the use of molecular criteria and targeted medicine is important. Equally important are the translation of molecular genetics into clinical practice and evidence-based therapy. The translation of medicine from the laboratory to clinical application needs to be carefully expedited. Indeed, molecular medicine has arrived.

This is the 14th volume in the series Tumors of the Central Nervous System. As in the case of the 13 previously published volumes, this volume mainly contains information on the diagnosis, therapy, and prognosis of brain and spinal cord tumors. Various aspects of a number of tumor types, including angiocentric glioma, pilomyxoid astrocytoma, pituicytoma, pediatric low-grade gliomas, meningiomas, and spinal cord tumors, are discussed in this volume.

Neuroblastoma is a malignant tumor of the sympathetic nervous system and occurs most often in children before the age of 5; it rarely occurs in adults. Amplification and overexpression of the MYCN proto-oncogene occur in approximately 20% of neuroblastomas and are associated with advanced stage disease, rapid tumor progression, and poor prognosis. Aberrant regulation of miRNA expression has disease, rapid tumor progression, and poor prognosis. Aberrant regulation of miRNA expression has been reported in a variety of cancers, including neuroblastoma. The molecular mechanism underlying the role of N-myc oncoprotein in miRNA expression in neuroblastoma tumorigenesis is explained.

Another subject discussed in detail in this volume is the development of central nervous system (CNS) metastasis from non-small cell lung carcinoma (NSCLC). Indeed, a significant proportion of NSCLC patients develop CNS metastasis, resulting in considerable morbidity and mortality. The discussion concentrates on providing currently available information on pathology, diagnostic imaging, and management of such brain metastases.

By bringing together a large number of experts (oncologists, neurosurgeons, physicians, research scientists, and pathologists) in various aspects of this medical field, it is my hope that substantial progress will be made against this terrible disease. It would be difficult for a single author to discuss effectively the complexity of diagnosis, therapy, and prognosis of any type of tumor in one volume. Another advantage of involving more than one author is to present different points of view on a specific controversial aspect of the CNS cancer. I hope these goals will be fulfilled in this and other volumes of this series. This volume was written by 20 contributors representing eight countries. I am grateful to them for their promptness in accepting my suggestions. Their practical experience highlights their writings, which should build and further the endeavors of the reader in this important area of disease. I respect and appreciate the hard work and exceptional insight into the nature of cancer provided by these contributors. The contents of the volume are divided into three subheadings: Pineal Tumors, Pituitary Tumors, and Spinal Tumors for the convenience of the reader.

It is my hope that the current volume will join the preceding volumes of the series for assisting in the more complete understanding of globally relevant cancer syndromes. There exists a tremendous, urgent demand by the public and the scientific community to address cancer diagnosis, treatment, cure, and hopefully prevention. In the light of existing cancer calamity, financial funding by governments must give priority to eradicating this deadly malignancy over military superiority.

I am thankful to Dr. Dawood Farahi and Philip Connelly for recognizing the importance of medical research and publishing through an institution of higher education. I am also thankful to my students for their contribution to the preparation of this volume.

Contents

1	Angiocentric Glioma, Pilomyxoid Astrocytoma, and Pituicytoma: New Entities in the World Health Organization Classification	1
	Jamie L. Odem and Douglas C. Miller	
2	Pediatric Low-Grade Gliomas: Diagnosis, Treatment, and Future Directions.....	13
	Angela J. Waanders, Jane E. Minturn, and Michael J. Fisher	
3	Kringle 1-5 Reduces Growth of Malignant Gliomas in Rats	25
	Ming-Chao Huang and Yi-Lo Lin	
4	Fluorescence Guided Resection and Photodynamic Therapy in Meningiomas	35
	Martin Hefti and Gord von Campe	
5	Classification of Meningiomas in the Cerebellopontine Angle.....	43
	Arasho Belachew, Tumul Chowdhury, Toma Spiriev, Nora Sandu, and Bernhard Schaller	
6	Postoperative Cerebral Hyperemia—Pathophysiology and Clinical Concerns.....	55
	Xiao-Yu Yang, Shou-Jing Zhou, and Hong-Zhi Xu	
7	Evidence-Based Approach for the Management of Cerebral Metastases from Lung Adenocarcinoma	67
	Mustafa Yavuz Samanci	
8	Intramedullary Spinal Cord Tumors: Long-Term Follow-Up After Surgery	79
	Antonino Raco, Marco Cimatti, and Massimo Miscusi	
	Index.....	95

Contents of Volume 1

- 1 Introduction**
- 2 Molecular Classification of Gliomas**
- 3 Glioblastoma: Endosialin Marker for Preicytes**
- 4 Glioma Grading Using Cerebral Blood Volume Heterogeneity**
- 5 The Role of Ectonucleotidases in Glioma Cell Proliferation**
- 6 Gliomas: Role of Monoamine Oxidase B in Diagnosis**
- 7 Glioma: Role of Integrin in Pathogenesis and Therapy**
- 8 Proton Magnetic Resonance Spectroscopy in Intracranial Gliomas**
- 9 Infiltration Zone in Glioma: Proton Magnetic Resonance Spectroscopic Imaging**
- 10 Malignant Gliomas: Role of E2F1 Trascription Factor**
- 11 The Role of Glucose Transporter-1 (Glut-1) in Malignant Gliomas**
- 12 Malignant Gliomas: Role of Platelet-Derived Growth Factor Receptor A (PDGFRA)**
- 13 Molecular Methods for Detection of Tumor Markers in Glioblastoma**
- 14 Role of MGMT in Glioblastoma**
- 15 Glioblastomas: Role of CXCL12 Chemokine**
- 16 Cell Death Signaling in Glioblastoma Multiforme: Role of the Bcl2L12 Oncoprotein**
- 17 Glioblastoma Multiforme: Role of Polycomb Group Proteins**
- 18 Glioblastoma Multiforme: Role of Cell Cycle-Related Kinase Protein (Method)**
- 19 Markers of Stem Cells in Gliomas**
- 20 Efficient Derivation and Propagation of Glioblastoma Stem- Like Cells Under Serum-Free Conditions Using the Cambridge Protocol**
- 21 Glioma Cell Lines: Role of Cancer Stem Cells**

-
- 22 Glioblastoma Cancer Stem Cells: Response to Epidermal Growth Factor Receptor Kinase Inhibitors**
 - 23 Low-and High-Grade Gliomas: Extensive Surgical Resection**
 - 24 Brainstem Gangliogliomas: Total Resection and Close Follow-Up**
 - 25 Glioblastoma: Temozolomide-Based Chemotherapy**
 - 26 Drug-Resistant Glioma: Treatment with Imatinib Mesylate and Chlorimipramine**
 - 27 Glioblastoma Multiforme: Molecular Basis of Resistance to Erlotinib**
 - 28 Enhanced Glioma Chemosensitivity**
 - 29 Malignant Glioma Patients: Anti-Vascular Endothelial Growth Factor Monoclonal Antibody, Bevacizumab**
 - 30 Aggravating Endoplasmic Reticulum Stress by Combined Application of Bortezomib and Celecoxib as a Novel Therapeutic Strategy for Glioblastoma**
 - 31 Targeted Therapy for Malignant Gliomas**
 - 32 Glioblastomas: HER1/EGFR-Targeted Therapeutics**
 - 33 Epidermal Growth Factor Receptor Inhibition as a Therapeutic Strategy for Glioblastoma Multiforme**
 - 34 Role of Acyl-CoA Synthetases in Glioma Cell Survival and its Therapeutic Implication**
 - 35 Malignant Glioma Patients: Combined Treatment with Radiation and Fotemustine**
 - 36 Malignant Glioma Immunotherapy: A Peptide Vaccine from Bench to Bedside**
 - 37 Malignant Glioma: Chemovirotherapy**
 - 38 Intracranial Glioma: Delivery of an Oncolytic Adenovirus**
 - 39 Use of Magnetic Resonance Spectroscopy Imaging (MRSI) in the Treatment Planning for Gliomas**
 - 40 Malignant Glioma Cells: Role of Trail-Induced Apoptosis**
 - 41 Long-Term Survivors of Glioblastoma**
 - 42 Glioblastoma Patients: P15 Methylation as a Prognostic Factor**

Contents of Volume 2

- 1 Introduction**
- 2 Gliomagenesis: Advantages and Limitations of Biomarkers**
- 3 Molecular Subtypes of Gliomas**
- 4 Glioblastoma: Germline Mutation of *TP53***
- 5 Gliomas: Role of the *TP53* Gene**
- 6 The Role of IDH1 and IDH2 Mutations in Malignant Gliomas**
- 7 Malignant Glioma: Isocitrate Dehydrogenases 1 and 2 Mutations**
- 8 Metabolic Differences in Different Regions of Glioma Samples**
- 9 Glioblastoma Patients: Role of Methylated MGMT**
- 10 Brain Tumor Angiogenesis and Glioma Grading:
Role of Tumor Bloods Volume and Permeability
Estimates Using Perfusion CT**
- 11 Vasculogenic Mimicry in Glioma**
- 12 Newly Diagnosed Glioma: Diagnosis Using Positron Emission
Tomography with Methionine and Fluorothymidine**
- 13 Role of Diffusion Tensor Imaging in Differentiation
of Glioblastomas From Solitary Brain Metastases**
- 14 I-TM-601 SPECT Imaging of Human Glioma**
- 15 Assessment of Biological Target Volume Using Positron
Emission Tomography in High-Grade Glioma Patients**
- 16 Skin Metastases of Glioblastoma**
- 17 Diffuse Low-Grade Gliomas. What Does
“Complete Resection” Mean?**
- 18 Quantitative Approach of the Natural Course of Diffuse
Low-Grade Gliomas**
- 19 Impact of Resection Extent on Outcomes in Patients
with High-Grade Gliomas**

-
- 20 Recurrent Malignant Gliomas: 5-Aminolevulinic Acid Fluorescence-Guided Resection**
 - 21 Glioma Surgery: Intraoperative Low Field Magnetic Resonance Imaging**
 - 22 Low-Grade Gliomas: Intraoperative Electrical Stimulations**
 - 23 Malignant Gliomas: Present and Future Therapeutic Drugs**
 - 24 Recurrent Malignant Glioma Patients: Treatment with Conformal Radiotherapy and Systemic Therapy**
 - 25 Glioblastoma: Boron Neutron Capture Therapy**
 - 26 Glioblastoma: Anti-Tumor Action of Cyclosporine A and Functionally Related Drugs**
 - 27 Glioblastoma Patients: Chemotherapy with Cisplatin, Temozolomide and Thalidomide.**
 - 28 Glioblastoma : Role of Galectin-1 in Chemoresistance**
 - 29 Glioma-Initiating Cells: Interferon Treatment**
 - 30 Glioblastoma : Antitumor Action of Natural and Synthetic Cannabinoids**
 - 31 Patients with Recurrent High-Grade Glioma: Therapy with Combination of Bevacizumab and Irinotecan**
 - 32 Monitoring Gliomas In Vivo Using Diffusion- Weighted MRI During Gene Therapy –Induced Apoptosis**
 - 33 High-Grade Gliomas: Dendritic Cell Therapy**
 - 34 Glioblastoma Multiforme: Use of Adenoviral Vectors**
 - 35 Fischer-F98 Glioma Model: Methodology**
 - 36 Cellular Characterization of Anti-VEGF and Il-6 Therapy in Experimental Glioma**
 - 37 Adult Brainstem Gliomas: Diagnosis and Treatment**
 - 38 Use of Low Molecular Weight Heparin in the Treatment and Prevention of Thromboembolic Disease in Glioma Patients**
 - 39 Brainstem Gliomas: An Overview**
 - 40 Tumor-Associated Epilepsy in Patients with Glioma**
 - 41 Chronic Epilepsy Associated with Brain Tumors: Surgical Neuropathology**
 - 42 Low-Grade Gliomas: Role of Relative Cerebral Blood Volume in Malignant Transformation**
 - 43 Angiocentric Glioma- Induced Seizures: Lesionectomy**

Contents of Volume 3

- 1 General Introduction**
- 2 Epidemiology of Primary Brain Tumors**
- 3 Brain Tumor Classification Using Magnetic Resonance Spectroscopy**
- 4 Cellular Immortality in Brain Tumors: An Overview**
- 5 Tumor-to-Tumor Metastases: Extracranial Tumor Metastasis to Intracranial Tumors**
- 6 Brain Metastases from Breast Cancer: Treatment and Prognosis**
- 7 Brain Metastasis in Renal Cell Carcinoma Patients**
- 8 Coexistence of Inflammatory Myofibroblastic Tumors in the Lung and Brain**
- 9 Breast Cancer and Renal Cell Cancer Metastases to the Brain**
- 10 Brain Metastases from Breast Cancer: Genetic Profiling and Neurosurgical Therapy**
- 11 Central Nervous System Tumors in Women Who Received Capecitabine and Lapatinib Therapy for Metastatic Breast Cancer**
- 12 Functional Role of the Novel NRP/B Tumor Suppressor Gene**
- 13 Brain Tumors: Diagnostic Impact of PET Using Radiolabelled Amino Acids**
- 14 Malignant Peripheral Nerve Sheath Tumors: Use of ¹⁸FDG-PET/CT**
- 15 Brain Tumors: Evaluation of Perfusion Using 3d-FSE-Pseudo-Continuous Arterial Spin Labeling**
- 16 Cerebral Cavernous Malformations: Advanced Magnetic Resonance Imaging**
- 17 Nosologic Imaging of Brain Tumors Using MRI and MRSI**
- 18 Oku: Brain Tumor Diagnosis Using PET with Angiogenic Vessel-Targeting Liposomes**

-
- 19 Frozen Section Evaluation of Central Nervous System Lesions**
 - 20 Clinical Role of MicroRNAs in Different Brain Tumors**
 - 21 Electrochemotherapy for Primary and Secondary Brain Tumors**
 - 22 Brain Tumors: Convection-Enhanced Delivery of Drugs (Method)**
 - 23 Brain Metastases: Clinical Outcomes for Stereotactic Radiosurgery (Method)**
 - 24 Noninvasive Treatment for Brain Tumors: Magnetic Resonance Guided Focused Ultrasound Surgery**
 - 25 Menard: Radioguided Surgery of Brain Tumors**
 - 26 Implications of Mutant Epidermal Growth Factor Variant III in Brain Tumor Development and Novel Targeted Therapies**
 - 27 Endoscopic Port Surgery for Intraparenchymal Brain Tumors**
 - 28 Intracranial Tumor Surgery in the Elderly Patients**
 - 29 Intracranial Hemangiopericytoma: Gamma Knife Surgery**
 - 30 Stereotactic Radiosurgery for Cerebral Metastasis of Digestive Tract Tumors**
 - 31 Malignant Brain Tumors: Role of Radioresponsive Gene Therapy**
 - 32 Brain Tumors: Quality of Life**
 - 33 Health Related Quality of Life in Patients with High-Grade Gliomas**
 - 34 Epilepsy and Brain Tumors and Antiepileptic Drugs**
 - 35 Familial Caregivers of Patients with Brain Cancer**
 - 36 Pain Management Following Craniotomy**
 - 37 Air Transportation of Patients with Brain Tumors**

Contents of Volume 4

- 1 Epidemiology of Primary Brain Tumors**
- 2 Supratentorial Primitive Neuroectodermal Tumors**
- 3 Adult Neurogenesis in Etiology and Pathogenesis of Alzheimer's Disease**
- 4 Epileptic and Supratentorial Brain Tumors in Children**
- 5 Breast Cancer Metastasis to the Central Nervous System**
- 6 Melanoma to Brain Metastasis: Photoacoustic Microscopy**
- 7 Extraaxial Brain Tumors: The Role of Genetic Polymorphisms**
- 8 Central Nervous System Germ Cell Tumor**
- 9 Microvascular Gene Changes in Malignant Brain Tumors**
- 10 Role of MicroRNA in Glioma**
- 11 Glioblastoma Multiforme: Cryopreservation of Brain Tumor-Initiation Cells (Method)**
- 12 Relationship Between Molecular Oncology and Radiotherapy in Malignant Gliomas (An Overview)**
- 13 High-Grade Brain Tumors: Evaluation of New Brain Lesions By Amino Acid PET**
- 14 Cyclic AMP Phosphodiesterase-4 in Brain Tumor Biology: Immunochemical Analysis**
- 15 Time-Resolved Laser Induced Fluorescence Spectroscopy (TRLIFS): A Tool for Intra-Operative Diagnosis of Brain Tumors and Maximizing Extent of Surgical Resection**
- 16 Molecular Imaging of Brain Tumors Using Single Domain Antibodies**
- 17 Quantitative Analysis of Pyramidal Tracts in Brain Tumor Patients Using Diffusion Tensor Imaging**
- 18 Differentiation Between Gliomatosis Cerebri and Low-Grade Glioma: Proton Magnetic Resonance Spectroscopy**

-
- 19 Peripheral Nerve Sheath Tumors: Diagnosis Using Quantitative FDG-PET**
 - 20 Tumor Resection Control Using Intraoperative Magnetic Resonance Imaging**
 - 21 Brain Tumors: Clinical Applications of Functional Magnetic Resonance Imaging and Diffusion Tensor Imaging**
 - 22 Trigeminal Neuralgia: Diagnosis Using 3-D Magnetic Resonance Multi-Fusion Imaging**
 - 23 Epilepsy-Associated Brain Tumors: Diagnosis Using Magnetic Resonance Imaging**
 - 24 Growth of Malignant Gliomas**
 - 25 Resection of Brain Lesions: Use of Preoperative Functional Magnetic Resonance Imaging and Diffusion Tensor Tractography**
 - 26 Paradigms in Tumor Bed Radiosurgery Following Resection of Brain Metastases**
 - 27 Rat Model of Malignant Brain Tumors: Implantation of Doxorubicin Using Drug Eluting Beads for Delivery**
 - 28 Electromagnetic Neuronavigation for CNS Tumors**
 - 29 Stereotactic Radiosurgery for Intracranial Ependymomas**
 - 30 Is Whole Brain Radiotherapy Beneficial for Patients with Brain Metastases?**
 - 31 Triggering Microglia Ontotoxicity: A Bench Utopia of a Therapeutic Approach?**
 - 32 Preoperative Motor Mapping**
 - 33 Intraoperative Monitoring for Cranial Base Tumors**
 - 34 Brain Tumors: Pre-Clinical Assessment of Targeted, Site Specific Therapy Exploiting Ultrasound and Cancer Chemotherapeutic Drugs**
 - 35 Headaches in Patients with Brain Tumors**
 - 36 Headache Associated with Intracranial Tumors**
 - 37 Patients with Brain Cancer: Health Related Quality of Life**
 - 38 Emerging Role of Brain Metastases in the Prognosis of Breast Cancer Patients**

Contents of Volume 5

- 1 Methylation in Malignant Astrocytomas**
- 2 Deciphering the Function of Doppel Protein in Astrocytomas**
- 3 Astrocytic Tumors: Role of Antiapoptotic Proteins**
- 4 Astrocytomas: Role of Wnt/ β -Catenin/Tcf Signaling Pathway**
- 5 Subependymal Giant Cell Astrocytoma: Role of mTOR Pathway and its Inhibitors**
- 6 Role of Progesterone Receptor Isoforms in Human Astrocytomas Growth**
- 7 Astrocytic Tumors: Role of Carbonic Anhydrase IX**
- 8 Development of Cysts in Pilocytic Astrocytomas: Role of Eosinophilic Granular Bodies (Method)**
- 9 Role of Synemin in Astrocytoma Cell Migration**
- 10 Diffuse Astrocytomas: Immunohistochemistry of MGMT Expression**
- 11 Central Nervous System Germ Cell Tumors: An Epidemiology Review**
- 12 RAF Genes and MAPK Activation in Pilocytic Astrocytomas**
- 13 Biomarker Discovery in Central Nervous System Neoplasms: Past, Present and Future**
- 14 Astrocytomas: Role of Taurine in Apoptosis Using Magnetic Resonance Spectroscopy**
- 15 Imaging of Hypoxia-Inducible Factor-1-Active Regions in Tumors Using a POS and ^{123}I -IBB Method**
- 16 Diffuse Low-Grade Astrocytomas: P53-Mediated Inhibition of Angiogenesis**
- 17 Spontaneous Regression of Cerebellar Astrocytomas**
- 18 Subependymal Giant Cell Astrocytoma: Gene Expression Profiling**

-
- 19 Time-Resolved Laser Induced Fluorescence Spectroscopy (TRLIFS): A Tool for Intra-Operative Diagnosis of Brain Tumors and Maximizing Extent of Surgical Resection**
 - 20 Magnetic Resonance-Guided Laser Interstitial Thermal Therapy for Brain Tumors**
 - 21 Nanotechnology-Based Therapy for Malignant Tumors of the Central Nervous System**
 - 22 Pilocytic Astrocytoma: Pathological and Immunohistochemical Factors Affecting Surgical Treatment and Surveillance**
 - 23 Pilomyxoid Astrocytoma: Chemotherapy**
 - 24 Astrocytomas: Predicting Survival and Recurrence Using Cerebral Blood Volume Measurements**
 - 25 Electronic Patient-Reported Outcome Monitoring (EPROM) in Brain Tumour Patients**
 - 26 Intra-Operative ICG Use in the Management of Hemangioblastomas**
 - 27 Hemangioblastoma Cysts: Diagnosis Using Fluorescence with 5-Aminolevulinic Acid**
 - 28 Hemangioblastoma-Stereotactic Radiosurgery**
 - 29 Gangliogliomas: Molecular Pathogenesis and Epileptogenesis**
 - 30 Epilepsy-Associated Gangliogliomas: Identification of Genes with Altered Expression**

Contents of Volume 6

- 1 General Introduction**
- 2 Pediatric Mixed Glioneuronal Tumors in the Spinal Cord**
- 3 Intradural Spinal Tumors: Classification, Symptoms, and Radiological Features**
- 4 Non-Dysraphic Intradural Spinal Cord Lipoma: Management Guidelines**
- 5 Malignant Astrocytomas of the Spinal Cord: Clinicopathologic Parameters**
- 6 Spinal Epidural Angiolipoma**
- 7 Spinal Cord Tumor Oligodendroglioma: Diagnosis**
- 8 Primary Spinal Oligodendroglioma: Diagnosis, Outcome, and Prognosis**
- 9 Pilomyxoid Astrocytoma of the Spinal Cord with Cerebrospinal Fluid and Peritoneal Metastasis**
- 10 Intraspinal Oncocytic Adrenocortical Adenoma: Diagnosis**
- 11 Chordomas of the Clivus and Upper Cervical Spine**
- 12 Spinal Teratoid/ Rhabdoid Tumor: Use of Diffusion - Weighted Imaging for Diagnosis**
- 13 Gangliogliomas of the Spinal Cord: Neuroimaging Correlations with Pathology, Controversies in Pathological Diagnosis, and Prognosis**
- 14 Surgery for Spinal Tumours**
- 15 Resection of Spinal Meningioma: Postoperative Focal Hyperemia**
- 16 Spinal Cord Hemangioblastomas: Surgical Management**
- 17 Spinal Radiosurgery: Delayed Radiation-Induced Myelopathy**
- 18 Metastatic Spine Disease: Indications, Timing, and Outcomes for Surgery and Radiation Therapy**

-
- 19 Sequence of Surgery, Radiotherapy, and Stereotactic Radiosurgery in the Treatment of Metastatic Spine Disease: Effects on Wound Healing**
 - 20 Treatment of Spinal Tumors with Cyberknife Stereotactic Radiotherapy**
 - 21 Recurrent Spinal Cord Cystic Astrocytomas: Treatment with Rhenium-186 Intracavitary Radiation**
 - 22 Embolization of Spinal Tumors**
 - 23 Embolization of Spinal Cord Tumours**
 - 24 Locomotor Recovery After Spinal Cord Transaction: Transplantation of Oligodendrocytes and Motoneuron Progenitors Generated From Human Embryonic Stem Cells**
 - 25 Malignant Primary Spinal Neoplasms: Total *En Bloc* Spondylectomy**

Contents of Volume 7

- 1 Meningiomas: Role of Semaphorin3A Protein in Antiangiogenesis**
- 2 Meningiomas: Role of Carbonic Anhydrase II**
- 3 Meningiomas: Role of Genetic Instabilities of the E-Cadherin Gene**
- 4 Intracranial Meningiomas: Role of Exogenous Hormones**
- 5 Meningiomas: Clinical Needs and Molecular Insights**
- 6 Meningioma: Urokinase Plasminogen Activator**
- 7 miR-200a Regulation of the Wnt Signaling in Meningioma Tumorigenesis**
- 8 Meningiomas: Determination of Subtypes Using Perfusion Magnetic Resonance Imaging**
- 9 Intracranial Meningioma in Mice: Noninvasive Bioluminescence Imaging**
- 10 Incidentally Discovered Meningiomas: Growth Rates and Patterns**
- 11 Cystic Papillary Meningioma: Diagnosis**
- 12 Meningioma Tumors: Detection of Subgroups**
- 13 Prognostic Parameters in Atypical and Malignant Meningiomas**
- 14 Sporadic Meningioangiomatosis: Diagnosis with Computed Tomography and Magnetic Resonance Imaging**
- 15 Atypical Meningioma: Distinguishing Features and Role of Adjuvant Radiation**
- 16 Gamma Knife Radiosurgery for Benign Meningioma: Significance and Therapeutic Potential**
- 17 Oncocytic Meningioma: Neurosurgery**
- 18 Intracranial Meningiomas: Treatment and Quality of Life**
- 19 Cavernous Sinus Meningiomas: Optimal Treatment**

-
- 20 Genetic and Clinical Features Associated with Recurrence in Atypical Meningioma**
 - 21 Recurrence and Progression in Meningiomas**
 - 22 Meningioma: Role of Erythropoietin Receptor in the Tumor Recurrence**
 - 23 Cyclin D1 Expression in Vestibular Schwannoma**
 - 24 Schwannomas: Role of Molecular Genetics and Epigenetic Mechanisms**
 - 25 Facial Nerve Schwannoma: Diagnosis Using Magnetic Resonance Imaging**
 - 26 Vestibular Schwannoma: Optimizing Tumor Growth Monitoring by Volume Measurements**
 - 27 Intermediate Nerve Schwannomas**
 - 28 Vestibular Schwannoma, Radiosurgery and Hydrocephalus**
 - 29 Solitary Vestibular Schwannoma: Decision Making of Treatments**
 - 30 Stereotactic Radiosurgery for Trigeminal Schwannoma: Tumor Control and Functional Preservation**
 - 31 Vestibular Schwannoma: Gamma Knife Radiosurgery**
 - 32 Vestibular Schwannoma: Gamma Knife Radiosurgery (Method)**
 - 33 Vestibular Schwannoma Surgery: Use of Fat Implant to Prevent Cerebrospinal Fluid Fistula**
 - 34 Retrosigmoidal Craniotomy for Vestibular Schwannoma Patients: Postoperative Cerebrospinal Fluid Leak**
 - 35 Vestibular Schwannoma Surgery: Histological Considerations and Operative Results**
 - 36 Vestibular Schwannomas: Treatment with Bevacizumab**
 - 37 Management of Vestibular Schwannoma Patients: Quality of Life Outcomes After Treatment**

Contents of Volume 8

- 1 Astrocytoma Cell Line: Role of Brain Natriuretic Peptide**
- 2 Malignant Brain Astrocytomas: Extent of Resection Affects Survival**
- 3 Medulloblastoma: Classification (A Review)**
- 4 Medulloblastomas: Clinically Important MicroRNA Genes**
- 5 Medulloblastoma: Role of OTX2 Transcription Factors**
- 6 Molecular Mechanisms of Chemoresistance in Medulloblastoma**
- 7 Extranodal Metastasis in Medulloblastoma**
- 8 Medulloblastoma: Therapy with Bortezomib/Tumor Necrosis Factor-Related Apoptosis-Inducing Ligand**
- 9 Standard-Risk Medulloblastoma: Hyperfractionated Radiotherapy**
- 10 Retinoma and Retinoblastoma: Genomic Hybridisation**
- 11 Cell Cycle Control by Ataxia Telangiectasia Mutated Protein Through Regulating Retinoblastoma Protein Phosphorylation**
- 12 Role of Survivin in Retinoblastoma: Diagnosis and Prognosis**
- 13 Retinoblastoma: The Role of Epigenetics**
- 14 Retinoblastoma: Disease, Diagnosis, Therapy and Management**
- 15 Long-Term Survivors of Retinoblastoma: Risk of Increased Second Malignancy**
- 16 New Cancers Among Long-Term Survivors of Retinoblastoma**
- 17 Chordoma: Role of Cam5.2**
- 18 Chordomas and Chondrosarcomas: Treatment with Particle Radiotherapy**
- 19 Skull Base Chordomas: Endonasal Endoscopic Transclival Approach**
- 20 Craniopharyngioma: Comparison Between Supra-Orbital Versus Endonasal Keyhole Approaches**

-
- 21 The Expanded Endoscopic Endonasal Approach
for Primary and Recurrent Craniopharyngiomas**
 - 22 Craniopharyngioma: The Role of Radiation**
 - 23 Cystic Craniopharyngiomas: Intratumoral Bleomycin Therapy**
 - 24 Anaplastic Oligodendroglioma Metastasized to Extraneural Sites**
 - 25 Recurrent Oligodendroglioma: Treatment with Bevacizumab**
 - 26 Ependymoma: An Overview**
 - 27 Ependymomas: Prognosis Based on Genetic Aberrations**
 - 28 Aberrant DNA Methylation in Ependymomas**
 - 29 Progressively Metastasizing Ependymoma: Genomic Aberrations**
 - 30 Extradural Ependymoma: Diagnosis Using Magnetic
Resonance Imaging**
 - 31 Primary Malignant Ependymoma of the Abdominal Cavity:
Diagnosis**
 - 32 Atypical Histologic Features and Patterns of Malignant
Evolution in Tanycytic Ependymoma**
 - 33 Intracranial Ependymoma: Role for Chemotherapy**

Contents of Volume 9

- 1 Neurolymphomatosis: Diagnosis, Treatment, and Outcome**
- 2 Primary Central Nervous System Lymphoma: Systemic Relapse**
- 3 Central Nervous System Recurrence in the Primary Mediastinal Large B-Cell Lymphoma: Treatment**
- 4 Primary Central Nervous System Lymphoma Resulting in Stroke and Leukoencephalopathy**
- 5 Primary CNS Lymphoma: Immunohistochemistry of Bcl-6 and Treatment with High-Dose Methotrexate**
- 6 Thiamine Deficiency Complicating the Treatment of Primary CNS Lymphoma**
- 7 Metastatic Brain Irradiation-Induced Lymphocytosis Predicts Efficacy of Radiotherapy**
- 8 Primary Central Nervous System Lymphoma: Treatment with High-Dose Methotrexate**
- 9 Paraneoplastic Syndromes in Primary CNS Lymphoma**
- 10 Supratentorial Primitive Neuroectodermal Tumor: Biology**
- 11 Outpatient Brain Biopsy and Craniotomy for Supratentorial Tumor**
- 12 Wrong-Site Craniotomy Prevention**
- 13 Diffuse Leptomeningeal Glioneuronal Tumors: Histology. Is it a New Entity?**
- 14 Temporomesial Glioneuronal Tumors: Epilepsy Surgery**
- 15 Rosette-Forming Glioneural Tumor: Conservative Management Strategy**
- 16 Ganglioneuroma: An Overview**
- 17 Ganglioglioma, mTOR Activation, and Epileptogenesis**

-
- 18 Gangliogliomas and Other Low Grade Neuronal Neoplasms of the Central Nervous System: Diagnosis, Treatment, and Prognosis**
 - 19 Adults Neuroblastoma Diagnosis**
 - 20 Proliferation of Neuroblasts in the Adult Brain: Role of Diversin**
 - 21 Subependymal Giant Cell Astrocytoma: Treatment**
 - 22 Acquired Retinal Astrocytoma**
 - 23 Presence of Both Ependymoma and Astrocytoma in the Same Patient: Diagnosis**
 - 24 Total Removal of Cavernous Hemangioma Using the Tonsillouveal Transaqueductal Approach (Method)**
 - 25 Cavernous Sinus Hemangiomas Treated with Gamma Knife Surgery**
 - 26 Linear Accelerator Radiosurgery for Cavernous Malformation**
 - 27 Treatment of Brainstem Hemangioblastomas**
 - 28 Craniopharyngiomas: An Overview**
 - 29 Radical Removal of Craniopharyngiomas**
 - 30 Neurogenesis Outside the Central Nervous System (An Overview)**
 - 31 Neurogenesis and Reproduction**

Contents of Volume 10

- 1 Papillary Tumor of the Pineal Region**
- 2 Pineal Region Tumors: Clinical Aspects**
- 3 Papillary Tumor of the Pineal Region: Diagnosis**
- 4 Pineal Parenchymal Tumors: Immunohistochemistry**
- 5 Pineal Parenchymal Tumors: Diagnostics and Prognosis**
- 6 Papillary Tumors of the Pineal Region: Diagnosis and Treatment**
- 7 Pineal Region Tumors: Optimal Neurosurgical Treatment**
- 8 Pituitary Tumors: Genetics and Heritable Predisposition**
- 9 Xanthogranulomas Associated with Pituitary Adenomas: Magnetic Resonance Imaging**
- 10 Pituitary Adenoma and Craniopharyngioma: An Overview**
- 11 Familial Pituitary Adenomas: An Overview**
- 12 Papillary Glioneuronal Tumor**
- 13 Solitary Fibrous Tumor**
- 14 Pituitary Adenomas: MCM2 Protein as a Cell Proliferation Marker**
- 15 Pituitary Adenomas: Role of Cyclin-Dependent Kinase Inhibitors**
- 16 Pituitary Tumorigenesis: Role of Regulation of Wee1 Kinase by MicroRNAs**
- 17 Pituitary Tumor Cells: Role of PKC α , PKC δ , and Pck ϵ Expression**
- 18 Pituitary Adenoma: Role of HMGA Proteins**
- 19 Pituitary Adenomas: Role of E-Cadherin in Tumor Invasion**
- 20 Pituitary Tumorigenesis: Role of the Wnt Signaling Pathway**

-
- 21 The Role of Aryl Hydrocarbon Receptor (AHR) and AHR-Interacting Protein (AIP) in the Pathogenesis of Pituitary Adenomas**
 - 22 Pituitary Tumors: Role of Pituitary Tumor-Transforming Gene-1 (PTTG1)**
 - 23 Pituitary Adenomas: Endoscopic Endonasal Transphenoidal Technique**
 - 24 Pituitary Adenoma Patients: Hypofractionated Cyberknife Radiosurgery (Method)**
 - 25 Transsphenoidal/Transcranial Surgery of Pituitary Adenomas: Prognosis-Related Occurrence for the Trigemino-Cardiac Reflex**
 - 26 Spinal Extradural Meningiomas**
 - 27 Spinal Cord Ganglioglioma**
 - 28 Spinal Angiolipoma: Diagnosis and Treatment**
 - 29 Spinal Cord Injury: Tissue Engineering Using Neural Stem Cells**
 - 30 Pediatric Spinal Tumors: Total Removal Using Laminotomy**
 - 31 Treatment of Metastatic Spinal Epidural Disease: Surgery Versus Radiotherapy**
 - 32 Metastatic Spinal Cord Compression from Synovial Sarcoma: Surgical Resection**
 - 33 Adult Spinal Intramedullary Ependymomas: Complete Resection**
 - 34 Spinal Intramedullary Astrocytomas: Prognostic Factors**

Contents of Volume 11

- 1 An Introduction to Brain Tumor Imaging**
- 2 Molecular Imaging of Brain Tumors**
- 3 Use of ^{11}C -4DST-PET for Imaging of Human Brain Tumors**
- 4 Diffusion Tensor Magnetic Resonance Imaging-Based Tractography for Glioma Surgery**
- 5 Angiocentric Glioma, Pilomyxoid Astrocytoma, and Pituicytoma: New Entities in the World Health Organization Classification**
- 6 Diffuse Intrinsic Pontine Gliomas in Children: Treatment (An Update)**
- 7 Gliomatosis Cerebri: Implications of Genetic Findings**
- 8 Impact of Bevacizumab Chemotherapy on Glioblastomas**
- 9 Glioblastoma Microvesicles Transport RNA and Proteins Promoting Tumor Growth**
- 10 Microrna in The Pathophysiology of Glioblastoma**
- 11 Surgical Management of Incident Low-Grade Gliomas**
- 12 Stereotactic Radiosurgery for the Treatment of Brain Metastases**
- 13 Local Control of Low-Volume Brain Metastases Using Stereotactic Radiosurgery**
- 14 Tumor Seeding After Stereotactic Brain Biopsy**
- 15 Intracranial Stereotactic Radiosurgery in High Risk Patients with Metastases from Radioresistant Primary Tumors**
- 16 Brain Metastases: Treatment with Stereotactic Iodine-125 Brachytherapy**
- 17 Stereotactic Radiosurgery for Skull Base Meningiomas**
- 18 Pediatric Spinal Cord Tumors**
- 19 Radiosurgery for Intramedullary Spinal Cord Tumors**

-
- 20 Novalis Spinal Radiosurgery**
 - 21 Etiology and Outcomes of Pediatric Spinal Cord Tumors**
 - 22 Selection of Elderly Meningioma Patients for Surgery Using a Clinical-Radiological Grading System as a Predictor of Outcome**
 - 23 Cranial Meningioma in Neurofibromatosis Type 2 Patients: Role of Mutations**
 - 24 Clasification of Meningioma Using Immunogenic Antigens**
 - 25 Treatment of Benign Meningiomas Using Radiosurgery**
 - 26 Necrotic Rhabdoid Meningiomas: Histopathological Diagnosis**
 - 27 Multiple Schwannomas: Diagnosis and Treatment**
 - 28 Treatment of Patients with Vestibular Schwanommas Using Gamma Knife Radiosurgery**
 - 29 Schwannomas in the Craniocervical Region: Complete Surgical Removal**
 - 30 Cranial Base Surgery of Hypoglossal Schwannomas**

Contents of Volume 12

- 1 Classification/Diagnosis of Brain Tumors Using Discriminant Function Analysis**
- 2 Neuronal Differentiation in the Adult Brain: CDK6 as the Molecular Regulator**
- 3 Prolactinomas: Role of VEGF, FGF-2 And CD31**
- 4 Human Brain Tumor Growth: Role of Aquaporins**
- 5 Molecular and Functional Characterization of Human Adipocytes**
- 6 Inhibition of Neuroblastoma Progression by Targeting Lymphangiogenesis: Role of an Endogenous Soluble Splice-Variant of VEGFR-2**
- 7 Hyperinsulinemia Tends to Induce Growth Without Growth Hormone in Children with Brain Tumors After Neurosurgery**
- 8 Childhood Brain Tumors: Proton Beam Therapy**
- 9 Pathogenesis of Medulloblastoma: Role of Molecular Genetic Alterations**
- 10 Medulloblastomas: Survival Differences Between Children and Adults**
- 11 Survivors of Childhood Cancer: Risk of Glioma and Meningioma Following Radiotherapy**
- 12 Survivors of Childhood Cancer: Risk of New Primary Neoplasms of the CNS**
- 13 Brain Tumor Typing and Therapy Using Combined Ex Vivo Magnetic Resonance Spectroscopy and Molecular Genomics**
- 14 Atlas-Based Segmentation of Tumor-Bearing Brain Images**
- 15 Nonthermal Irreversible Electroporation as a Focal Ablation Treatment for Brain Cancer**
- 16 The Role of Whole Brain Radiation Therapy for Metastatic Brain Tumors**

-
- 17 Craniotomy for Intracranial Tumors: Role of Postoperative Hematoma in Surgical Mortality**
 - 18 Pituitary Adenomas: Treatment Using the Endonasal Approach**
 - 19 Intracranial Tumors: Treatment with Endoscopic Neurosurgery**
 - 20 Incidence of Recraniotomy for Postoperative Infections After Surgery for Intracranial Tumors**
 - 21 Transsphenoidal Surgery for Non-Adenomatous Tumors: Effect on Pituitary Function**
 - 22 Treatment of Brain Tumors: Electrochemotherapy**
 - 23 Comparison of Navigated Transcranial Magnetic Stimulation to Direct Electrical Stimulation for Mapping the Motor Cortex Prior to Brain Tumor Resection**
 - 24 The Role of Glutathione and the Glutathione-Linked Enzyme Systems in Brain Tumor Drug Resistance**
 - 25 Drugs for Primary Brain Tumors: An Update**
 - 26 Neurosurgical Treatment for Brain Metastases: Clinical Features, Operative Strategies, Recurrence and Survival**
 - 27 Image-Guided Radiosurgery for Hypothalamic Hamartomas**
 - 28 Image-Guided Stereotactic Radiosurgery for Spinal Pathology**
 - 29 Image Guided Stereotactic Radiosurgery for Optic Nerve Sheath Meningiomas**
 - 30 Brain Metastases: The Application of Stereotactic Radiosurgery and Technological Advances**
 - 31 Treatment of Chordomas by Stereotactic Radiosurgery with a Linear Accelerator: Comparison with Other Modes**

Contents of Volume 13

- 1 Embryonal Tumor: Molecular Characterization**
- 2 Oligodendroglial Tumors: Intra-Arterial Chemotherapy**
- 3 Metastatic Oligodendroglioma: Diagnosis with Fine-Needle Aspiration Cytology**
- 4 Management of Hemangiopericytoma**
- 5 Role of Cyclooxygenase-2 in the Development and Growth of Schwannomas**
- 6 Adult Primary Gliosarcoma: Epidemiology**
- 7 Mesenchymal Chondrosarcoma in the Central Nervous System: Histological Diagnosis**
- 8 Supratentorial Primitive Neuroectodermal Tumors (PNETs)**
- 9 Retinoblastoma and Reproductive Decision-Making**
- 10 Trigeminal Neuralgia with Cerebellopontine Angle Tumors**
- 11 The Concept of a Preeniche for Localization of Future Metastases**
- 12 Computer Systems for Cell Counting in Histopathologic Slides of Tumours of the Central Nervous System: Advantages and Limitations**
- 13 Intraoperative Ultrasonography in Tumor Surgery**
- 14 Advantage of Intraoperative Power Doppler Ultrasonography for Intracranial Tumors**
- 15 Intraoperative Ultrasound in Neurosurgical Oncology - Scope and Utility**
- 16 Resection of Brain Tumors: Intraoperative Confocal Microscopy Technology**
- 17 Brainstem Cavernomas, Accessible Lesions: Surgery**
- 18 The Role of Surgical Resection for Metastatic Brain Tumors**
- 19 Factors Responsible for Local Recurrence of Brain Metastasis**

-
- 20 Role of MMP2 in Brain Metastasis**
 - 21 Differentiating Choroid Plexus Tumors from Metastatic Carcinomas: use of Inwardly Rectifying K⁺ Channel KIR7.1 and Excitatory Amino Acid Transporter-1**
 - 22 Alexander Disease: Role of Glial Fibrillary Acidic Protein**
 - 23 Lipoma: An Overview**
 - 24 Tumefactive Demyelination**
 - 25 Immunotherapies for Brain Cancer: From Preclinical Models to Human Trials**
 - 26 The Role of Hyaluronic Acid and Its Receptors in the Growth and Invasion of Brain Tumors**
 - 27 Neonatal Hypoxic-Ischemic Brain Damage: Human Umbilical Cord Blood Mononuclear Cells Transplantation**
 - 28 Pathological Angiogenesis: An Overview**
 - 29 Use of Mobile Phones and Brain Cancer Risk in Children?**

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Angiocentric Glioma, Pilomyxoid Astrocytoma, and Pituicytoma: New Entities in the World Health Organization Classification

Jamie L. Odem and Douglas C. Miller

Contents

Abstract	1
Introduction	2
Angiocentric Glioma	2
Introduction and Historical Context.....	2
Clinical Presentation and Imaging.....	2
Histopathology.....	3
Genetic Abnormalities.....	4
Differential Diagnosis.....	4
Treatment and Prognosis.....	5
Pilomyxoid Astrocytoma	5
Introduction and Historical Context.....	5
Clinical Presentation/Imaging.....	5
Histopathology.....	6
Genetic Abnormalities.....	7
Differential Diagnosis.....	7
Prognosis and Treatment.....	7
Pituicytoma	8
Intro/Historical Context.....	8
Clinical Presentation/Imaging.....	8
Histopathology.....	9
Genetic Abnormalities.....	9
Differential Diagnosis.....	9
Prognosis and Treatment.....	10
References	10

Abstract

The 2007 World Health Organization (WHO) Classification of CNS Tumors added three new entities to its compendium of primary CNS neoplasms which are similar in that they consist of relatively monomorphous spindle cells: angiocentric glioma (AG), pilomyxoid astrocytoma (PMA), and pituicytoma. This chapter describes the clinical features, histopathology, diagnosis, treatment, and prognosis of these three newly codified entities. Angiocentric glioma, a cortically-based neoplasm that causes intractable epilepsy in young people, shares many similarities with ependymoma. However, these similarities were not enough to classify it as an ependymoma subtype, and AG is under “Other Neuroepithelial Tumors” in the new classification. A WHO Grade I tumor, it is an indolent, slow-growing neoplasm characterized by bipolar cells with spindled nuclei that tend to cluster around intratumoral vessels (giving this tumor its name). The primary differential diagnosis is cortical ependymoma but also includes pilomyxoid astrocytoma. Angiocentric glioma is usually cured with surgical excision, even in subtotally resected tumors. Pilomyxoid astrocytoma is categorized as an aggressive variant of pilocytic astrocytoma encountered mainly in very young children but may present in older children or adults. PMA is usually located in the optic chiasm/hypothalamic region. Histologically, PMA is a strikingly

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monomorphous tumor composed of bipolar cells in a richly myxoid background which lacks the biphasic pattern of pilocytic astrocytoma and in which the tumor cells form ependymoma-like perivascular pseudorosettes. Considerations in the differential of PMA include pilocytic astrocytoma with a myxoid component, angiocentric glioma, and infiltrating glioma. Because PMA behaves more aggressively, following a biopsy diagnosis, it is treated with chemotherapy for low-grade gliomas with or without gross total resection, depending on tumor burden. Last, pituitaryoma is listed as a separate entity under “Tumors of the Sellar Region” and is a solid tumor of the posterior pituitary stalk. Thought to be derived from pituicytes, the glial cells of the pituitary stalk and neurohypophysis, this tumor presents in adults as an intrasellar or suprasellar mass causing visual or hormonal disturbances. Histopathologically, pituitaryomas are composed of spindled, bipolar cells which arrange themselves in a characteristic storiform pattern. The differential diagnosis of pituitaryoma includes pituitary adenoma, spindle cell oncocyoma, and granular cell tumor. Pituitaryoma is cured by surgical excision and is WHO Grade I.

Introduction

The most recent World Health Organization (4th Edition, 2007) “Blue Book” for central nervous system (CNS) tumors adds three new entities to the list of tumor types, all of which are similar in that they are composed mostly of monomorphous spindle cells: angiocentric glioma, pilomyxoid astrocytoma, and pituitaryoma. All three of these entities are relatively newly described (within the last 15 years). As case reports and small case series have accumulated, it has become clearer how these entities should be formally classified, thus leading to their adoption into the WHO classification scheme. This chapter will review the historical context, clinical presentation including radiographic findings, histopathology, genetic abnormalities, differential diagnosis, and treatment of these three entities.

Angiocentric Glioma

Introduction and Historical Context

Angiocentric glioma first came to the forefront of the literature in 2005, with two nearly-simultaneous case series (Lellouch-Tubiana et al. 2005; Preusser et al. 2006), although Wang et al. had previously described this entity in 2002 as angiocentric bipolar astrocytoma (Wang et al. 2002). These case series described cortically-based tumors with superficial subcortical extension in young patients with intractable epilepsy, usually partial seizures. At the time of this writing, there have been only 44 cases of angiocentric glioma described (Pokharel et al. 2011). Histologically, these tumors are composed of spindled, monomorphous cells which cluster around vessels, somewhat like cells of most ependymomas, and lack atypia. They typically express Epithelial Membrane Antigen (EMA) in a cytoplasmic dot-like positive pattern similar to ependymomas, and the perivascular pattern and this immunohistochemical feature have engendered speculation that these are a variant of ependymoma, although the WHO classification places them outside of the ependymoma group. Angiocentric glioma is characterized as WHO Grade I as it is usually cured by surgical resection, even in some cases by subtotal resections. Synonyms for this entity include monophasic angiocentric glioma and angiocentric neuroepithelial tumor (ANET).

Clinical Presentation and Imaging

Angiocentric gliomas affect mainly young adults (mean age 17 years), but have been described in younger children and older adults (up to 70 years of age) (Brat et al. 2007). To date, all described examples have been located peripherally in the cerebral hemispheres, most often the temporal lobe, with parietal and frontal locations also being common (Lellouch-Tubiana et al. 2005; Preusser et al. 2006; Koral et al. 2012). (We have seen one example which histologically resembled angiocentric glioma but which was located in the dorsal midbrain’s tectal region. The diagnosis for this

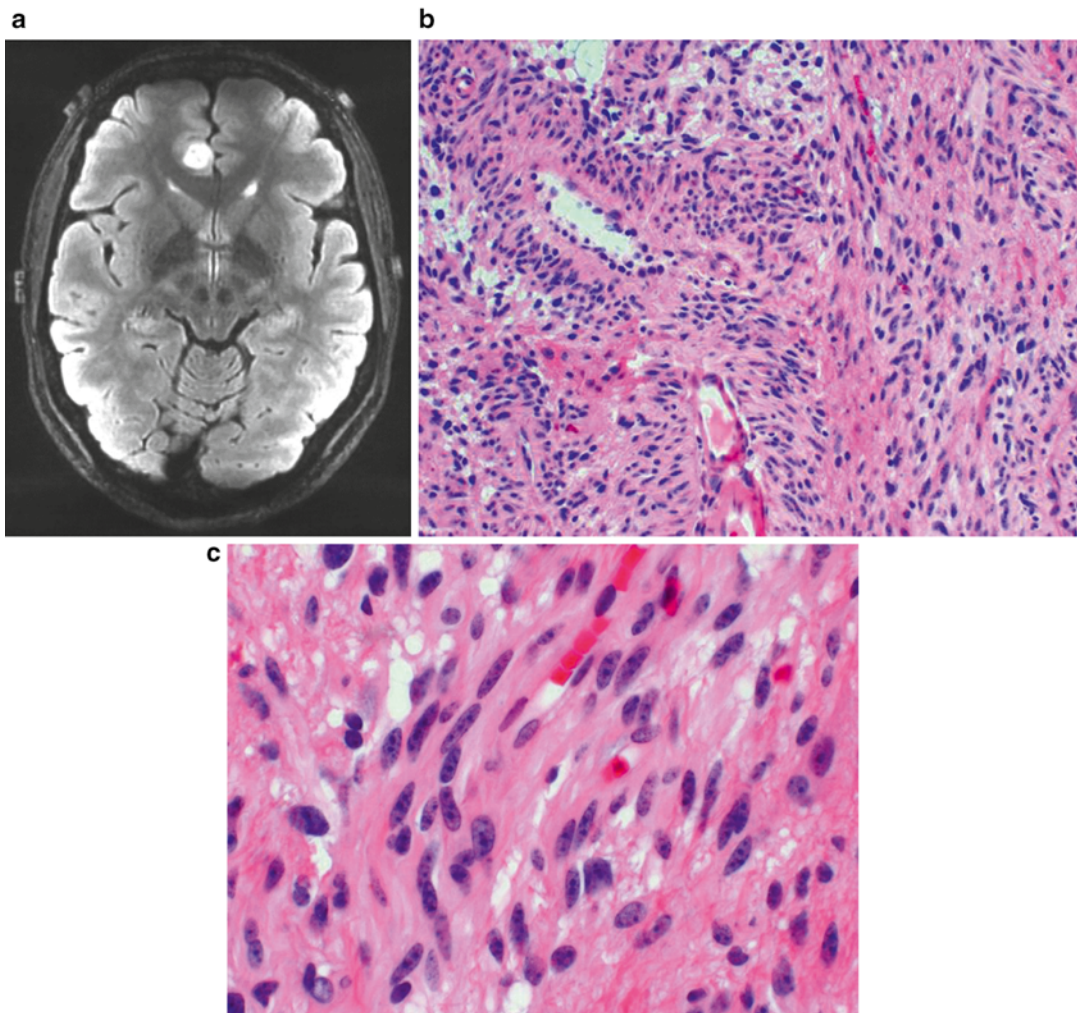


Fig. 1.1 Angiocentric glioma: (a) Axial, CUBE (3d TSE) image. Cortical-based well-demarcated lesion in the right cingulate gyrus with bright signal. (b) Bipolar spindle cells orient around the vasculature, recapitulating perivascular

pseudorosettes seen in ependymoma, H&E, 100X. (c) The nuclei are angular to spindled, showing very little pleomorphism. H&E, 400X (*MRI Image Courtesy Dr. Tarik Tihan, H&E Images Courtesy Dr. Tony Yachnis*)

tumor has been debated among several neuropathologists to whom it has been shown). The typical cortical location results in chronic, intractable partial seizures, and almost all angiocentric gliomas are resected from patients with a history of chronic epilepsy. Imaging shows a diffuse nonenhancing cortical lesion with extension into the subcortical white matter which expands the parenchyma rather than forming a discrete mass (Brat et al. 2007) (Fig. 1.1a). Magnetic Resonance Imaging (MRI) shows a cortical tumor which often has rim-like hyperintensity on T1-weighted imaging. T2-weighted imaging shows subcortical hyperin-

tensity which may have a stalk-like extension up to the border of the closest lateral ventricle. The rim-like T1 hyperintensity is present in 62% of cases and can be a helpful imaging feature to distinguish this tumor from other epileptogenic cortical neoplasms such as dysembryoplastic neuroepithelial tumors (DNTs) (Koral et al. 2012).

Histopathology

Angiocentric gliomas are diffuse, monomorphous tumors composed of bipolar spindle cells

which have a striking orientation around the vasculature, giving the tumor its name. Mitotic figures are rarely seen and MIB-1 immunostains demonstrate low proliferative activity (1–5%). Anaplasia is absent. The neoplastic cells are composed of a small, angular to spindle nucleus with indistinct lightly eosinophilic cytoplasm (Fig. 1.1b, c). The elongated cells may arrange themselves in a loose, streaming pattern around the vessels or in a more palisaded configuration similar to the perivascular pseudorosettes seen in ependymomas (Fig. 1.1b). A subset of tumors may also show subpial palisading. Angiocentric glioma is a diffuse tumor that may infiltrate the surrounding white or gray matter. One of the original descriptions of this entity indicated that neurons with synaptophysin immunoreactivity may be part of the neoplasm, hence the early name angiocentric neuroepithelial tumor (ANET) (Lellouch-Tubiana et al. 2005), but this is controversial and others have suggested such neurons are native cells entrapped by the infiltrating glial cells (Wang et al. 2005; Brat et al. 2007).

Immunohistochemically, these tumors are positive for antibodies to glial markers such as glial fibrillary acidic protein (GFAP), Vimentin, and S-100 protein, and also have the mentioned dot-like cytoplasmic positivity for epithelial membrane antigen (EMA). The EMA staining pattern is similar to that seen in ependymoma, and the ultrastructural evidence of ciliated microvilli with projecting microvilli also supports an ependymal origin for this tumor (Pokharel et al. 2011; Preusser et al. 2007). The interpretation of synaptophysin immunopositivity is controversial, as mentioned above. Other negative immunohistochemical markers include chromogranin, neurofilament protein, and p53.

Genetic Abnormalities

There is very little literature on the cytogenetic and molecular abnormalities in angiocentric glioma, and so far with the small number of cases described there has been no consistent pattern of such abnormalities. Preusser et al. (2007) showed one tumor to have a copy number gain of 11p11.2 and another

tumor to have a 6q24–6q25 deletion. Tatevossian et al. (2010) described an angiocentric glioma with a 6q23 deletion involving the *MYB* gene. The significance of these genetic abnormalities is uncertain and further studies are needed.

Differential Diagnosis

The clinical and neuroimaging differential diagnosis for angiocentric glioma can be complex, with the main contenders for diagnosis being other cortical epileptogenic neoplasms. Indeed, much of the literature to date has emphasized the importance of considering angiocentric glioma in the differential diagnosis of a resection specimen from a patient with intractable epilepsy, along with cortical dysplasia, ganglioglioma, dysembryoplastic neuroepithelial tumor (DNT), diffuse astrocytoma, cortical ependymoma, pleomorphic xanthoastrocytoma (PXA), pilocytic astrocytoma and pilomyxoid astrocytoma (Pokharel et al. 2011; Lum et al. 2008; Shakur et al. 2009). Diagnosis rests on histopathological features, which quickly eliminate much of the above differential but clearly one must exclude true ependymomas and perhaps pilocytic astrocytomas with careful histological analysis and immunostains.

On ordinary (H&E) morphologic grounds, DNT and ganglioglioma can be ruled out, as can focal cortical dysplasia (unless coexistent with the neoplasm). But the differential diagnosis with these other entities may become difficult. Angiocentric glioma may have solid, infiltrative areas that resemble diffuse astrocytoma. However, the perivascular nature of the tumor should be striking enough to eliminate the possibility of diffuse astrocytoma. As previously stated, angiocentric glioma shares several features with ependymoma, including the striking angiocentricity, perivascular pseudorosettes, and the dot-like cytoplasmic EMA positivity. In fact, the distinction between cortically-based ependymoma and angiocentric glioma may be impossible, as pointed out by Lum et al. (2008). As for pilocytic and pilomyxoid astrocytoma, these tumors share the elongate piloid cell shape with angiocentric glioma, and pilomyxoid astrocy-

toma may have perivascular pseudorosettes. However, angiocentric glioma lacks the biphasic pattern of pilocytic astrocytoma or the extensive myxoid background seen in pilomyxoid astrocytoma (Pokharel et al. 2011; Lum et al. 2008). Furthermore, these tumors have different typical locations: pilomyxoid astrocytoma almost always occurs in the hypothalamic/third ventricle region rather than the cortex.

Treatment and Prognosis

Treatment is generally recommended to be surgical excision without adjuvant therapy. Studies have shown that this usually resolves the patient's seizures, unless the tumor is subtotally resected (Shakur et al. 2009). Even subtotally resected tumors are thought not to recur in the sense of progressive growth, although experience remains limited and so this conclusion should be stated with caution. Of note, in this regard, is that despite the so-called low-grade nature of this tumor, there have been reports of angiocentric gliomas with higher MIB1-labeling indices (Pokharel et al. 2011; Wang et al. 2005), and one such case did recur as an anaplastic astrocytoma, illustrating that this tumor may evolve into a more aggressive lesion. However, the majority of these neoplasms are slow-growing and indolent. Gross total resection is considered the treatment of choice, but failure to achieve gross total excision may justify radiation therapy (Shakur et al. 2009).

Pilomyxoid Astrocytoma

Introduction and Historical Context

The term pilomyxoid astrocytoma (PMA) was first coined by Tihan et al. in 1999 as a distinct subset of hypothalamic/optic pathway pilocytic astrocytomas mainly occurring in infants. Since then, such tumors have also been described in adults. PMA is recognized by the WHO as a subcategory of pilocytic astrocytoma (PA) with a worse prognosis, ie it has more frequent local recurrence and CSF metastases than its ordinary

pilocytic counterpart. For that reason, it has been given WHO Grade II. It typically occurs in infants who present with failure to thrive, and has mainly been described along the optic chiasm, in the third ventricle and in the hypothalamus. Histologically, pilomyxoid astrocytomas are monomorphous tumors which lack the biphasic pattern and Rosenthal fibers of typical pilocytic astrocytomas and only rarely contain eosinophilic granular bodies.

PMA is often treated with partial resection, followed by chemotherapy and/or radiation therapy, although precise treatment modalities are still debated. Cottingham et al. (1996) was the original descriptor of PMA (called at that time pilocytic astrocytoma in infants) and indicated that PMA may "mature" into PA, which has since been demonstrated. Furthermore, recent literature has also been exploring the concept of the "intermediate" PMA, or intermediate pilomyxoid tumor (IPT) which has overlapping features of PMA and PA.

Clinical Presentation/Imaging

PMA mainly affects infants and young children with a median age at presentation of 10 months (Louis et al. 2007) as compared with a median age of 14 months for patients with PA. Rarely, adults, usually in the 3rd decade, are diagnosed with PMA, and there has been one report of a 77-year old with this entity (Toyoda et al. 2009). Infants with PMA typically present with failure to thrive, developmental delay, and altered consciousness (Tihan et al. 1999). In older children, clinical presentation may include headaches, nausea, and disorientation (Brat et al. 2007). One report describes PMA presenting with spontaneous hemorrhage, although the exact cause of the hemorrhage remains unclear (Gottfried et al. 2002). Since then, intratumoral hemorrhage has been described not infrequently in PMA (Linscott et al. 2008).

PMA is classically a tumor of the chiasmatic/hypothalamic area, but PMA has been described in other locations, including the cerebellum, the thoracic and cervical spinal cord, basal ganglia, thalamus, lateral ventricle, and 4th ventricle

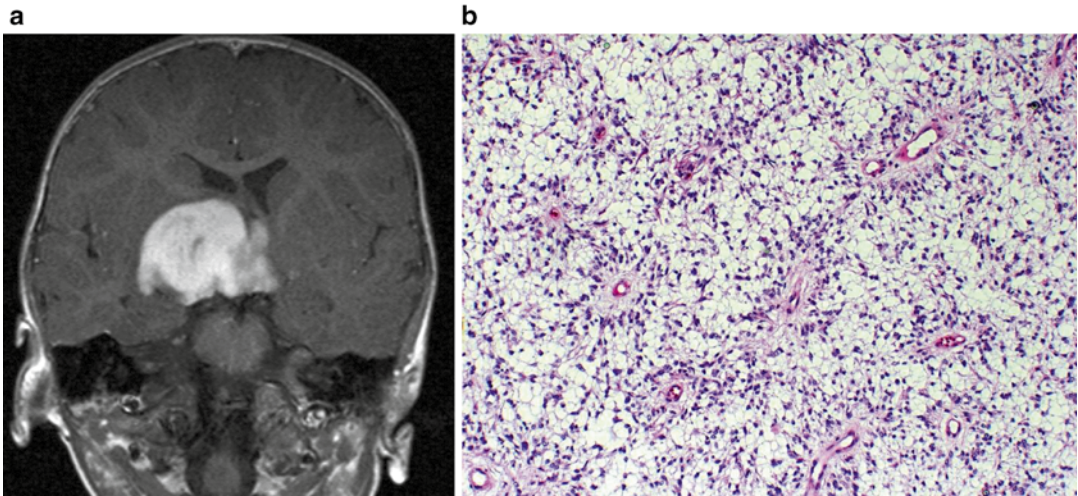


Fig. 1.2 Pilomyxoid astrocytoma: (a) T1-weighted MRI, post-contrast, coronal view. A mainly solid, enhancing mass near the optic chiasm compresses the right lateral ven-

tricle. (b) Bipolar cells in a markedly myxoid background arrange themselves in loose, fibrillar perivascular arrangements. H&E, 100X (*Images courtesy of Dr. Tony Yachnis*)

(Forbes et al. 2011; Matsuzaki et al. 2010; Omura et al. 2008). In general, these tumors show a predilection for central, supratentorial brain parenchyma, in contrast to the usual cerebellar location of PA (Lee et al. 2011). However, one series found that nearly half of PMAs occurred in other locations, highlighting the need for increased awareness for PMA in non-classic locations (Linscott et al. 2008).

MRI findings in PMA show a solid enhancing lesion with homogeneous T2 signal intensity (Fig. 1.2a). Aside from the classic chiasmatic/hypothalamic location, other features that are often present include hydrocephalus, tumor extension into deep white and gray matter, a small cystic component, and leptomeningeal dissemination. In contrast, PA typically does not enhance except for the mural nodules, more often has a predominant cystic component, and is more frequently in a cerebellar location (Arslanoglu et al. 2003; Lee et al. 2011).

Histopathology

Pilomyxoid astrocytomas are mainly solid tumors composed of bland bipolar (piloid) cells that often show an angiocentric arrangement,

although these arrangements are looser and more fibrillar than a typical ependymal pseudorosette. The cells are embedded in a lightly fibrillar and rich myxoid background (Fig. 1.2b). PMAs lack the biphasic pattern seen in PA, nor do they exhibit Rosenthal fibers. Eosinophilic granular bodies are rarely seen. The piloid cells are elongate, with a regular, hyperchromatic nucleus and lightly eosinophilic cytoplasm. In solid areas, nuclear pleiomorphism may be more striking. Necrosis may be present, as may focal peripheral infiltration of the parenchyma, but neither characteristic is so prominent as to meet the criteria for infiltrating astrocytoma (Tihan et al. 1999; Brat et al. 2007). Both PA and PMA may show vascular hyperplasia.

Immunohistochemically, the MIB-1 labeling indices are regionally variable within a PMA and range from 2 to 5%, which is not significantly different from the MIB-1 labeling of PA (0–4%), although PA does not usually demonstrate the regional variability. Otherwise, PMA has strong and diffuse staining for GFAP, which highlights the myxoid and fibrillar background as well as the tumor cells. The tumor cells are vimentin positive, but antibodies to synaptophysin, chromogranin, and neurofilament protein are usually negative (Tihan et al. 1999; Brat et al. 2007).

However, Miller (2009) described several cases of PMA with positive neuronal markers, indicating that PMA may be a type of glio-neuronal tumor. Mixed glio-neuronal staining and ultra-structural features have also been reported by others (De Chadarevian et al. 2006; Fuller et al. 2001). The tumor cells are variably immunoreactive for nestin, and Nagaishi et al. (2011) indicate that strong nestin staining correlates with a higher MIB-1 labeling index and a higher rate of recurrence.

Genetic Abnormalities

As with angiocentric glioma, there is little literature on the genetic abnormalities in PMA, and there is no consistent genetic abnormality at the time of his writing. Two patients with neurofibromatosis type 1 (NF1) have been described with PMA, although the exact association is unclear (Jimenez et al. 2010). (Of course there is a well-known association of PA with NF1, particularly in the optic chiasm/hypothalamic region.) There has been one report of a derivative chromosome 17 (the same chromosome that is affected in NF1) which carried and disrupted the BCR gene (Melendez et al. 2006). These authors postulate that the BCR gene may be constitutively active (similar to the BCR-ABL fusion protein and its constitutive activation) and result in tumorigenesis. Two array-comparative genomic hybridization studies have been performed. The first, a preliminary study by Komotar et al. (2004), found no genetic abnormalities in PMA. The second, by Jeon et al. (2008) found no significant difference in the number of copy gains or losses in PMA as compared with PA, although a clustering analysis was able to separate the tumors into distinct subgroups.

Differential Diagnosis

The main differential diagnosis for pilomyxoid astrocytoma includes pilocytic astrocytoma with a myxoid component, angiocentric glioma, and infiltrating astrocytoma. Purely on clinical

grounds, angiocentric glioma may be ruled out because of its cortical location and presentation with epilepsy. PMA may be differentiated morphologically from PA with a myxoid component because PMA is much more homogeneous and lacks the biphasic pattern of PA, as well as Rosenthal fibers. Eosinophilic granular bodies, often present in PA, are only rarely present in PMA. As for infiltrating astrocytoma, this is also a morphologic distinction: PMA may infiltrate the brain parenchyma at the periphery of the lesion, but does not have the extensive infiltrative nature of an infiltrating astrocytoma.

Prognosis and Treatment

As previously mentioned, PMA has a worse prognosis compared to PA, with more frequent local recurrence and leptomeningeal spread as well as lower overall survival. Therefore, PMA is usually treated more aggressively than PA. Tihan et al. (1999) in their original (retrospective) description of PMA indicated that 76% of patients with PMA had local recurrence after a variety of treatment modalities (13 of 17; two of these cases also had CSF tumor spread) with 2-year overall survival of 39%. The PA group, by comparison, had a 2-year overall survival of 100%. A subsequent study by Komotar et al. (2004, 2005) compared 21 hypothalamic/chiasmatic PMAs with 42 PA's in the same location. PMA had a 76% local recurrence rate, a progression-free survival time of only 26 months, and an overall survival time of 63 months. In comparison, the PA group had a 50% local recurrence rate, a progression-free survival time of 147 months, and an overall survival time of 213 months. Furthermore, 14% of PMAs had CSF dissemination, which did not occur in any PAs, and one third of PMA patients died of disease, compared with 17% of PA patients.

There is no current standard of care in the treatment of PMA. Ideally, PMA is treated with gross total resection (GTR). However, most chiasmatic/hypothalamic tumors are not amenable to GTR and are treated with partial resection

followed by chemotherapy and/or radiation (Komotar et al. 2005). Because of the effects on the CNS of radiotherapy in very early life, chemotherapy is the adjuvant treatment of choice in infants, and radiotherapy is used after the age of 3–5 years. Chemotherapy may be used to delay radiotherapy in infants who have recurrent or persistent disease. Tsugu et al. (2009) indicate that cisplatin/carboplatin-based chemotherapy regimens may be useful for initial treatment of PMA and intermediate pilomyxoid tumors (see next paragraph). However, in patients with leptomeningeal dissemination, frontline chemotherapy is often ineffective. Terasaki et al. (2012) describe a case of a 5-year old child with leptomeningeal disease who responded to craniospinal irradiation and temozolimide after failed frontline chemotherapy.

Of interest is the concept of a “maturing” PMA or one that, upon recurrence, shows more features of pilocytic astrocytoma, its less aggressive counterpart. This concept is closely related to the idea of the “intermediate pilomyxoid tumor,” (IPT) which shows histologic features of both PA and PMA. Recent reviews have indicated that, indeed, IPTs are not rare, and that it is unclear if these intermediate tumors carry the worse prognosis of pure PMA (Johnson et al. 2010), although Forbes et al. (2011) demonstrated that pilomyxoid tumors of the cerebellum do have a higher rate of recurrence and leptomeningeal disease than cerebellar PAs. Because of the uncertain significance of pilomyxoid features, the pathologist should note such features when they are present. Of note, PMAs do not always mature towards a less aggressive neoplasm. Paraskevopoulos et al. (2011) reported a case of a 12 year old girl with PMA with leptomeningeal dissemination that recurred in 3 months as glioblastoma.

Pituicytoma

Intro/Historical Context

Pituicytoma is a term that previously was once used synonymously with granular cell tumor or

pilocytic astrocytoma of the posterior pituitary, infundibulum, or suprasellar region. However, Brat et al. (2000) defined criteria for pituicytoma, and this tumor is now restricted to a distinct clinicopathologic entity that the WHO has recognized as a “Tumor of the Sellar Region,” separate from its previous synonyms. At the time of this writing, there are 54 described cases of pituicytoma, including one incidental finding at autopsy. Alternate, but less preferred names, for pituicytoma include infundibuloma and posterior pituitary astrocytoma.

Clinical Presentation/Imaging

Pituicytomas are located in the neurohypophysis (posterior pituitary) or infundibulum and may be intra- or suprasellar. Because of their location, they may result in compression of the optic chiasm, causing visual disturbances, or hypopituitarism. Hyperprolactinemia (resulting in decreased libido and amenorrhea) may occur when the cytoplasm of the neoplastic pituicytes engulfs normal infundibulum, causing interruption of the normal dopamine release pathway (Brat et al. 2000; Islamian et al. 2012). Although pituicytoma is mainly a tumor of adults, there has been one described case in an 11-year old girl (Yilmaz et al. 2011).

Imaging demonstrates pituicytomas as predominantly solid masses of the sellar, suprasellar, or (less commonly) parasellar region. Pituicytoma has a similar appearance with MRI to pituitary adenoma: a demarcated lesion with marked homogeneous enhancement with contrast administration (Fig. 1.3a). Pituicytomas are usually isointense on T1-weighted imaging, and iso- to hyperintense on T2-weighted imaging. Bony remodeling and cystic change are rarely described, and these tumors are rarely if ever calcified, unlike craniopharyngiomas which also occur here, half of which have significant calcification. Radiologically, the differential diagnosis of this lesion will almost always include pituitary macroadenoma and meningioma, and distinction with pituicytoma is not possible based on imaging (Hammoud et al. 2010; Chu et al. 2011).

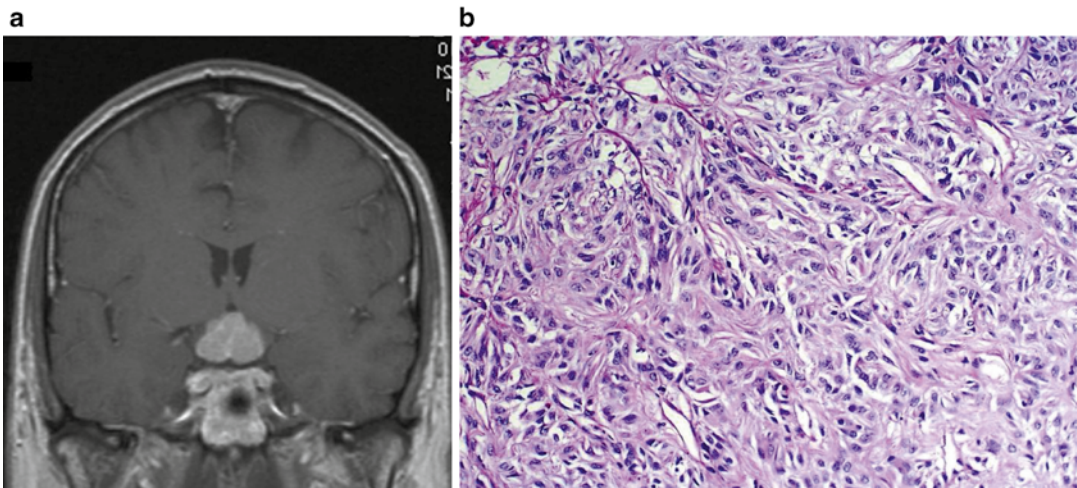


Fig. 1.3 Pituicytoma: (a) MRI, coronal view. Pituicytoma, indistinguishable on imaging from pituitary adenoma, presents as a demarcated, solid sellar and suprasellar

mass. (b) Histologically, pituicytoma forms intersecting fascicles of bland, moderately-sized pituicytes. H&E, 400x (*Images courtesy of Dr. Tony Yachnis*)

Histopathology

Pituicytomas are circumscribed, solid tumors composed of bipolar (piloid) cells thought to be derived from pituicytes, a type of specialized glial cell (Fig. 1.3b). These tumors may locally infiltrate surrounding pituitary, but are otherwise non-infiltrative. The bipolar cells form intersecting fascicles, often in a storiform pattern. Pituicytes have abundant eosinophilic cytoplasm and moderately-sized, oval nuclei which lack pleiomorphism (Fig. 1.3b). The cells lie in a glioma-like delicate fibrillary background which may be very vascular. Pituicytomas are more cellular than the normal neurohypophysis. Mitotic figures are rare or absent.

Immunohistochemically, pituicytes are immunopositive for glial fibrillary acidic protein (GFAP), vimentin, and S-100 protein, although their immunoreactivity may be variable, especially for GFAP. Neuronal markers such as synaptophysin, chromogranin, and neurofilament protein are immunonegative (Miller 2009). Antibody stains for pituitary hormones are also negative. Importantly, the eosinophilic cytoplasm of pituicytoma is not granular and has only minimal positivity with Periodic-Acid-Schiff (PAS) stains, unlike granular cell tumor and oncocy-

toma, in which the PAS-positivity is striking (Brat et al. 2000), and which can also be found as an infundibular mass. Interestingly, Phillips et al. (2010) demonstrated strong, diffuse Bcl-2 immunopositivity and suggested that mutations in Bcl-2 may provide clues to tumorigenesis.

Genetic Abnormalities

There is no known consistent genetic abnormality in pituicytoma. An array-comparative genomic hybridization study of one pituicytoma showed a gain on 5p, and losses on 1p, 14q, and 22p. Although these copy gains and losses are different from the majority of genetic abnormalities in pituitary adenomas, case reports have reported individual pituitary adenomas with the same abnormalities, suggesting that pituicytoma is a distinct, but related, entity (Phillips et al. 2010).

Differential Diagnosis

The main considerations for the clinical differential diagnosis of pituicytoma include pituitary adenoma, spindle cell oncocytoma, and granular cell tumor. Histologically the tumors have no

resemblance to pituitary adenomas, their cells do not have the finely granular oncocyctic cytoplasm seen in the oncocytomas or granular cell tumors, and ultrastructurally they lack the abundant lysosomes or mitochondria that are found in these two entities respectively (Brat et al. 2007). In small biopsies, one must be careful not to overdiagnose a distorted fragment of non-neoplastic neurohypophysis as a pituicytoma. As the cells in the neurohypophysis are also glia, and will be immunopositive for GFAP, vimentin, and S100, the distinction may rest in part on proliferative activity (MIB1) as well as on the recognition in the normal neurohypophysis of the dilated nerve endings (Herring Bodies) from which the neurohypophysial hormones are released (Miller 2009).

Prognosis and Treatment

Pituicytoma is treated with surgical resection. Because of its location, its vascularity, and its propensity to adhere to (but not invade) surrounding structures, gross total resection may not be possible. Studies indicate that even with only partial resection, pituicytomas regrow only slowly. Secci et al. (2012) describe a recurrence rate of 21% in subtotally resected tumors, with a mean time to recurrence of 24 months (range 5–60 months). Islamian et al. (2012), on the other hand, describe a recurrence rate of 39% in subtotally resected tumors, with time to recurrence ranging from 3 to 84 months. Several patients have undergone post-operative adjuvant radiation therapy, but the follow-up data are insufficient to indicate the role of radiotherapy in pituicytoma (Islamian et al. 2012). Malignant transformation or leptomeningeal spread has not been reported.

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Pediatric Low-Grade Gliomas: Diagnosis, Treatment, and Future Directions

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and Michael J. Fisher

Contents

Abstract.....	13
Introduction.....	14
Diagnosis.....	15
Treatment and Outcomes.....	16
Future Directions.....	20
References.....	22

Abstract

Low-grade gliomas are the most common brain tumor in children, accounting for 30–50% of central nervous system tumors in the pediatric age group. They can arise anywhere along the neuraxis and are a heterogeneous group of neoplasms with treatments and outcomes that vary widely by age, tumor location, and extent of surgical resection. Treatment ranges from expectant observation to surgery, chemotherapy, and radiotherapy as single or combined modality therapies. Pediatric low-grade gliomas share a unique association with several inherited genetic conditions, including neurofibromatosis type 1 and tuberous sclerosis. Children with these genetic disorders have a markedly increased risk of developing low-grade gliomas during childhood, and the prognosis and treatment choices may differ substantially from children with identical tumors who do not carry these genetic mutations. Long-term overall survival of all children with low-grade gliomas is very good; therefore, the potential late effects of treatment must be considered carefully in making treatment decisions. As our understanding of the genetic aberrations in low-grade gliomas evolves, novel molecular targets are being identified that may allow for more precise, tumor-directed drug delivery.

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Introduction

Low-grade gliomas are the most common type of brain tumor in children. They can occur anywhere within the central nervous system (CNS) including the cerebellum (15–25% of all pediatric CNS tumors), cerebral hemisphere (10–15%), deep midline structures (10–15%), optic pathway (5%), spinal cord (<5%), and brain stem (2–4%) (Sievert and Fisher 2009; Scheinemann et al. 2009). They comprise a heterogeneous group of tumor histologies. Classification is based on which normal cell they most closely resemble and includes astrocytomas, oligodendrogliomas, and mixed glial-neuronal tumors. Grading is determined by a number of factors including presence of necrosis, giant cells, mitosis, endothelial proliferation, hyperchromatic nuclei, and pleomorphic cells. Low-grade gliomas encompass World Health Organization (WHO) grade I and grade II tumors.

As a group, low-grade gliomas in children tend to be indolent. Unlike their adult counterparts, malignant transformation is rare. Dissemination is also uncommon, occurring at diagnosis in approximately 5% and at the time of progression in up to 10% of patients (Perilongo et al. 2003). Pilocytic astrocytomas (WHO grade I) and diffuse fibrillary astrocytomas (WHO grade II) account for the majority of pediatric low-grade gliomas. Pilocytic astrocytomas (WHO grade I) are considered the classic pediatric low-grade glioma and most commonly occur in the 0–14 year age range. Although they can arise anywhere in the central nervous system, they are more frequently located in the cerebellum, hypothalamus/optic pathway, dorsally exophytic brain stem, and spinal cord. In contrast, only 10% of diffuse fibrillary astrocytomas occur before the age of 20 years (Dolecek et al. 2012). These tumors are more often located in the supratentorial region, deep midline structures, and the cervicomedullary region of the brain stem. Although dissemination has been reported with primary tumors in all CNS locations and most low grade glioma histologies, it appears to occur more commonly with pilocytic astrocytomas and primary tumors of the

hypothalamic-chiasmatic region (Perilongo et al. 2003).

Other low-grade glioma histologies are less common in children. Pilomyxoid astrocytoma (WHO grade II) describes a subset of pilocytic tumors with more aggressive clinical behavior. These tumors tend to occur in the hypothalamus and more often arise in very young children with mean age of 21 months (Johnson et al. 2010). Although they are also WHO grade II, pleomorphic xanthoastrocytomas are less aggressive than diffuse fibrillary astrocytomas, with a clinical phenotype similar to pilocytic astrocytomas. Gangliogliomas (WHO grade II) are often discovered as an incidental finding in children with seizures and localize to the temporal lobes. Subependymal giant cell astrocytomas (WHO grade I) occur in the ventricles and are almost exclusively seen in children with tuberous sclerosis syndrome. Oligodendrogliomas (WHO grade II) are uncommon, accounting for only 2% of brain tumors in children less than 14 years of age.

Low-grade gliomas present in a similar manner as other brain tumors, although often with a prolonged duration of symptoms prior to diagnosis. Symptoms are either due to obstruction of the ventricles resulting in increased intracranial pressure or dependent on the location of the tumor. With the former, symptoms include headaches (particularly in the morning), nausea, vomiting, and lethargy with findings of decreased upward gaze, sixth cranial nerve palsies, and papilledema on neurological examination. In addition to symptoms of increased intracranial pressure, low-grade gliomas of the cerebellum often present with ataxia and dysmetria. Symptoms of cortical tumors are dependent on which lobe is involved; for example, temporal lobe tumors present with seizures while frontal lobe tumors are often associated with behavioral changes. Hypothalamic gliomas can present with failure to thrive, diabetes insipidus, precocious puberty, delayed growth, or other endocrine abnormalities. In addition, because of the proximity of the hypothalamus to the optic chiasm, visual field deficits can occur. Children with optic pathway gliomas, frequently present with visual acuity loss, nystagmus, strabismus, head-tilt, proptosis,

and optic nerve pallor on examination. Of note, multifocal involvement of the optic pathway is found almost exclusively in children with neurofibromatosis type 1 (NF1). These patients account for up to 70% of all optic pathway gliomas in children. The presentation of low-grade gliomas of the brain stem is dependent on tumor location but often includes lower cranial nerve deficits (dysphagia, dysarthria, abnormal breathing) and long tract signs (hemiparesis, spasticity, hyperreflexia, extensor plantar response). Children with low-grade gliomas in the spinal cord typically present with back or neck pain, extremity weakness, gait disturbance, and/or kyphoscoliosis.

This chapter will focus on diagnosis, current treatment strategies, outcomes, and future treatment directions in the most common pediatric low-grade gliomas.

Diagnosis

Despite their clinical heterogeneity, pediatric low-grade gliomas demonstrate common features on magnetic resonance imaging (MRI), including hypointensity on T1-weighted images and hyperintensity on T2-weighted or fluid attenuation inversion recovery (FLAIR) sequences (Figs. 2.1 and 2.2). Enhancement patterns on post-gadolinium sequences are variable. Pilocytic

astrocytomas tend to be well-circumscribed, with a brightly contrast-enhancing mural nodule and often a large cystic component (Fig. 2.1). In contrast, diffuse fibrillary astrocytomas are less circumscribed and typically exhibit minimal enhancement. It is important to distinguish low-grade brain stem gliomas (e.g. dorsally exophytic, cervicomedullary, and focal) (Fig. 2.2) from the highly aggressive diffuse intrinsic pontine brain stem gliomas. The latter expand the brain stem, have non-discrete borders and do not enhance to a large extent. Low-grade gliomas of the spinal cord tend to be focal, well-circumscribed lesions (Fig. 2.2), often associated with a syrinx.

In most cases, surgical biopsy is performed to obtain tumor tissue for histological diagnosis, and more recently for characterization of genetic alterations. On rare occasions, diagnosis can be made without obtaining tissue because of characteristic appearance of the tumor on MRI; examples include optic pathway gliomas in the setting of NF1 and most tectal gliomas. Extent of resection is determined with postoperative MRI, which should be performed within 24–48 h after surgery in order to better distinguish residual tumor from post-operative changes. Although uncommon, if leptomeningeal dissemination is suspected, MRI of the entire spine (Fig. 2.2) and lumbar puncture for CSF cytology should be performed.

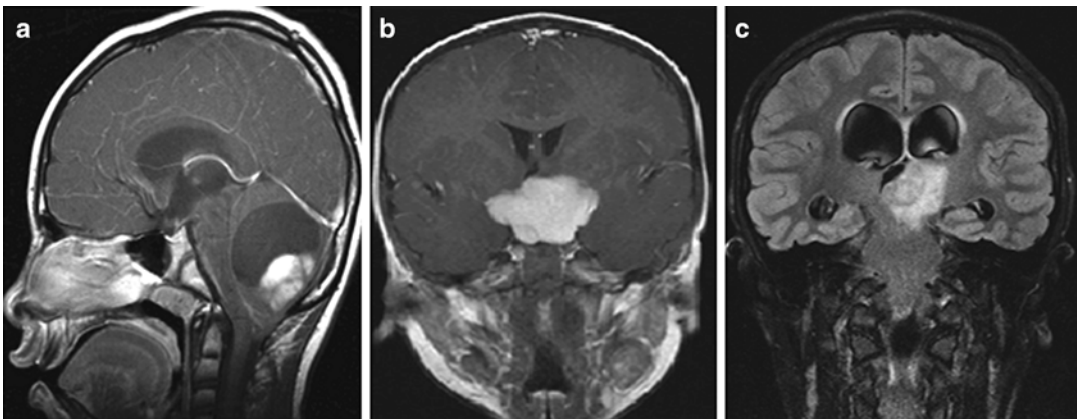


Fig. 2.1 Magnetic resonance images of three different pediatric low-grade gliomas. **(a)** Cerebellar pilocytic astrocytoma (sagittal post-gadolinium sequence) with mural nodule and large cystic component. **(b)**

Hypothalamic/optic chiasm glioma (coronal post-gadolinium sequence). **(c)** Thalamic pilocytic astrocytoma (coronal fluid attenuation inversion recovery sequence)

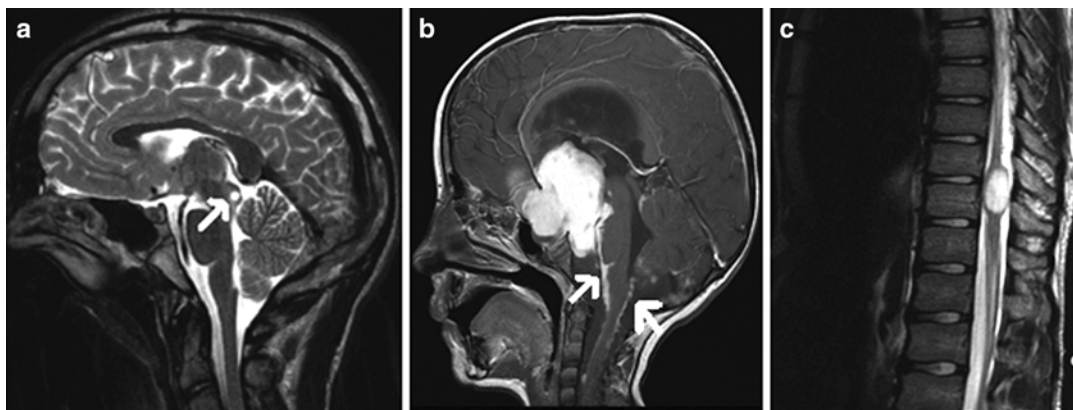


Fig. 2.2 Magnetic resonance images of three different pediatric low-grade gliomas. **(a)** Tectal glioma (sagittal T2 sequence, marked by *arrow*). **(b)** Disseminated pilocytic astrocytoma (sagittal post-gadolinium sequence)

with large hypothalamic primary and leptomeningeal seeding (*arrows*). **(c)** Thoracic spinal cord diffuse fibrillary astrocytoma (sagittal T2 sequence)

Despite WHO criteria, histopathological classification of gliomas can be difficult. Although partly due to intra-tumoral heterogeneity, more often, the difficulty lies with the experience of the individual neuropathologist and lack of molecular markers for accurate classification. Post-hoc neuropathology review of specimens from a large pediatric high-grade glioma trial resulted in 70 of the 250 cases being reclassified from high-grade to low-grade (Gilles et al. 2008). Although this study was conducted before the current WHO criteria, it highlights the subjective nature and risk for error in classification.

Treatment and Outcomes

Complete surgical resection is the most consistent prognostic factor for prolonged progression-free (PFS) and overall survival (OS) for pediatric low-grade gliomas, and is associated with 10-year overall survival rates of 90% or greater and rare tumor recurrences (Sievert and Fisher 2009). A significantly higher rate of tumor progression is reported in children with subtotal resections. In one report, the 10-year PFS after subtotal resection was only 55% (Fisher et al. 2001). Similarly, a large (660 children), prospective, cooperative intergroup trial, performed by the Children's Cancer Group and Pediatric Oncology Group,

reported a 5-year progression-free survival of 45–65% in subjects with residual tumor of any size (Shaw and Wisoff 2003). A recent single institution review of spinal cord low-grade glioma outcomes also supports extent of surgical resection as the most important prognostic factor for PFS; 5-year PFS was 88% in children with complete resection, but only 34% after subtotal resection (Scheinemann et al. 2009). Although diffuse infiltrating histology is associated with an increased risk of progression and/or recurrence compared to pilocytic astrocytomas, treatment decisions are still primarily driven by tumor location, as this is the best indicator of expected extent of resection.

Low-grade gliomas in the cerebellum and superficial cerebral hemispheres are most amenable to definitive surgical therapy. If a gross total resection is achieved, adjuvant treatment is not indicated. Post-operative management consists of observation with close clinical and radiographic follow-up. The treatment decision for children with subtotal resection is less straightforward. If the child tolerated the initial surgery with minimal morbidity, a second surgery in the immediate post-operative period to attempt a more definitive resection can be considered. Alternatively, given the generally indolent nature of these tumors, serial follow-up with brain MRI at regular intervals, and deferring the decision for

repeat resection or adjuvant therapy until there is evidence of tumor progression, is an accepted approach. If further therapy is indicated and repeat surgery is not feasible or would result in unacceptable morbidity, adjuvant chemotherapy should be considered for young children, with radiation therapy reserved for older children. The role of adjuvant therapy along with specific treatment regimens will be discussed below. Overall survival outcomes are excellent for children with low-grade gliomas in these locations (greater than 90% at 10 years). Ten year PFS is 82% in cerebellar tumors and 79% in cerebral hemisphere tumors, and this rises to virtually 100% if total resection was achieved on initial surgery (reviewed in Sievert and Fisher 2009).

Low-grade gliomas located in the supratentorial midline structures are more difficult to manage because extent of resection is often limited. Biopsy of these tumors is essential, to guide treatment and inform prognosis. This is particularly important in unilateral and bilateral tumors of the thalamus, in which a significant proportion are high-grade in histology (Puget et al. 2007). As these tumors may have mixed histology with both low-grade and high-grade areas, it is important to sample the part of the tumor that has the most malignant characteristics. A single institution retrospective review of 54 children with unilateral thalamic tumors demonstrated a mean overall survival of greater than 60 months (low-grade glioma) versus 21 months (high-grade glioma) (Puget et al. 2007). Positron emission tomography (PET) can be used to help guide stereotactic biopsy. Imaging with amino acid tracers may have an advantage over fluorodeoxyglucose (FDG), because of the high FDG uptake in pilocytic astrocytomas (Fisher and Phillips 2006). Although aggressive resection has historically been deferred because of the risk of significant neurological compromise, recent data suggests that greater than 90% resection can be achieved in almost half of unilateral thalamic gliomas, and that extent of resection is a significant prognostic factor (Puget et al. 2007). However, almost one third of patients were neurologically worse following aggressive surgical resection. Following biopsy or attempted resection, if symptoms are

minimal, observation until radiographic progression is a reasonable approach. Thalamic low-grade gliomas are usually associated with a worse progression free and overall survival in comparison to tumors in the cerebellar region, likely related to extent of surgical resection as well as anatomical location. Moreover, children with bithalamic low-grade gliomas have a reported worse overall survival compared to those with unilateral thalamic tumors (Puget et al. 2007).

Historically, following biopsy, focal radiation therapy (45–54 Gy) was considered as front-line therapy for supratentorial midline low-grade gliomas. Data looking at the impact of radiotherapy on overall survival is conflicting. Reported 10-year progression-free survival rates for children receiving radiation therapy have typically been as high as 65–90% (Sievert and Fisher 2009). More recent studies suggest that immediate postoperative irradiation, when compared to cytotoxic chemotherapy or observation alone, does not result in improved progression free survival in children with residual low-grade gliomas (Kortmann et al. 2010). In this context, the role of radiotherapy needs to be carefully considered, particularly weighing possible benefit against the known neurocognitive effects. Multiple studies confirm the adverse effect of whole brain irradiation on IQ in survivors of pediatric brain tumors (reviewed in Sievert and Fisher 2009). Other neurocognitive effects of radiotherapy to the brain include memory impairment and attentional difficulties, and are not limited to young children. In addition, radiation is associated with hormone deficits, secondary malignant neoplasms, vasculopathy and stroke.

Given the risks of radiotherapy, there is an increasingly prominent role for adjuvant chemotherapy in children with progressive low-grade gliomas, particularly those less than 12 years of age, who are at highest risk for radiation-related neurocognitive impairment. A summary of clinical trials using multi-agent chemotherapy for low-grade glioma is listed in Table 2.1.

The most widely used combination therapy is carboplatin and vincristine. In a large multi-institutional trial enrolling 78 children, reduction

Table 2.1 Multi-agent chemotherapy for pediatric low grade glioma

Drugs	N	Objective response ^a	PFS
Vincristine/Actinomycin D (Packer et al. 1988)	24 ^b	37.5%	~48% (4 years)
Carboplatin/Vincristine (Packer et al. 1997)	78	56%	68% (3 years)
Carboplatin/Vincristine (Ater et al. 2012)	137	50%	39% (5 years EFS)
Carboplatin/Vincristine (Ater et al. 2008)	127 ^{b, c}	61%	69% (5 years EFS)
TPCDV (Prados et al. 1997; Mishra et al. 2010 ^b)	42	36%	45% (3 years)
	33 ^b	n/a ^b	30% (5 years) ^b
			23% (15 years) ^b
TPCV (Ater et al. 2012)	137	52%	52% (5 years EFS)
Cisplatin/Etoposide (Massimino et al. 2002)	34	70%	78% (3 years)
Cisplatin/Etoposide (lower-dose regimen) (Massimino et al. 2010)	37	65%	65% (3 years EFS)
Procarbazine/Carboplatin, Etoposide/Cisplatin, Vincristine/Cyclophosphamide (Laithier et al. 2003)	85 ^b	60%	34% (5 years)

Abbreviations: *PFS* progression-free survival, *EFS* event-free survival, *T* thioguanine, *P* procarbazine, *C* CCNU (Lomustine), *D* dibromodulcitol, *V* vincristine

^aComplete, partial, and minor response

^bHypothalamic/optic pathway glioma only

^cNeurofibromatosis type 1 only

in tumor size or disease stabilization occurred in the majority of patients with a 3-year PFS rate of 68% (Packer et al. 1997). The primary limitation to this drug combination is that many children, up to 40% in some reports, develop an anaphylactoid hypersensitivity reaction to carboplatin with increased exposure to the drug (Sievert and Fisher 2009). An alternative regimen using the combination of thioguanine, procarbazine, lomustine (CCNU), dibromodulcitol, and vincristine (TPCDV) is also effective against progressive low-grade gliomas, with a reported 3-year PFS of 45% (Prados et al. 1997). However, the 15-year PFS rate of the same TPCDV regimen in children with hypothalamic/optic pathway low-grade gliomas is only 23% (Mishra et al. 2010). A modification of this regimen (eliminating dibromodulcitol, now “TPCV”) was directly compared to carboplatin/vincristine in a randomized clinical trial within the Children’s Oncology Group. Results from this trial demonstrate a non-statistically significant trend toward improved event-free survival with the TPCV regimen over carboplatin/vincristine, though both regimens had equal response rates and both allowed delay in radiotherapy (Ater et al. 2012).

Chemotherapy options are limited for children with tumors that do not respond to either of these

two, commonly used, front-line chemotherapy regimens. Cisplatin plus etoposide has been evaluated in a small group of children with unresectable low-grade gliomas with a reported 3-year progression free survival of 78 and 65% using a lower dose regimen (Massimino et al. 2010). Cisplatin is not associated with the same rate of allergic reactions as carboplatin; however, there is a risk of hearing loss with cisplatin, which is a particular issue in children with optic pathway or hypothalamic tumors, who are already at risk for visual impairment.

Single agent, low-dose weekly vinblastine, has demonstrated disease stabilization in 42% of patients who were started on it for recurrent or refractory disease (Bouffet et al. 2012). Several chemotherapy regimens used for adult and pediatric high-grade glioma have been studied in recurrent/refractory low-grade glioma with varying success. Single agent temozolomide demonstrated activity in a subset of pediatric low-grade gliomas, but provides only limited disease stabilization with a 4 year PFS rate of 17% (Gururangan et al. 2007). Bevacuzimab-based therapy, commonly used in adult high-grade gliomas, demonstrates activity in limited early trials of multiply recurrent childhood low-grade glioma and warrants further study and attention (Hwang et al. 2012).

Hypothalamic/optic tract low-grade gliomas represent a unique category of low-grade glioma in pediatrics. Up to 70% of these tumors occur in the setting of neurofibromatosis type 1 (NF1). Most are pilocytic astrocytomas, although other low-grade glioma histologies can occur. Fortunately, most of these tumors are indolent; in fact, less than half of those associated with NF1 will become symptomatic and require treatment (Fisher et al. 2012). Favorable prognostic factors include a concurrent diagnosis of NF-1 and tumor location, with tumors of the optic nerves and chiasm associated with a better visual outcome than tumors involving the hypothalamus and optic tracts/radiations (Fisher et al. 2012; Sievert and Fisher 2009). Management of these patients is focused primarily on the goal of preserving vision. In most cases, newly diagnosed patients are observed with short-interval MRI and ophthalmologic follow-up. Indications for treatment include a documented decline in visual acuity or significant tumor progression on MRI scan with associated symptoms and signs. Treatment may be considered at diagnosis for patients with severe visual impairment, extensive tumor, and/or tumor involving the hypothalamus and optic tracts/radiations. Surgical resection is usually contraindicated due to the risk of worsening vision loss, compromise of the hypothalamic-pituitary axis, and risk of stroke. However, resection of unilateral optic nerve tumors is considered when there is absent or severe impairment of vision and painful or disfiguring proptosis. In addition, large chiasmal/hypothalamic tumors that exert mass effect on surrounding structures and obstruct the third ventricle with resultant hydrocephalus may benefit from surgical debulking. When treatment is indicated, chemotherapy is usually pursued initially. Carboplatin/vincristine is the preferred front line regimen, as TPCV is avoided in children with NF1 due to the increased risk of secondary malignancies in this patient population with exposure to the alkylating agents procarbazine and CCNU. Focal radiation to the optic pathway/hypothalamus may result in worsening of vision and permanent endocrine dysfunction. In addition, the risks of cerebrovascular disease, secondary malignant neoplasms,

and neurocognitive deficits are particularly high for those with NF1 (Sievert and Fisher 2009). Therefore, radiotherapy is reserved for older children without NF1 or those with refractory progressive tumors that have failed multiple chemotherapy regimens.

Low-grade gliomas of the brainstem include dorsally exophytic, cervicomedullary, and tectal gliomas. Diffuse intrinsic pontine gliomas, even if low-grade in histology, are not included in this category as they have an especially aggressive clinical course and poor overall survival. Surgical resection is the primary treatment for children with dorsally exophytic brainstem gliomas. As these tumors arise from the floor of the fourth ventricle, a thin layer of tumor often remains. However, the majority of patients remain progression-free after resection (Rutka and Kuo 2004). Cervicomedullary low-grade gliomas tend to grow slowly. As such, they are usually managed conservatively with close observation. Surgical resection should be considered with radiographic or clinical progression. However, gross total resection often cannot be achieved without unacceptable risk (Rutka and Kuo 2004). Adjuvant therapy is recommended for tumors that progress following surgery. Tectal gliomas are indolent tumors and rarely cause functional impairment. They typically present with hydrocephalus, and therefore are managed with endoscopic third ventriculostomy or placement of a ventriculoperitoneal shunt. Biopsy and further therapy is pursued only for radiographic or clinical progression (Rutka and Kuo 2004).

There are few reports on spinal cord low-grade glioma in children, and a consistent approach to treatment does not exist. The majority of low-grade gliomas of the spinal cord are located in the cervical or thoracic cord and are WHO grade I histology. Progression free and overall survival rates are reportedly comparable to intracranial tumors; however, controversy remains regarding the role of adjuvant therapy for residual spinal cord tumors after surgery. Extent of resection has the greatest impact on PFS; unfortunately, orthopedic and neurologic surgical morbidity can be unacceptably high (Scheinmann et al. 2009). Carboplatin/vincristine, TPCV, and vinblastine

have been used at initial presentation and/or at recurrence with reported activity. Chemotherapy appears to warrant further study as an approach to improve PFS, minimize the need for aggressive surgical attempts, and delay or eliminate the use of radiation therapy (Scheinmann et al. 2009).

There is no standard management for disseminated low-grade glioma. Although biopsy is indicated to confirm diagnosis, aggressive resection is rarely pursued, as gross total resection of the primary tumor is not curative in and of itself. The selection of radiation or chemotherapy is made based on similar factors (age and tumor location) as for non-metastatic tumors. Objective tumor responses and prolonged stable disease can be achieved; however both PFS and OS are worse compared with isolated tumors (Perilongo et al. 2003).

Future Directions

Newer treatments are needed for the many children each year who suffer considerable morbidity and premature death from progressive low-grade gliomas. Treatment options are limited for children with tumors that cannot be completely surgically resected. Studies of adjuvant chemotherapy and radiation therapy reveal significant rates of early and late progression. Very young children and infants present an even more difficult treatment challenge as they are at higher risk for significant, neurocognitive treatment-related deficits. Newer radiation treatment strategies are being assessed for their ability to minimize long-term toxicity while maintaining efficacy. Stereotactic fractionated radiotherapy appears to have similar efficacy compared with traditional radiotherapy approaches, however, whether it minimizes the neurocognitive risks is not yet clear (Sievert and Fisher 2009). Proton radiotherapy is currently under evaluation for this population (Brower et al. 2013).

Intensification of treatment with traditional chemotherapy agents does not improve survival (Fouladi et al. 2003), likely secondary to biological reasons. Cytotoxic chemotherapy targets rap-

idly dividing cancer cells. Conversely, pediatric low-grade gliomas often display indolent growth patterns. Despite this, their location can make even sub-centimeter growth potentially devastating with neurological sequela such as loss of vision or disruption of hypothalamic function. Moreover, for long-term survivors, there is significant risk of adverse neurocognitive and neuroendocrine outcomes. Survivors who had tumors located in the cerebral hemisphere, were younger at diagnosis, and/or had hydrocephalus requiring a shunt are more likely to have lower cognitive performance on IQ scales (Reimers et al. 2007). Even survivors of pediatric low-grade glioma of the cerebellum, treated with surgery alone, have long-term cognitive and adaptive-behavioral impairment (Ris et al. 2008).

Improvements in pediatric low-grade glioma therapy have previously been slow to advance, and only recently, has the biology underlying these tumors begun to be molecularly characterized, and potential therapeutic targets identified. Historically, glioma classification has relied on histological characterization. As, large-scale sequencing efforts utilizing whole-genome, transcriptome, and targeted high-throughput sequencing of tumors have become available, recurrent genetic alterations driving aberrant cell signaling pathways have been identified. In 2008/2009, the identification of an oncogenic *KIAA1549-BRAF* fusion gene, resulting in constitutive BRAF activation was described in the majority of pilocytic astrocytomas and first implicated BRAF in pediatric low-grade glioma development (Pfister et al. 2008; Sievert et al. 2009). Additional activating mutations of BRAF, including *BRAF*^{V600E} in WHO grade II pleomorphic xanthoastrocytomas and diffuse astrocytomas (Zhang et al. 2013), new *KIAA1549-BRAF* fusion variants, and novel *BRAF* fusions and mutations in non-cerebellar pilocytic astrocytomas (Jones et al. 2013) further validate BRAF as an oncogenic driver in low-grade gliomas. More recently, recurrent mutations, fusions, and duplications affecting the tyrosine kinase domain in the fibroblast growth factor receptor (*FGFR1*) were identified in diffuse grade II (Zhang et al. 2013) and non-cerebellar pilocytic (Jones et al. 2013)

astrocytomas, which, like mutated *BRAF*, result in dysregulation of the mitogen-activated protein kinase (MAPK) and PI3K pathways (Fig. 2.3). The convergence of multiple, mutually exclusive genomic alterations in multiple pediatric low-grade glioma subtypes on a single, common signaling pathway (MAPK), confirm the importance of this signaling pathway in low-grade glioma development. Inhibitors of *BRAF* and of the MAPK pathway are currently in early phase clinical trials in pediatrics. Although these new targeted agents may be promising, they will likely need to be combined with traditional chemotherapy regimens or with other pathway inhibitors, as adult clinical trials reveal that drug resistance may develop rapidly with selective targeted inhibitors (Salama and Flaherty 2013). In addition, it has been demonstrated that some RAF-

specific inhibitors paradoxically activate (instead of inhibit) the MAPK pathway (Sievert et al. 2013; Heidorn et al. 2010; Hatzivassiliou et al. 2010); defining mechanisms of resistance is under investigation. Regardless, this creates an additional safety concern when conducting clinical trials in children with slow-growing tumors who may require treatment with these medications for a prolonged period (months to years).

Neurofibromatosis type 1 (NF-1) is caused by a germline, inherited or sporadic, inactivating mutation of the *NF1* gene on chromosome 17, whose protein product neurofibromin negatively regulates RAS, functioning as a tumor suppressor. Therefore, inactivation of neurofibromin causes increased activation of the RAS/RAF/MEK/MAPK and the PI3K/AKT pathways (Fig. 2.3). There are a number of different

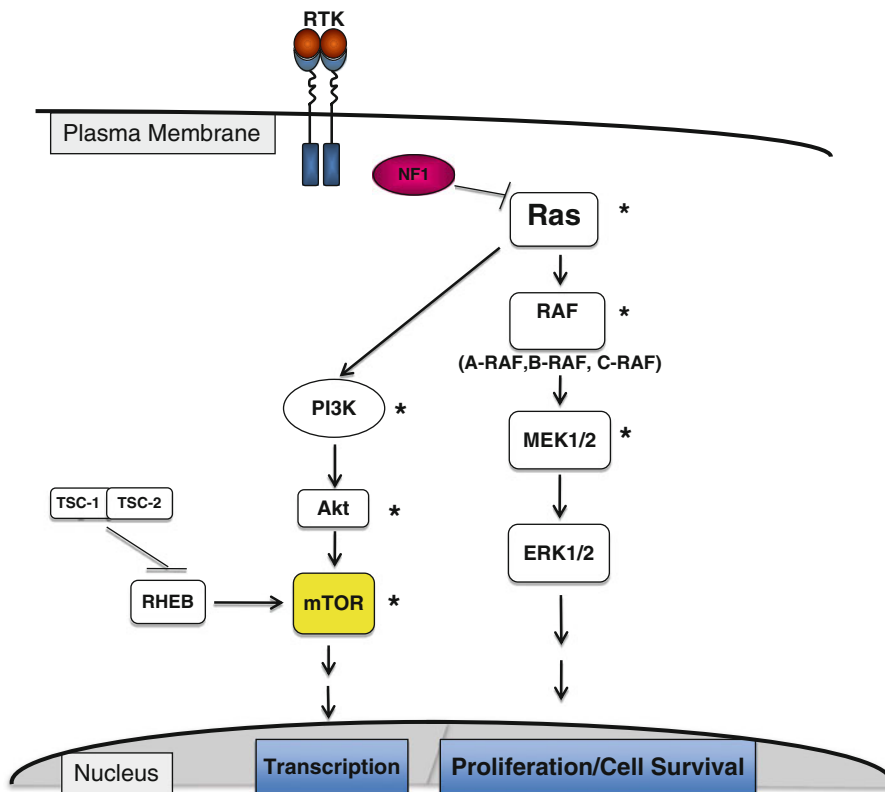


Fig. 2.3 Schematic diagram of the Ras-mediated signal transduction pathways (RAS/RAF/MEK/MAPK and RAS/PI3K/AKT) commonly dysregulated in low-grade gliomas. Distinct mutations of different effector mole-

cules along the pathway (see text for detail) provide unique targets for therapeutic intervention. (asterisks identify targets for which drug inhibitors are currently in clinical trial)

neurofibromin mutations described, most resulting in truncation of the protein. SNP-microarray analysis of pediatric low-grade gliomas demonstrates that loss of NF1 and activating BRAF mutations are mutually exclusive (Pfister et al. 2008; Sievert et al. 2009; Jones et al. 2013), which further supports the importance of the MAPK pathway in glioma biology. The mTOR pathway, downstream of the PI3K pathway, also appears to play a role in low-grade glioma, especially in the context of inherited predisposition syndromes. Children with Tuberous Sclerosis (TS) can have biallelic inactivation in either of the TS tumor suppressor genes (TSC1 or TSC2) that result in direct activation of mTOR pathway, and mTOR inhibition has demonstrated efficacy in treating TS-associated astrocytomas (Franz et al. 2013). In addition, up-regulation of mTOR has been noted in NF1-associated glioma mouse models (Banerjee et al. 2011). Inhibitors of mTOR are currently in clinical trials for children with TS and subependymal giant cell astrocytoma and for children (with or without NF1) with recurrent or refractory low-grade glioma (Franz and Weiss 2012).

Future therapies for pediatric low-grade gliomas are currently under development with biologic rationale. Collaborative basic and translational research efforts are identifying the molecular characteristics and biochemical signatures for this diverse set of tumors. Targeted agents carry the potential of improved tumor control with limited toxicity; however, the burden of proving their effectiveness in a low-grade tumor will be much higher as the overall survival for these children is more favorable than those with high-grade tumors. Goals of therapy should focus not only on improving progression free survival but on minimizing the morbidity to long-term survivors.

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Kringle 1-5 Reduces Growth of Malignant Gliomas in Rats

3

Ming-Chao Huang and Yi-Lo Lin

Contents

Abstract	25
Introduction	26
Angiogenesis	26
Importance of Angiogenesis in Gliomas.....	26
Tumor Vasculature Has Tumor Origin.....	27
Antiangiogenesis in Treating Malignant Gliomas.....	27
Kringle 1-5	29
The Structure and Antiangiogenic Effect of the Kringle.....	29
Discussion	30
K1-5 Inhibits Tumor Angiogenesis.....	30
K1-5 Decreases Hypoxic Area of the Glioma.....	30
Lower VEGF Expression in K1-5 Treated Glioma.....	30
Inhibition Effect of K1-5 on Glioma Cell Invasion.....	31
K1-5 Treated Animals Had Longer Survival.....	31
Conclusion.....	31
References	33

Abstract

Malignant gliomas is a highly vascularized and invasive tumor. The prognosis is still extremely poor despite advances in the cancer therapy. The growth and spread of gliomas within the host tissue depends upon their ability to stimulate new vessel formation, the angiogenesis, to get nutrients and remove waste products. Therefore, blocking the action of angiogenic factors, or utilizing angiogenic suppressors to inhibit angiogenesis, constitute a new strategy to suppress tumor growth. A representative drug is Avastin (Bevacizumab). It has been proved by FDA to treat patients with recurrent glioblastoma. It is a protein that acts against vessel growth factor. However, there are many vessel growth factors and Avastin only target one of them. Therefore the inhibition effect for vessel growth can be compensated by other angiogenic factors. Other disadvantages such as drug resistance and increased vascular cooption of the host vasculature have been reported. Kringle 1-5 (K1-5), discovered by Cao, is also a protein aimed on the vessels. Compared with Avastin, K1-5 acts directly on endothelial cells and leads endothelial cell death. The tumor inhibition effect

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of K1-5 was proven in a murine hepatocellular carcinoma model. In this chapter the therapeutic effect of K1-5 in malignant gliomas was discussed. We demonstrated that K1-5 is effective in the treatment of malignant gliomas. It is effective not only in the inhibition of vascular growth but also in limiting tumor invasiveness and improving tumor hypoxia, and then lead to a longer survival.

Introduction

Antiangiogenic therapy has been a new choice for cancer therapy during the past decade. The theory is based on the angiogenesis-dependent tumor growth. Therefore, suppressing angiogenesis may decrease tumor growth. Among various proangiogenic factors, vascular endothelial growth factor (VEGF) is a major angiogenic factor. Agents target VEGF signaling axis, such as VEGF neutralizing antibody and agents that acts on VEGF receptor, has become a main interest for many cancer treatments. These drugs demonstrate some improvement in certain type of tumors, however, the overall benefit was modest. The most promising antiangiogenic drug currently available for treating recurrent glioblastoma, one of the most frustrating diseases, is bevacizumab. However, there are still some controversies regarding using bevacizumab for this brain tumor. Besides VEGF axis, another target is endothelial cells because they are relative stable targets comparing to tumor cells, in either genetically (mutation) or drug resistance aspect. K1-5, discovered by Cao et al. (1999), is currently the most potent endogenous angiogenesis inhibitor in the laboratory. It is so far the most potent antiangiogenic agent in the kringle family in inhibiting endothelial proliferation, migration, and tumor growth (in animal models). However, studies on K1-5 are still limited. Therefore we were interested in its effect in treating malignant gliomas. We implanted C6 glioma cells into the rat brains. K1-5 was administrated via carotid artery 5 days later. The promising treatment results in this rat glioma model were published in Neurosurgery in 2012 (Lin et al. 2012). In this chapter, we discussed what we learn from K1-5 in a rat glioma model and highlight the main difference between

K1-5 and anti-VEGF therapy, especially in its ability to reduce tumor invasion and hypoxia.

Angiogenesis

Angiogenesis is a process for the formation of new vessels. It includes proliferation of endothelial cells and alignment of these cells to form a tube structures (vasculogenesis), elongation and branching of the newly formed vessels into the tissues (angiogenesis), and the differentiation of these vessels into arteries (arteriogenesis). The process of angiogenesis is controlled by positive proangiogenic factors as well as negative anti-angiogenic factors. The proangiogenic factors include VEGF, fibroblast growth factor (FGF), epidermal growth factor (EGF), endostatin, tumstatin, canstatin, etc. Normally, these two effects are balanced to maintain the stability of vascular structure.

Angiogenesis is of key importance in keeping the growth and spread of malignant tumors. Tumor cells can secrete proangiogenic factors to stimulate the formation of new blood vessels. They can also stimulate surrounding cells to secrete these factors to reach the same goal. For example, a glioma cell can secrete VEGF into surrounding tissues. These VEGF can bind to the VEGF receptor in the endothelial cell membranes. After a series of relay proteins, these signals reach the cell nucleus to stimulate cell mitosis as well as activation of certain genes for the production of even more VEGF. These newly formed vessels can in one hand feed the tumors with nutrient and oxygen, and on the other hand to allow tumors cells to invade nearby tissues as well as to metastasize distantly. Therefore, blocking angiogenesis can theoretically inhibit the growth and migration of malignant tumors.

Importance of Angiogenesis in Gliomas

Malignant glioma is one of the most commonly seen malignant central nervous system tumors. In gliomas, angiogenesis plays a crucial role in its growth and the development. The growth and

spread of gliomas within the host tissue depends upon their ability to obtain nutrients and remove waste products. Malignant gliomas initiate angiogenesis and neovascularization by disrupting the normal homeostasis between angiogenic and antiangiogenic factors. The level of angiogenesis is correlated with the aggressiveness of gliomas and is a useful marker for patient's prognosis (Birlik et al. 2006).

Structurally, the tumor vessels in gliomas are dilated, tortuous, and disorganized (Fig. 3.1a). The rapid proliferation of endothelial cell causes variable thickness of vessel walls with multiple layers of endothelial cells to obliterate the vascular lumen. Therefore, although there is dense vascular structure in malignant gliomas, the actual tissue perfusion is poor leading to tumor necrosis. In fact, multiple layers of endothelial cells and tumor necrosis are the characteristic pathological features of a glioblastoma (Jain et al. 2007). Furthermore, the leakage of tumor vasculature and the recruitment of pericytes result in the impairment of the blood-brain barrier (BBB) and then cause vasogenic oedema and increase of interstitial fluid pressure. In general, long-term corticosteroid treatment is required to control vasogenic edema. However, numerous side effects may accompany chronic corticosteroid usage.

Tumor Vasculature Has Tumor Origin

In Dec. 2010, two research groups identified that some blood vessels in tumors are originated from tumor cells. Wang et al. (2010) demonstrated that stem-like cells (CD133) without endothelial lineage characters (CD144 vascular endothelia-cadherin, maker for endothelial progenitors) are capable of differentiating into endothelial progenitor cells while retaining its tumor stem cell characters at the same time (CD133⁺/CD144⁺) when promoted by extrinsic signals such as co-culturing with tumor-derived endothelial cells. A subset of CD133⁺/CD144⁺ cell is also observed in the collected human glioblastoma specimens. Although this subset is a very small fraction (1.4%), it may provoke the progression of glioblastoma. Ricci-Vitiani et al. (2010) analyzed endothelial cells from glioblastoma with remarkable angiogenesis, 20–90% of these endo-

thelial cells show glioblastoma-specific chromosomal aberration, which may indicate that these endothelial cells have tumor origin.

Antiangiogenesis in Treating Malignant Gliomas

The prognosis of malignant gliomas is extremely poor despite extensive efforts at surgery (such as adding intraoperative neuromonitoring and 5-aminolevulinic acid (5-ALA) to increase tumor resection), radiotherapy (such as adding volumetric-modulated radiation treatment technique (RapidArc), an intensity modulated radiation therapy (IMRT) technique and a 3D conformal technique), chemotherapy (such as adding intraoperative Gliadel and postoperative temozolamide). One of the reasons for its poor prognosis is that these tumors are “smart” tumor which can obtain resistance to the therapeutic agents and these survived tumor cells (most likely cancer stem cells) may become more malignant and regrowth (recurrent) (Chen et al. 2012). Therefore, the future therapy for malignant gliomas relies on finding new targets or new combination of treatment.

Among many targets that are currently under investigation, antiangiogenesis has gained the most attention because malignant gliomas is highly vascularized. Since 2000s, a wide range of antiangiogenic factors has been approved by FDA for clinical use and clinical trials for cancer therapy. Glioblastoma has been in trials of some drugs approved for other indications. This included: Erlotinib (phase II, approved for lung cancer in China, 2004), Sorafenib (phase II, approved for kidney cancer in the US, 2005), and lenalidomide (phase II, approved for myelodysplastic syndrome in the US, 2005).

Currently, bevacizumab was the only antiangiogenic drug proved by FDA to use in recurrent glioblastoma. Bevacizumab is a neutralizing antibody for VEGF. In two single-arm trials, bevacizumab was proven to be effective in treating recurrent glioblastoma with a median response duration of 4.2 and 3.9 months, respectively. Based on the impressive positive results of bevacizumab, the FDA granted accelerated approval of its use as a single agent in recurrent glioblastoma

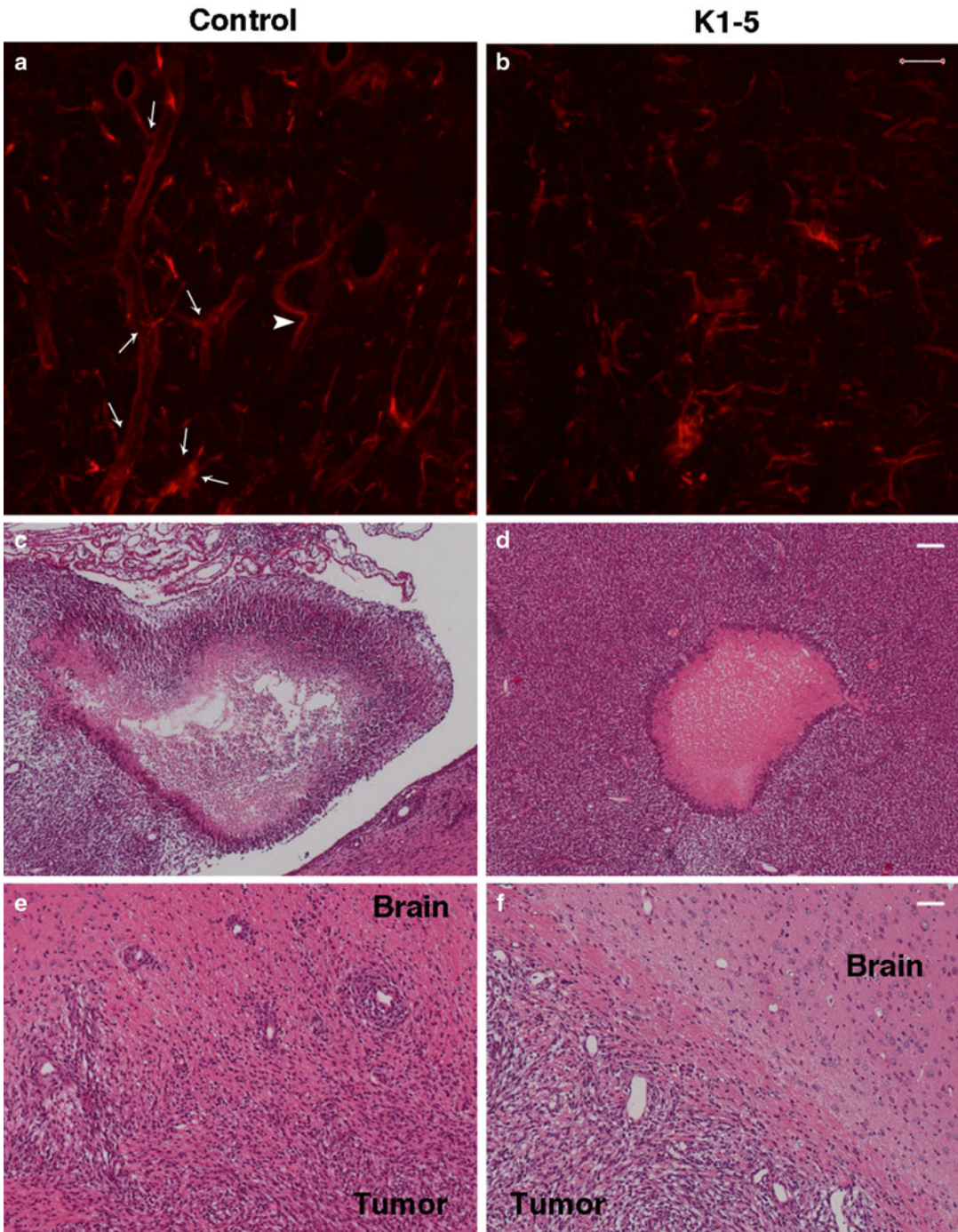


Fig. 3.1 The normalization of the tumor vasculature by K1-5 may lead to the inhibition of tumor invasion and decrease necrotic area. (a, b): in the hotspot areas (the area with the highest density of microvessels), the blood vessels are torturous (arrowhead) with many branches (arrows) in the control tumor (a). In contrast, the diameter of vessel lumen, the vessel density, the number of vessel branching were all decreased in the K1-5 treated tumor (b).

(c, d): Large areas of pseudopalisading necrosis in the control tumor (c); less necrotic area is observed in the K1-5 treated tumor (d). (e, f): in the control group (e), the invasion of tumor cells along the blood vessels (perivascular infiltration) is prominent and the tumor-brain margin is ill-defined. In the K1-5 treated tumor (f), the invasion is inhibited and the tumor-brain margin is well-defined. Scale bar (a, b): 20 μ m, (c, d): 100 μ m, (e, f): 50 μ m

in May 2009. Bevacizumab has also been showed to be effective in prolonging median survival when combining with chemotherapy or radiotherapy (8.2–9.7 and 12.5 months, respectively). It is effective at delaying the progression of brain tumors, too. Beside its antiangiogenetic effect, a recent study by von Baumgarten et al. (2011) has demonstrated a potential indirect antitumoral effect in high dose of bevacizumab.

However, there are some concerns. Similar to chemotherapy and all other targeted therapies, malignant gliomas develop resistance to anti-VEGF signaling axis. During the past few years, growing clinical evidences indicate that the therapeutic effects of antiangiogenic drugs are not as promising as expected. The most significant improvement is the reduction of brain edema, according to MRI T2 image (Pope et al. 2006). It is likely, at least partly, due to the inhibition of VEGF effect in the progression of brain edema (Lucio-Eterovic et al. 2009), since VEGF, in addition to its antiangiogenic activity, can increase the vessel permeability and thus lead to cerebral edema. However, there are other agents that can reduce brain edema in a much lower cost. It is thus doubtful to use such an expensive drug in the cost effectiveness perspective. In addition to brain tumor, the use of Avastin in other cancers is also disappointing. In 2010, FDA revoked the approval of the breast cancer indication for Avastin because it is not beneficial for life quality and survival time in these patients. It was first approved at 2004, and in Nov. 18, 2011, FDA announced their decision that Avastin does not appear to be safe and effective in breast cancer patients. Beside anti-VEGF axis, clinical trials targeting other angiogenesis agents such as epidermal growth factor receptor (EGFR) or platelet-derived growth factor receptor (PDGFR), which are also overexpressed in glioblastomas, were not promising (Stupp et al. 2006).

Kringle 1-5

Obviously, new targets beyond VEGF or single angiogenic factor are needed. In contrast to targeting VEGF signaling axis, endogenous angiogenesis inhibitors act directly on endothelial

cells may display a broad-spectrum inhibitory effect (Cao 2011). However, less attention was paid to endogenous angiogenesis inhibitors despite that many of them have been discovered for more than a decade. Kringle 1-5 (K1-5), discovered by Cao et al. (1999), is currently the most potent endogenous angiogenesis inhibitor in the laboratory.

The Structure and Antiangiogenic Effect of the Kringle

In 1994, O'Reilly et al. (1994) isolated a 38 kDa protein from a subclone of Lewis lung carcinoma and named it angiostatin. It is a circulating angiogenesis inhibitor in serum and urine of mice bearing a murine Lewis lung carcinoma. Amino acid sequence analysis revealed that angiostatin was identical to an internal fragment of plasminogen. This is a potent inhibitor of angiogenesis which showed specifically inhibition to endothelial cells, not to other cell type. The mechanism and their signaling events remain unclear. Proteolytic fragments of human plasminogen are similar to the murine angiostatin and both showed strong inhibition on the proliferation of capillary endothelial cells. Structurally, plasminogen contains five disulfide-linked structures, known as kringle domains, in total; angiostatin contains the first three or four kringle domains (K1-3, or K1-4). Later on, Cao et al. (1996) made recombinant fragments of multiple and individual kringle domains and compared their inhibition effect on endothelial cell proliferation. The function of kringle family is different from its intact parental molecule, plasminogen, in inhibiting endothelial proliferation. This inhibition effect of K1 was greater K4, and K5 was greater than angiostatin. In 1999, Cao et al. (1999) further made recombinant K1-5. Significant inhibition of the vascular tree by K1-5 was proved in the chick chorio-allantoic membrane, the mouse corneal micro-pocket, and murine fibrosarcoma. Structurally, K1-5 contains angiostatin (K1-4) and most of the K5 domain; functionally, it has the same potency in the inhibition of endothelial cell proliferation in cell culture when compared to the combination of K1-4 and K5. Furthermore, K1-5 can achieve

the same effect at lower doses and has a longer half-life when compared to angiostatin (Galaup et al. 2005; Schmitz et al. 2007). One mechanism of K1-5 is its ability to bind to endothelial ATP synthase, which activates the caspase-apoptotic pathway and results in endothelial cell apoptosis (Veitonmaki et al. 2004). The antiangiogenic ability in tumor was also reported in murine hepatocellular carcinomas (Schmitz et al. 2007) and human breast tumors in nude mice (Galaup et al. 2005). In glioma models, angiostatin and K5 have been shown to have an effect on inhibiting angiogenesis and migration (Kirsch et al. 1998; Perri et al. 2005). We also demonstrated that K1-5 exerted promising results in the rat glioma model with inhibition of tumor growth, tumor spreading, and prolongation of survival (Lin et al. 2012).

Discussion

K1-5 Inhibits Tumor Angiogenesis

Abnormal vascularization is a hallmark of gliomas, functionally and structurally. Morphologically, tumor blood vessels are dilated and tortuous, which may lead to poor efficiency of supplying oxygen and nutrients. Moreover, the uneven vessel lumen may obstruct drug delivery, which decreases therapeutic efficacy. In addition to the excessive proliferation of endothelial cells, pericytes and basement membranes are also abnormal. These leaky vessels result in the increase of interstitial fluid pressure. In the K1-5 treated tumor, the vessels were straighter, and the vessels length and density were also decreased significantly. The distribution of vessel size shifted to smaller diameter ($<20\text{ }\mu\text{m}$: ~60%). In contrast, the majority of C6 gliomas had vessel diameter in $20\text{--}80\text{ }\mu\text{m}$. Therefore, K1-5 is potent in trimming and normalizes tumor vessels in our model (Fig. 3.1b).

K1-5 Decreases Hypoxic Area of the Glioma

Another concern of antiangiogenic agents is that severely destroyed vasculature may compromise oxygen delivery which leads to tumor hypoxia,

which in turn can lead to an increase in tumor invasion (de Groot and Milano 2009). The aggressiveness of malignant brain tumor has been associated with tumor hypoxia (Evans et al. 2004). Tumor hypoxia leads to resistance of tumor treatment, the induction of genetic instability, the progression of aggressive tumor phenotypes (Reynolds et al. 1996; Harris 2002), the migration and invasion of glioma cells (Gessler et al. 2010), the recruitment of bone marrow-derived vascular modulatory cells (Du et al. 2008), and survival of glioblastoma stem cells (Heddleston et al. 2009). In K1-5 treatment tumor, we observed significant reduction of hypoxic area and HIF-1 α expression, which causes less pseudopalisading necrosis in the treated animals (Fig. 3.1c, d). Antibodies against VEGF or VEGFR-2 were also effective at inhibiting tumor and vessel growth; however, these agents promote a strong tumor invasion and hypoxia (Lamszus et al. 2005; de Groot and Milano 2009). Since K1-5 directly acts on endothelial cells by binding to endothelial ATP synthase which triggers the caspase-apoptotic pathway, it is likely that the proliferating endothelial cells are killed and non-proliferating endothelial cells (or “stable” vessels) remain. Therefore, it is reasonable to speculate that the remaining normalized vessels could improve tissue perfusion thus limit the devastating effect of hypoxia.

Lower VEGF Expression in K1-5 Treated Glioma

Another characteristic of glioma tissues is a high level of VEGF which is mainly responsible by tumor hypoxia. High level of VEGF has been associated with high-grade malignancy (Chaudhry et al. 2001) and is a poor prognostic factor for patients (Jain et al. 2007). Furthermore, VEGF can lead to vasogenic edema, which also relate to a poor prognosis. Currently, bevacizumab is the most prominent drugs targeting VEGF. Unlike bevacizumab that targets VEGF directly, no evidence showed that K1-5 act on VEGF ligands or receptors. Instead, the lower VEGF expression in the K1-5 treated group may be contributed by the improvement of tumor hypoxia. The decreased production of HIF

can cause a reduction of VEGF levels. Taken together, the decrease of tumor hypoxia and VEGF expression can lead to a better prognosis other than its direct antiangiogenic effect.

Inhibition Effect of K1-5 on Glioma Cell Invasion

One of the characters of malignant gliomas is diffuse parenchymal infiltration. It grows fast and infiltrate aggressively into neighboring areas of the brain tissue, which destroy neurological function, lead to poor survival, and worsen the quality of life. The ill-defined tumor area complicates thorough surgical resection. Therefore, the finding of lower degree of tumor cell invasion in the K1-5 treated group is noteworthy (Fig. 3.1e, f). As one of the main concerns of bevacizumab is that it increases the diffuse infiltration of glioma (Keunen et al. 2011), which induces a more aggressive tumor phenotype, as mechanisms of upregulation of pro-invasive molecules or alternative pro-angiogenic factors to avoid VEGF pathway (Lucio-Eterovic et al. 2009). In contrast, K1-5 may restrain the tumor cell infiltration by obliterating tumor vascular growth. Because tumor cells favor to migrate along vessels and proliferate at or near vascular branches (Farin et al. 2006), K1-5 might impede tumor proliferation and perivascular infiltration by abolishing angiogenesis and pruning the hypervascularity. However, we have to point out that human malignant gliomas not only infiltrate the adjacent tissue along the vessels, they also infiltrate extensively along white matter tracts. We do not know if K1-5 inhibits those vessel-independent infiltrating glioma cells. Recently, Zhang et al. (2012) demonstrated the combination of bevacizumab with angiostatin appears to decrease the bevacizumab-induced invasion in a glioma model. This may provide new insight of combination treatment.

Furthermore, it has been documented that angiogenesis and infiltration of human gliomas may be associated with the function of macrophages (Nishie et al. 1999). Human angiostatin (K1-3) has been shown to inhibit monocyte/macrophage migration (Perri et al. 2007b). Besides endothelial cell and macrophages, angiostatin also inhibits the migration of cancer cells (Perri et al.

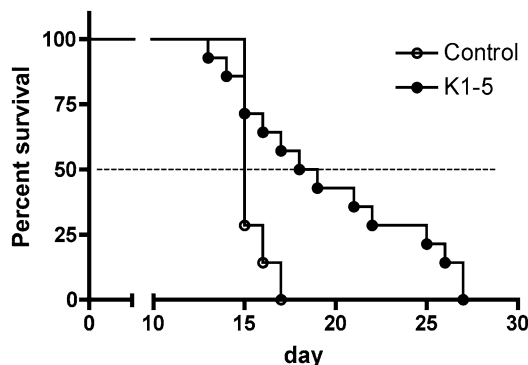


Fig. 3.2 K1-5 prolongs the survival. No animal in the control group survived longer than 17 days. In contrast, eight of the animals survived longer than 17 days after K1-5 treatment. The increase in survival is statistically significant ($p < 0.05$)

2007a). In a glioma model, Perri et al. (2005) demonstrated that the anti-tumorigenic effect of K5 relied not only on inhibiting endothelial cells but also on suppressing the recruitment of tumor-infiltrating macrophages. Therefore, reduction of macrophages might contribute to the inhibition of tumor-associated macrophage angiogenesis. Although we don't have direct evidence, it is speculated that K1-5 may possess similar function as angiostatin and K5. Further studies characterizing this function are warranted.

K1-5 Treated Animals Had Longer Survival

The application of K1-5 resulted in a significant decrease in tumor volumes and a longer survival time (Fig. 3.2). Recently Jain's lab demonstrated that vascular normalization can increase tumor perfusion and improve survival in patients who responds to anti-VEGF treatment (Sorensen et al. 2012). Because K1-5 is potent in normalizing tumor vessel, longer survival may be achieved. Further studies for longer treatment period are needed.

Conclusion

In Conclusion our result has shown that K1-5 is a potential drug for treating malignant gliomas. In spite of its direct antiangiogenic effect by

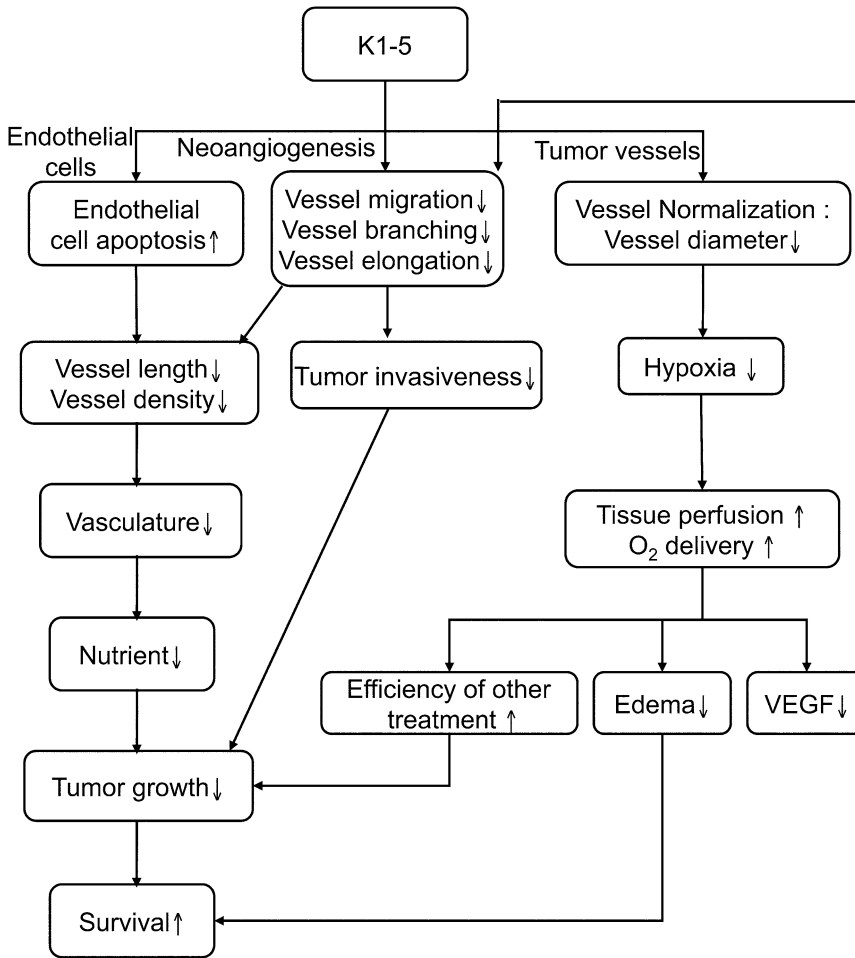


Fig. 3.3 Summary of K1-5 effects on inhibition malignant glioma. Three parallel/coexisting mechanisms involved in the processes according to the angiogenesis stage. The endothelial pathway: the action of K1-5 on endothelial cells results in cell apoptosis and decreasing vessel numbers. The lack of nutrition eventually starves tumor cells. The second pathway is by the action of anti-neovascularization. The inhibition of vessel branching and

migration decreases the tumor invasion at the same time, which reduces the tumor growth. The third pathway represents the effects of K1-5 on the normalization of tumor vessels. By normalizing the vasculature, the tissue hypoxia is improved and therefore the tissue perfusion improved. Therefore, the expression of VEGF is downregulated and tissue edema is decreased. Furthermore, the normalized microvessels are beneficial for other tumoricidal treatments

decreasing tumor vasculature and tumor, more importantly, it can normalize tumor vessels to improve tissue perfusion and thus decreasing tumor hypoxia. By decreasing tumor hypoxia, it can decrease tumor growth, tumor invasion, and VEGF production. The decrease of VEGF production itself may have the beneficial effect such as decreasing cerebral edema to improve neurological status and patient survival (Fig. 3.3).

However, the effect of K1-5 is still in its preliminary experimental status. More studies on different animal models and its exact mechanism are warranted before jumping into conclusions.

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Fluorescence Guided Resection and Photodynamic Therapy in Meningiomas

4

Martin Hefti and Gord von Campe

Contents

Abstract	35
Introduction	36
Photodynamic Diagnosis (PDD) in Meningiomas	36
5-Aminolevulinic Acid	37
Intratumoral Heterogeneity and Fluorescence Intensity in Meningiomas After 5-ALA Pretreatment	37
Photodynamic Therapy (PDT) in Meningiomas	38
References	40

Abstract

Meningiomas represent 30% of all intracranial neoplasms. They are predominantly slow growing extra axial brain tumours arising from arachnoidal cells and usually have an excellent prognosis if completely removed. However, radical tumour removal near vital functional structures and areas of infiltration is not always possible and histologically “benign” meningiomas may exhibit aggressive behaviour like infiltration and early recurrence. Biological markers that would make areas of infiltration visible by fluorescence might therefore have a significant impact on patient survival and quality of life.

5-Aminolevulinic acid (5-ALA) is a precursor in the cellular heme biosynthesis. The application of exogenous 5-ALA leads to an intracellular accumulation of protoporphyrin IX (PpIX), causing these PpIX saturated cells to become fluorescent and photosensitive under light of an appropriate wavelength. 5-ALA induced PpIX fluorescence has the ability to define infiltration zones into dural structures and bone and mark residual meningioma tissue. It has the potential to facilitate meningioma resection and to individualize the extent of dural resection for each patient in order to prevent morbidity.

Specific intracellular accumulation of PpIX can be used for photodynamic therapy (PDT). Induction of selective apoptosis, reduction of

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tumour vessel density and no risk of secondary carcinogenesis make PDT an ideal treatment option for meningiomas. Due to the variable potentials for PpIX accumulation within different meningioma subtypes, further research is required to ensure sufficiently intense fluorescence to enable PDT in these cases.

Introduction

Meningiomas are predominantly slow growing extra axial brain tumours arising from arachnoidal cells. They have a female predominance of middle to late adult life and make up nearly 30% of all intracranial neoplasms (Louis et al. 2007). Ninety percent of these lesions are of WHO grade I and have an excellent prognosis once the tumour is completely removed (Simpson 1957).

The importance of radical resection on the outcome of patients with meningiomas regardless of histological grade is well established. Non-radical resections show a 4.2-fold higher mortality during the second to the 15th postoperative year as compared to patients in whom the tumour was completely removed (Kallio et al. 1992). However, complete tumour removal is often not achieved since infiltration into bone and dura is hard to judge intraoperatively and complex topographical arrangements of vital functional structures frequently make surgery challenging (Hashemi et al. 2010). Accordingly, recurrent tumour is sometimes found at multiple sites, indicating tumour remnants left behind during a previous resection (Philippon and Cornu 1991).

Not only atypical and anaplastic meningiomas, but also “benign” grade I meningiomas may present with an aggressive behaviour such as penetration of the arachnoid membrane, destruction of bone and regrowth of residual tumour. This specific aggressive “subtype” within grade I meningiomas is indistinguishable in vivo from the more benign grade I meningioma (Pfisterer et al. 2007) and the infiltrating portion is easily missed in all types as demonstrated by the up to 20% recurrence rate of these tumours.

The extent of resection according to the Simpson grading system (Simpson 1957) is currently the standard by which surgical quality is established and continues to be used as a basis for clinical outcome. Small tumour remnants may be impossible to identify on MRI scans, and dural infiltration, bone invasion and arachnoid border destruction can be missed even with intraoperative imaging and navigation techniques. Biological markers that would make these areas of infiltration visible by fluorescence might therefore have a significant impact on patient survival and quality of life.

Photodynamic Diagnosis (PDD) in Meningiomas

Several case reports as well as small illustrative series where fluorescence was used for resection guidance in meningioma patients have been published. The reports were encouraging: a strong, sometimes homogenous fluorescence marked the tumour in most cases, regardless of histopathological subtype, tumour grading or proliferation indices (Kajimoto et al. 2007).

Visualizing the main tumour bulk in meningiomas can usually be achieved macroscopically, or with the use of an operating microscope, without adjuvant visualisation tools such as fluorescence. Without a pressing need to more specifically mark neoplastic tissue, fluorescence guided resection in meningiomas was hardly ever published. Using 20 mg/kg body weight of 5-aminolevulinic acid (5-ALA) 4 h prior to surgery to induce protoporphyrin IX (PpIX) accumulation in their meningioma patients, Kajimoto, Morofuji and others demonstrated positive fluorescence not only within the main tumour bulk, but also within areas of meningeal tumour infiltration and areas of bone invasion, regardless of tumour grading (Kajimoto et al. 2007; Morofuji et al. 2008). In accordance with the well known finding that dura infiltration shown in enhanced MRI and CT scans as “dural tail” mostly represents venous congestion (Kawahara et al. 2001), they

could elegantly demonstrate that within the areas of “hypertrophic dura” only limited fluorescence is encountered, and that these regions of fluorescence could be clearly distinguished from non-tumoral hypertrophic dura (Kajimoto et al. 2007).

More interesting still was the fact that corresponding areas of tumour invasion into bony structures of the skull showed positive fluorescence, visible through a specially fitted operating microscope. Although in most cases of meningiomas hyperostosis is due to tumour infiltration (Pieper et al. 1999), invasion of meningioma cells into bony structures has also been demonstrated in patients without radiographically identified hyperostosis (Pieper et al. 1999). The extent of bone involvement in meningioma patients is therefore impossible to delineate reliably with neuronavigation or intraoperative imaging. Moreover intraoperative pathological diagnosis cannot be accurately performed, since bone specimens need demineralization prior to histological examination. Bone invasion however directly affects outcome in meningioma patients (Gabeau-Lacet et al. 2009): an accurate visualization of tumour infiltration within these hyperostotic areas would therefore facilitate complete resection and avoid unnecessary openings of nasal sinuses and pneumatized bone at the skull base, thus directly impacting outcome. Tumour specific biomarkers, such as 5-ALA induced PpIX fluorescence, could therefore provide a new visualisation modality, as shown by preliminary data and quantitative fluorescence analysis of bone invasion within selected patients (Morofuji et al. 2008) (Kajimoto et al. 2007).

5-Aminolevulinic Acid

5-Aminolevulinic acid (5-ALA) is a precursor in the cellular heme biosynthesis, regulated by a negative feedback mechanism through the control of ALA synthetase by free heme. The negative feedback mechanism can be overcome by providing the cells with an excess amount of

exogenous 5-ALA, resulting in intracellular accumulation of fluorescent porphyrins – mainly PpIX –, which causes the cells to become fluorescent and photosensitive.

Cellular uptake of 5-ALA, increased PpIX synthesis, reduced PpIX conversion into heme and low availability of iron are probably the most relevant factors affecting the potential amount of PpIX accumulation within a cell.

Available data show that even though all these factors play a role in the heme synthesis in meningioma cells, the highest impact on PpIX accumulation stems from the lowered conversion of PpIX into heme by the reduced activity of the enzyme ferrochelatase (FCH) (Hefti et al. 2011).

Intratumoral Heterogeneity and Fluorescence Intensity in Meningiomas After 5-ALA Pretreatment

Differences in fluorescence intensity after 5-ALA application cannot only be seen between different meningiomas (Bekelis et al. 2011), but also on an intra-tumoral level (Hefti 2011). Even histologically confirmed tumour tissue can show no visible level of fluorescence, but still accumulate significant PpIX amounts to allow a distinction from the surrounding normal tissue when a more sensitive method than the human eye is used for detection (Bekelis et al. 2011). This phenomenon has already been extensively described in high grade gliomas, where it is mostly due to infiltration, necrosis, and areas of lower malignancy (Hefti et al. 2008). The exact mechanisms for the variable PpIX accumulation between meningioma cells within the same tumour are however not yet understood.

Differences in FCH activity and synthesis were shown to be the main reason for varying concentrations of 5-ALA induced PpIX accumulation between two different benign meningioma cells lines (Hefti et al. 2011). Yet, there may be multiple other reasons for these differences outside of the actual tumour biology, including tumour vascularity and hemodynamics.

In terms of tumour biology, meningiomas are genetically heterogeneous entities that display different patterns of chromosomal changes, with the presence of more than one tumour cell clone in almost half of the cases (Sayagues et al. 2004). These molecular changes may precede histopathological changes and therefore have clinical implications in as far as removal of the more malignant areas of a heterogeneous tumour has significant impact on patient survival (Scheck et al. 1996).

The question whether heterogeneity in fluorescence intensity and molecular heterogeneity correlate to some degree is crucial in defining the usefulness of 5-ALA induced fluorescence for resection control in meningiomas (Whitson et al. 2011). For, if 5-ALA induced PpIX fluorescence would not only show infiltration reliably (Fig. 4.1), but intensity would also coincide with more malignant tumour areas (with extremely low or no fluorescence in benign areas), 5-ALA induced fluorescence would be a promising tool for resection guidance, indeed.

Photodynamic Therapy (PDT) in Meningiomas

Although Simpson Grade I and II resections can be achieved with microsurgical techniques in deep seated skull base lesions, sometimes complemented by endoscopy and wide bone removal, extensive surgery in these areas often comes at a price of new neurological deficits. Eloquent anatomical structures, the brainstem, cranial nerves (to which the tumour is often adherent), blood vessels or dural sinuses are regularly compromised (Hashemi et al. 2010). Alternative strategies are therefore used to achieve local tumour control after intentional subtotal resection, mostly in the form of radiation therapy either as stereotactic radiosurgery or using radionuclides (Bartolomei et al. 2009). While generally yielding satisfactory to good results, high-energy irradiation has the potential of causing extensive DNA damage within the targeted cells, triggering mutations and carcinogenesis. As meningioma patients have a reasonable 10–15 year survival

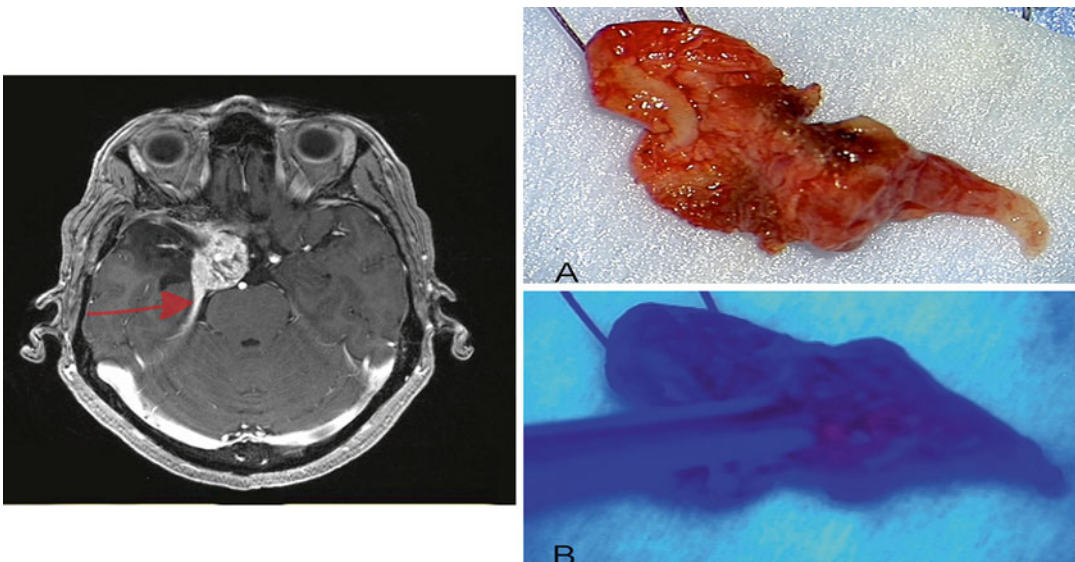


Fig. 4.1 Axial T1 sequence with Gadolinium enhancement of a medial sphenoid meningioma infiltrating the cavernous sinus and the tentorial fold. The red arrow shows the area resected as “dural tail” according to intraoperative navigation. Dural tail under white light (a) and illuminated with blue light (b) (the patient was pre-treated

with 20 mg/kg ALA 4 h previous to surgery). The lesser part of the dural tail shows active fluorescence. Fluorescent tissue could be histopathologically correlated to tumor infiltration, whereas non-fluorescent tissue showed no tumour infiltration

rate, that potential may eventually lead to tumour induction in itself.

Photodynamic therapy (PDT) consists of the administration and tumour uptake of a photosensitizer, followed by irradiation with light of an appropriate wavelength. PDT has the potential to produce selective necrosis and apoptosis of tumour cells without causing damage to the surrounding normal tissue (Dougherty 1993). It thus offers a very selective treatment modality, and consequently seems ideal for brain tumours that often infiltrate normal brain and/or are seated in close vicinity to eloquent areas. As photosensitizers tend not to accumulate in cell nuclei, but at the inner mitochondrial wall in the case of 5-ALA mediated PpIX accumulation, PDT has a low potential for causing DNA damage, chromosomal mutations or carcinogenesis. Experimental and clinical studies demonstrated that PDT complements the existing traditional tumour therapies like surgical resection, radiation therapy and chemotherapy (Schmidt et al. 2004). Compared to other adjuvant therapies, resistance to PDT has not been encountered (Fisher et al. 1995). Additionally, the effect of PDT is not only related to apoptosis and/or necrosis, but also to tumour vessel damage leading to secondary cell death (Fisher et al. 1995). PDT could reduce the necessity for aggressive and potential hazardous surgery in the vicinity of vital structures and carries no risk of secondary carcinogenesis in a disease with a relevant life expectancy. It does not interfere with established therapies, and targets tumour vascular structures. These characteristics would make meningiomas an ideal candidate for PDT.

Early preclinical studies on PDT in brain tumours begun in 1972 with glioma cell cultures (Diamond et al. 1972); subsequent clinical studies on PDT in brain tumours were published in growing numbers over the last three decades. Clinical as well as experimental studies almost exclusively involve malignant gliomas and their recurrences with the odd exception of solitary anaplastic meningioma cases published as illustrative reports (Kostron et al. 1988; Stylli and Kaye 2006).

Although experimental studies demonstrated PDT efficacy on benign as well as on malignant primary meningioma cell cultures (Marks et al. 1992; Malham et al. 1996), enthusiasm for PDT in a less “dismal” brain tumour than high grade gliomas remained low.

This is partly due to the specific side effects of PDT, like increased intracranial pressure, oedema and prolonged skin photosensitization. These side effects depend on the type of photosensitizer, its concentration, the duration of light exposure and the light intensity used (e.g.: heamatoporphyrin derivates (HPD), meta-tetrahydroxyphenylchlorin (mTHPC, Foscan)) (Kostron 2010).

5-ALA has the advantage of an improved “tumour to normal cell” uptake ratio and a skin photosensitization duration of only a few hours, as compared to other exogenous porphyrins. This translates into less or no damage to surrounding structures and significantly less brain oedema (Lilge and Wilson 1998), making it an ideally suited photosensitizer for intracranial lesions.

Insufficient 5-ALA induced PpIX accumulation in major tissues of mesoderm origin was demonstrated in 1992 (Kennedy and Pottier 1992). Meningiomas are neoplasms derived from the arachnoidal cap cells and thus considered by most of mesoderm origin. Consistent with these findings, later in vitro studies showed inadequate concentrations of PpIX in meningioma cells after pre-treatment with 5-ALA in various concentrations (Tsai et al. 1999). To make matters even worse, PpIX is less phototoxic than the exogenous HPD or mTHPC, resulting in lower photodynamic effects at comparable concentrations and light doses (Madsen et al. 2002). Low PpIX concentrations and low phototoxic potential made susceptibility to PDT in 5-ALA pre-treated meningioma cells marginal (Tsai et al. 1999). Meningiomas were therefore not considered a suitable target for 5-ALA based PDT (Tsai et al. 1999).

However, reports of consistent and frequent PpIX induced fluorescence in meningiomas after 5-ALA application encouraged further basic research on PDT in these tumours. It was found that the inadequate PpIX synthesis after 5-ALA

pre-treatment of HBL-52 meningioma cells was related to a high ferrochelatase (FCH) activity. In comparison, PpIX synthesis was abundant in a benign meningioma (BEN-MEN-1) of low FCH activity. HBL-52 cells were unresponsive to PDT, whereas over 95% of the BEN-MEN-1 cells showed loss of vitality at light doses as low as 2 J/cm² (Hefti et al. 2011). Our findings show that FCH activity plays a crucial part in the accumulation of PpIX in meningioma cells, and that when the FCH activity is low, PDT for meningioma is feasible even under physiological conditions. This data further implies that FCH inhibiting substances could ensure sufficiently intense fluorescence after 5-ALA administration to allow local PDT of tumour remnants too risky to remove surgically, even in meningiomas normally not prone to PpIX accumulation.

Skull base meningiomas infiltrating the cavernous sinus, the orbit, the tentorial fold, Dorello's canal or other anatomical regions where neurological functional integrity is at risk could be treated with PDT without the hazard of future dedifferentiation or carcinogenesis. 5-ALA based PDT could be a useful adjunct in the treatment of meningiomas, but as with PDD, the question whether heterogeneity in fluorescence intensity and molecular heterogeneity correlate to some degree will be key to the usefulness of PDT in these challenging tumours.

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Classification of Meningiomas in the Cerebellopontine Angle

5

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Contents

Abstract	43	Chemotherapy	51
Introduction	44	Location in Relation to Surgical Treatment	51
Anatomy	44	Conclusion	51
Pathology	45	References	51
History	45		
Types and Classification	45		
Premeatal Tumor Location.....	46		
Retromental Tumor Location	47		
Risk Factors	47		
Symptoms	48		
Natural History	48		
Neuroimaging	48		
Trigemino-cardiac Reflex	49		
Treatment Options	49		
Surgery	49		
Observation	50		
Radiation Therapy	50		

Abstract

Meningiomas represent the second most common type of neoplasm of the cerebello-pontine angle (CPA). Their relationship to critical neural or vascular structures of the CPA is variable and they present with different signs and symptoms. The retromental located CPA meningiomas show a significantly larger tumour size and the diagnosis is made later compared to premeatally located CPA meningiomas. Before the operation, a reduction of the facial nerve function and hearing function is present significantly more often in premeatal CPA meningiomas. The clinical appearance of the retromental CPA meningiomas is dominated by cerebellar symptoms. Both pre-operative and postoperative impairment of facial nerve and auditory function prevailed in the premeatal group. The topological classification of CPA-meningiomas according to their location anterior or posterior to the internal auditory canal is important, because the clinical presentation, the surgical strategy to be applied, and the functional outcome of critical neural structures as well as the occurrence

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of the trigeminocardiac reflex differ between the two subtypes. These substantial differences in relation to meningioma location within the CPA provide substantial evidence for the paradoxical observation that premeatal CPA meningiomas have a significantly worse postoperative functional outcome compared to retromental meningiomas although premeatal meningiomas become already symptomatic earlier and at smaller sizes.

For this reason, their relationship to critical neural or vascular structures is quite variable, and they manifest with a wide range of signs and symptoms (Langmann et al. 1990; Schaller et al. 1999b). With regard to clinical presentation and surgical management, CPA meningiomas may represent an even greater challenge than the more predictable vestibular schwannomas which always originate within the CPA. Therefore, we present modified classification of meningioma on the basis of above mentioned factors.

Introduction

Generally, posterior fossa meningiomas have presented a formidable challenge throughout the history of neurosurgery. Some of these, especially those arising from petroclival region, were considered “inoperable” up until the 1970s. However, with the advent of the newer modalities including operating microscope, computer tomogram (CT scan) and magnetic resonance imaging (MRI), microsurgical techniques as well as the development of various skull base approaches, it became possible now to resect the majority of posterior fossa meningiomas quite safely.

Meningiomas comprise of approximately 12.5% of all intracranial neoplasms and occur at frequency of 5–8% in the cerebellopontine angle (CPA) (Langmann et al. 1990; Schaller et al. 1999a; Schaller 2003a, b; Spiriev et al. 2012). Next to vestibular schwannomas (VS), meningiomas represent the second most common tumor in the CPA with a prevalence of 3–13% of all CPA lesions. The term CPA meningioma, although widely used, is somewhat imprecise in regard to the dural site of attachment, because many of these tumors in fact originate outside of the CPA. Meningiomas are believed to arise from cap cells of arachnoid villi, the site of cerebrospinal fluid absorption which are distributed throughout the intracranial space, predominantly in relation to veins and dural sinuses. Meningiomas which involve the CPA may originate from the posterior surface of the temporal bone adjacent to the internal auditory canal (IAC), from the meninges of the IAC itself, or from the inferior surface of the tentorium cere-

Anatomy

The posterior fossa, deeply concave, is formed by the occipital, the petrous and mastoid portions of the temporal bone. It is separated from the middle fossa in the median line by the basilar suture and on each side by the superior border of the petrous portion of the temporal bone. This serves for the attachment of the tentorium cerebelli, is grooved externally for the superior petrosal sinus, and at its inner extremity presents a notch, upon which rests the fifth nerve. Its circumference is bounded posteriorly by the grooves for the lateral sinuses. In the centre of this fossa is the foramen magnum, bounded on either side by a rough tubercle, which gives attachment to the odontoid ligaments; and a little above these are seen the internal openings of the anterior condyloid foramina. In front of the foramen magnum is the basilar process, grooved for the support of the medulla oblongata and pons Varolii, and articulating on each side with the petrous portion of the temporal bone, forming the petro-occipital suture, the anterior half of which is grooved for the inferior petrosal sinus, the posterior half being encroached upon by the foramen lacerum posterius or jugular foramen. Above the jugular foramen is the IAC. Behind the foramen magnum are the inferior occipital fossae, which lodge the lateral lobes of the cerebellum, separated from one another by the internal occipital crest, which serves for the attachment of the falx cerebelli, and lodges the occipital sinuses. These fossae are surmounted, above, by the deep transverse grooves for the lodgment of the lateral sinuses.

Pathology

Meningioma arise from one of the layers of the meninges (Samii and Ammirati 1991; Samii and Draf 1989; Schaller 2005). These are usually benign, slow growing tumors and do not spread into normal brain tissue (Samii and Ammirati 1991; Samii and Draf 1989). A meningioma usually grows inward causing pressure on the brain or spinal cord. It may grow outward toward the skull, causing the skull to thicken (Samii and Ammirati 1991; Samii and Draf 1989; Schaller 2005). Some of them may also contain cysts, small deposits of calcium or tightly packed bunches of blood vessels (Schaller et al. 1995, 1999a).

History

Harvey Cushing was the first to use the term “meningioma” in 1922, although this tumor was known under various names since the seventeenth century by French surgeon Antoine Louis, who called it fungus durae matris (Kaye 2009). Various description and terms as well as site of origins followed thereafter (Bondy and Lee Ligon 1996; Okonkwo and Laws 2009). Meningiomas are now known to arise from

arachnoidal cap cells of the leptomeninges. It was in 1915 as Harvey Cushing sided with Cleland and considered that meningeal tumors arose from arachnoidal cap cells. Seven years later, Harvey Cushing proposed the term “meningioma” for these tumors and the term achieved global acceptance. Working together with his student, Percival Bailey, at the Peter Bent Brigham Hospital, he adopted a histopathological classification system that listed four variants: meningothelial, fibroblastic, angioblastic and osteoblastic. Later, Bailey and Bucy expanded the initial histopathological classification scheme of meningiomas in such a manner as the new WHO histological classification is almost the same despite new discoveries made by molecular biology, biochemistry and technology from the last years (Okonkwo and Laws 2009).

Types and Classification

The above mentioned WHO classification scheme recognizes currently 15 variations of meningiomas according to their cell type as seen under a microscope. These variations are called meningioma subtypes; the technical term for these cell variations is histological subtypes.

World Health Organization (WHO) meningioma classifications

WHO Grade I—Benign	WHO Grade II—Atypical	WHO Grade III—Malignant
Meningiothelial	Chordoid	Papillary
Fibrous (fibroblastic)	Clear cell	Rhabdoid
Transitional (mixed)	Atypical	Anaplastic
Psammomatous		
Angiomatous		
Microcystic		
Secretory		
Lymphoplasmacyte-rich		
Metaplastic		

The vast majority of intracranial meningiomas (92%) have a benign histology, whereas remaining 8% show atypical or malignant features. For all meningiomas, the most common histopathological subtype is the meningotheliomatous type (63%), followed by transitional (19%), fibrous (13%), and

psamomatous (2%) meningiomas (Kondziolka et al. 1999a, b). Fortunately, less than 10% of meningiomas are malignant (Park et al. 2009).

In fact, meningiomas are the second most common tumor which arises in the CPA second most after vestibular schwannomas (Schaller and

Buchfelder 2006; Schaller et al. 2007, 2008). CPA meningiomas arising from the dura of the posterior surface of the pyramid, lateral to the trigeminal nerve, are defined as CPA meningiomas (Gardner and Robertson 1988). In accordance with others, we also consider those meningiomas that arise from adjacent areas but have their main bulk in the CPA, as CPA meningiomas (Hitselberger and Gardner 1968; Schaller and Baumann 2003; Spiriev et al. 2011). Therefore, meningiomas constitute the second most frequent tumor of the CPA after the vestibular schwannomas and compromise 6–15% of all the CPA tumors (Hitselberger and Gardner 1968; Schaller and Baumann 2003; Spiriev et al. 2011).

It is known that the origin of meningiomas is the arachnoid villi that are primarily found along major venous sinuses, especially around the superior sagittal sinus. Meticulous histological study has shown that these arachnoid villi can also be within the IAC and may serve as a site of origin for premeatal meningiomas. Nager and Masica (1970) found that arachnoid villi were distributed not only along the dural sinuses and in the gasserian envelopes but also along the greater superficial petrosal nerve, within the IAC, around the geniculate ganglion of the facial nerve, and within the jugular foramen. Guzowski et al. (1976) histologically examined 200 randomly selected temporal bones and confirmed the presence of arachnoid granulations around the petrous apex, near the trigeminal impression, and in the sulcus for the greater superficial petrosal nerve. Although they could not find true arachnoid granulations in the IAC, there were small clusters of arachnoid epithelium that could also serve as an origin of meningioma.

The variable attachment sites and direction of the growth determine the high heterogeneity of CPA meningiomas and impose different clinical presentations as well as operative challenges. In relation to IAC, these can be broadly divided into two groups; premeatal and retromental meningiomas (Gardner and Robertson 1988). This classification has both the operative as well as the prognostic significance—the more medially located tumors are related to greater surgical complexity and to poorer outcome. CPA meningiomas may develop therefore primarily anterior

or posterior to the IAC and may extend either in a cranial or caudal direction (Samii and Ammirati 1991; Samii and Draf 1989; Schaller et al. 1995). Therefore, the orientation of CPA meningiomas to surrounding structures largely depends upon its site of origin (Jackler and Lalwani 1992; Samii and Draf 1989; Schaller et al. 1995). From a surgical standpoint, the pre- and retromental locations of CPA meningiomas correspond to the CPA cisterns, which have an anatomical distinction in a superior and inferior part. The superior part of the CPA cisterns, in which premeatal meningiomas originate, contains the distal anterior inferior cerebellar artery (AICA), the labyrinthine artery and the distal branches of the basilar artery (Spiriev et al. 2011). Retromental meningiomas, mainly located in the inferior part of the cerebellopontine cistern harbor the nerves of the jugular foramen (IX–XI) and the proximal posterior inferior cerebellar artery (PICA) (Spiriev et al. 2011). For this reason, there exist different clinical manifestations as well as operative challenges. In addition, different types of intraoperative hemodynamic perturbations may be anticipated during resection of these meningiomas.

Premeatal Tumor Location

Increased frequency of tinnitus and vertigo herald a more intimate relationship between the tumor and the stato-acoustic complex as compared to CPA meningiomas posterior to the IAC. This may be explained because premeatal CPA-meningiomas compress, stretch and displace the facial-cochlear nerve complex and accompanying vascular structures posteriorly (Castellano and Ruggiero 1953; Langmann et al. 1990; Samii and Ammirati 1991). Sudden hearing loss, a common presenting symptom in vestibular schwannomas (Schaller et al. 1999b), is rare in CPA meningiomas. This may be influenced by the different tumor histology with less intimate connection between the tumor and the neurovascular bundle in case of CPA-meningiomas particularly within the IAC. In regard to the risk of intra-operative hearing loss, it is generally accepted that loss of auditory

function is more likely due to an interruption of the vascular supply of the cochlea rather than to mechanical factors like stretching or tension of the cochlear nerve. This vascular lesion is caused either by a vasospasm of the labyrinthine artery, induced by mechanical irritation of the neuro-vascular bundle or by direct injury and coagulation of labyrinthine vessels during dissection of a more vascularized tumor. Therefore, it is not surprising that the postoperative facial nerve function as well as hearing is substantially worse in premeatal group of tumors in comparison to retromeatal tumors. This might simply reflect the increased risk of inadvertent surgical damage to the neurovascular bundle in anteriorly located tumors because the mostly posteriorly displaced stato-acoustic complex imposes a continuous obstacle during tumor removal.

Retromeatal Tumor Location

Retromeatal CPA meningiomas have a much more favourable outcome in comparison to vestibular schwannomas or premeatal CPA meningiomas. Thus, they form a distinct clinical and surgical entity reflected by a number of specific features. Decreased facial or cochlear nerve function is uncommon in these tumors. The signs of cerebellar compression, which are restricted to retromeatal CPA meningiomas, represent the main clinical characteristics. The oligosymptomatic presentation of retromeatal tumors may be explained by a larger cistern that accommodates tumor expansion (Samii and Ammirati 1991) and is reflected in the larger size of tumors becoming symptomatic. In general, retromeatal meningiomas are spared from the neurovascular complex of VII and VIII nerves by an intact arachnoid layer. They only displace, but do not infiltrate the facial-cochlear nerve complex. As described by us earlier (Schaller 2003a, b; Schaller et al. 1999a), the Vth cranial nerve was frequently displaced to the superior surface of the tumour, and the lower cranial nerves (IX–XI) were often distorted towards the foramen magnum. For these reasons, auditory and facial nerve function were found to be preserved postoperatively. Preoperative lower cranial nerves

dysfunctions may be anticipated in this group and its severity sometimes determine the extubation of the patients postoperatively. In addition, impaired cough/gag reflex may impose greater chances of aspiration risks under general anesthesia.

Risk Factors

As noted earlier, meningiomas most often occur in people between the ages of 30 and 70 (Kaye 2009; Okonkwo and Laws 2009). Predilection for pediatric population is less likely. In comparison to men, women are two times more prone for having meningioma (Kaye 2009; Okonkwo and Laws 2009; Schaller 2005) and they are also three times more likely to develop malignant meningioma (Schaller 2005). Spinal meningiomas occur 10 times more frequently in women than in men (Schaller 2005).

Exposure to ionizing radiation, especially high doses, has been associated with a higher incidence of intracranial tumors, particularly, meningiomas. There is also evidence indicating a connection between meningiomas and low doses of radiation as well.

The genetic disorder Neurofibromatosis type 2 (NF2) is other condition which usually associated with higher risk of developing meningioma (Schaller 2003a, b). Patients with NF2 also tend to develop more malignant or multiple meningiomas (Schaller 2003a, b).

Per the Brain Science Foundation, a number of studies have suggested a correlation between meningiomas and hormones. Such findings include the following (Schaller et al. 1996):

- Increased occurrence of meningioma in women
- The detection of such hormones as estrogen, progesterone and androgen in some meningiomas
- A link between breast cancer and meningioma
- A connection between meningioma growth, menstrual cycles and pregnancy.

Symptoms

Because meningiomas commonly are slow-growing tumors, they often do not cause noticeable symptoms until they are quite large. Some CPA meningiomas may remain asymptomatic for a patient's lifetime or be detected incidentally when a patient has a neuroimaging for unrelated symptoms. Symptoms are typically present a mean of 2.5–5 years before the actual diagnosis (Schaller et al. 1995, 1999a). The presenting signs and symptoms depend on the size and location/extent of the tumor as described earlier here and can be related to brainstem compression, cerebellar compression, cranial nerve involvement as well as increased intracranial pressure caused by mass effect or hydrocephalus (Schaller et al. 1995, 1999a).

Brainstem compression can result in long-tract signs, spastic weakness, and somatosensory changes in 15–50% of patients (Schaller et al. 1995, 1999a). Cerebellar compression can be related to dysmetria, ataxia, and unsteady gait and can be present in as many as 70% of patients (Samii and Draf 1989; Schaller et al. 1996). Headache and papilledema are signs of increased intracranial pressure and may be related to a large size of a tumor or to an obstruction in the flow of cerebrospinal fluid (CSF) that causes hydrocephalus (Samii and Draf 1989; Schaller and Buchfelder 2006; Schaller et al. 2007).

It is difficult to differentiate small premeatal meningiomas from vestibular schwannomas preoperatively (Schaller 2003a, b). The clinical symptoms caused by premeatal meningiomas are mostly identical to those caused by vestibular schwannomas and other lesions that occupy the IAC. Most of the cases initially manifest a hearing problem (Schaller et al. 1995, 1999a). The subtle difference is that facial nerve symptoms are more likely to occur with meningiomas than with vestibular schwannomas when the size is small (Schaller et al. 1995, 1999a). Needless to say, the facial nerve schwannoma should also be considered when the patient presents with facial nerve symptoms.

Pre- and retromeatal meningiomas have typical clinical development and symptomatology (Schaller et al. 1995, 1999a). Premeatal tumors are diagnosed earlier and consequently are smaller. Their clinical presentation is with trigeminal signs and facial and cochlear nerve signs, whereas retromeatal meningiomas present with cerebellar signs (Schaller et al. 1995, 1999a).

Natural History

Although most CPA meningiomas grow slowly and have a low mitotic rate, clinical benefit has been reported in many case series with either tumor regression or stasis after radiotherapy; however, these results have not been confirmed in randomized trials. Oya et al reported on the natural history of meningiomas (Oya et al. 2011). The prospect of benign meningioma growth is an important factor to consider in their proper management. Approximately 40% of 273 meningiomas (in 244 patients) grew within a 4-year period. Lack of calcification, hyperintensity in T2 MRI, and peritumoral edema were predictors of growth in follow up. In addition, age younger than 60 years and tumor size larger than 25 mm (diameter) were also associated with a greater risk.

Neuroimaging

Tumor features and extension can be best appreciated on the contrast-enhanced magnetic resonance imaging (MRI). On T1-weighted MRI studies, meningiomas are mostly isointense to slightly hypointense relative to brain parenchyma. On T2-weighted MRI studies, they show higher intensity than that of vestibular schwannomas. Contrast enhancement is usually homogenous in both meningiomas as well as vestibular schwannomas. The radiological differential diagnosis between cerebellopontine angle meningiomas and vestibular schwannomas is based on several criteria. Usually, meningiomas are centered away

from the IAC and have broad contact with the petrous bone or the tentorium. The angle between the tumor and the pyramid is obtuse and the IAM is not widened. Although secondary invasion of the IAC might be observed in 10–22%, primary IAC meningiomas are very rare. Signal intensity of these masses on MRI will not contribute to the accurate radiographic diagnosis of the premeatal versus retromeatal meningioma. Both lesions are isointense to hypointense on T1-weighted MR images and are of variable signal intensity on T2-weighted MR images. They will also both brightly enhance after administration of contrast medium. Vestibular schwannomas that originate from the IAC comprise approximately 90% of CPA tumors (Brackmann and Bartels 1980). In this context, when a patient is found to have an enhancing mass in the IAC, it is usually assumed to be a vestibular schwannoma. Most of the reported cases of premeatal meningiomas were initially suspected to be vestibular schwannomas (Caylan et al. 2000; Haught et al. 1998; Hodgson and Kingsley 1995; Ishikawa et al. 1999; Langmann et al. 1990; Singh et al. 1975; Zeitouni et al. 1997).

Nonetheless, there are some radiographic findings that should raise the suspicion of CPA meningioma. In vestibular schwannoma, the facial nerve usually lies anterosuperiorly to the tumor and is therefore encountered late in surgery (Schaller et al. 1999a). This relationship is lost in CPA meningiomas, because the facial nerve may lie along the posterior tumor edge and can be injured early in surgery (unless care is taken to identify it). Meningiomas in the IAC also have a tendency to involve adjacent nerve tissues or bones. Nager and Masica (1970) showed, by histological examination, that meningiomas located in the IAC can invade the labyrinth and cochlea by following their individual nerve fibers to their ends. Meningiomas can also infiltrate widely into surrounding petrous bone marrow spaces and air cells. Conversely, dilatation of the IAC due to bone erosion is a more common radiologic finding with vestibular schwannomas and extensive bone invasion is unusual. Thus, the presence of bone invasion around the IAC is suggestive the diagnosis of meningioma.

Intraoperative MRI also is used during surgery to guide tissue biopsies and tumor removal. Magnetic resonance spectroscopy (MRS) is used to examine the tumor's chemical profile and determine the nature of the lesions seen on the MRI.

Trigemino-cardiac Reflex

Significant decrease in heart rate (HR) and mean arterial blood pressure (MABP) occurs during tumor dissections near the trigeminal nerve at the brainstem (Arasho et al. 2013; Schaller 2004; Schaller et al. 1999b, 2009), also in CPA meningiomas (Schaller et al. 1999b). This phenomenon is called trigemino-cardiac reflex (Arasho et al. 2013; Schaller 2004; Schaller et al. 1999b, 2009) and is described in more detail earlier in this book series (Arasho et al. 2013). Therefore all patients in whom the meningioma is removed in the CPA should undergo continuous monitoring of MABP and HR both during surgery and for at least 24 h after the procedure. It could be shown that premeatal location as well as abrupt and sustained traction on the trigeminal nerve is more prone to elicit the TCR in CPA than smooth and gentle traction (Sandu et al. 2015). A noninvasive, temporary pacemaker that can respond rapidly to abrupt changes in HR should be considered for patients with known cardiac risk factors and pre-existing higher vagal tone to prevent the occurrence of the TCR (Arasho et al. 2013; Schaller 2004; Schaller et al. 2009; Sandu et al. 2015).

Treatment Options

Surgery

CPA meningiomas primarily are benign tumors, frequently with defined borders and often enabling complete surgical removal (Schaller 2003a, b). The goal of surgery is to remove the CPA meningioma completely, including the fibers that attach it to the coverings of the brain and bone. However, complete removal can carry potential risks that may be significant, especially when the tumor has invaded brain tissue or surrounding veins.

Although the goal of surgery is to remove the tumor, the first priority is to preserve or improve the patient's neurological functions. With patients for whom total removal of the tumor carries significant risk of morbidity (any side effect that can cause decreased quality of life as described earlier), it may be better to leave some of the tumor in place and observe future growth with regular imaging studies (Schaller 2003a, b). In such cases, the patient will be observed over a period of time with regular examinations and MRIs, while for other patients, radiation therapy may be deemed the best approach. It is common for patients to undergo pre-operative embolization of the tumor to ensure safety during the surgical procedure.

Observation

Observation over a period of time may be the appropriate course of action in patients who meet the following criteria (Schaller 2003a, b):

- Patients with few symptoms and little or no swelling in the adjacent brain areas.
- Patients with mild or minimal symptoms who have a long history of tumors without much negative effect on their quality of life.
- Older patients with very slow-progressing symptoms.
- Patients for whom treatment carries a significant risk.
- Patients who choose not to have surgery after being offered alternate treatment options.

Radiation Therapy

Radiation therapy uses high-energy X-rays to kill cancer cells and abnormal brain cells, and to shrink tumors. Radiation therapy may be an option in the following cases (Kondziolka et al. 1999a, b):

- It is mainly used for small (<3 cm in diameter) residual or recurrent lesions when surgery is considered to carry a significantly high risk of morbidity.
- It has been advocated as an effective management strategy for small meningiomas and for meningiomas involving the skull base or the cavernous sinus.
- It is used primarily to prevent tumor progression.
- In a recently published series, the long-term follow up after radiosurgery was reported; a tumor control rate of 94% was found after an average of 103 months.

Standard External Beam Radiotherapy uses a variety of radiation beams to create a conformal coverage of the tumor while limiting the dose to surrounding normal structures. The risk of long-term radiation injury with modern delivery methods is very low. Newer techniques of delivery aside from 3-dimensional conformal radiotherapy (3DCRT) include intensity-modulated radiotherapy (IMRT).

Proton Beam Treatment employs a specific type of radiation in which protons, a form of radioactivity, are directed specifically to the tumor. The advantage is that less tissue surrounding the tumor incurs damage.

Stereotactic Radiosurgery (such as Gamma Knife, Novalis and Cyberknife) is a technique that focuses the radiation with many different beams on the target tissue. This treatment tends to incur less damage to tissues adjacent to the tumor. Currently, there is no data to suggest one delivery system is superior to another in terms of clinical outcome. Each has its advantages and disadvantages.

- This modality of treatment is reserved for malignant cases after failure of surgery and radiotherapy to control the disease.
- The main drugs studied include temozolomide, which had no effect against recurrent meningiomas in phase 2 study, and hydroxyurea (ribonucleotide reductase inhibitor); RU-486 (synthetic anti-progestin); and interferon-alpha. The last 3 drugs also showed disappointing results. A recently published prospective phase 2 study of irinotecan (CPT-11) also failed to demonstrate any efficacy.
- The combination of interferon alpha and 5-fluorouracil synergistically reduces meningioma cell proliferation in culture and warrants further investigation.
- The role of targeted chemotherapy to block the tumorigenic pathways of meningiomas at specific sites is being extensively investigated (Milker-Zabel et al. 2009).

- Molecules to block specific growth factors or enzymes are being developed. Atypical meningioma (WHO grade II) and anaplastic meningioma (WHO grade III) showed increased fatty acid synthase (FAS) expression. FAS inhibitor (cerulein) decreased meningioma cell survival in vitro. Thus, increased FAS expression in human meningiomas represents a novel therapeutic target for the treatment of unresectable or malignant meningiomas (Haase et al. 2010; Schaller et al. 2007, 2008).

Chemotherapy

Chemotherapy is rarely used to treat CPA meningioma, except in atypical or malignant subtypes that cannot be adequately treated with surgery and/or radiation therapy. The current experience with chemotherapy is disappointing:

Location in Relation to Surgical Treatment

The two subtypes of location present with distinct clinical features require different treatment strategy and have different prognoses. The more medially located the tumor, the greater is the surgical challenge and the poorer is the outcome.

Conclusion

The topological classification of CPA-meningiomas according to their location anterior or posterior to the ICA is important, because the clinical presentation, the surgical strategy to be applied, and the functional outcome of critical neural structures as well as the TCR differ between the two subtypes. Our results provide substantial evidence for the paradoxical observation that premeatal meningiomas have a significantly worse postoperative functional outcome compared to retromental meningiomas although premeatal meningiomas become symptomatic earlier and at smaller sizes.

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Postoperative Cerebral Hyperemia—Pathophysiology and Clinical Concerns

6

Xiao-Yu Yang, Shou-Jing Zhou, and Hong-Zhi Xu

Contents

Abstract	55
Introduction	56
Pathophysiology	56
Cerebral Autoregulation and Brain Tumor.....	56
Postoperative Hypertension	57
Carbon Dioxide (CO ₂).....	58
Hemodilution	58
Anesthetic Agents	59
Sympathetic Overactivity	59
Surgery Related Factors	60
Clinical Complications	60
Postoperative Brain Tissue Oxygenation/ Metabolism Monitor	61
Transcranial Doppler.....	61
Jugular Bulb Venous Oximetry	61
Near-Infrared Spectroscopy	61
Management of Postoperative Cerebral Hyperemia	62
Blood Pressure Control	62
Maintain Normocapnia	62
Esmolol	63
Indomethacin.....	63
Conclusion and Perspectives	63
References	63

Abstract

There is considerable controversy and a surprising lack of data regarding cerebral hemodynamic status right after craniotomy. Cerebral hyperemia occurs frequently after brain tumor resection procedures. Therefore, it needs more clinical concerns. Initially, we start with reviewing studies focusing on postoperative cerebral hyperemia, outlining the changes of cerebral blood flow and brain metabolism in the immediate postoperative period. Afterwards, we review the mechanisms currently thought to be responsible for postoperative cerebral hyperemia. Impaired cerebral autoregulation, postoperative hypertension, hypercapnia and adaption to hypocapnia, hemodilution, the effect of anesthetics, sympathetic overactivity and surgery related factors are discussed respectively. Lastly, we introduce several available monitoring techniques in the clinical setting, possible preventions and therapies of cerebral hyperemia after craniotomy. Further understanding of its pathophysiology will allow the rational development of prophylactic and therapeutical interventions of postoperative cerebral hyperemia, and eventually improve patient safety as well as outcome.

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Introduction

The brain is the organ that is most sensitive to ischemia when blood supply becomes limited. On the other hand, the brain is also very vulnerable to excessive blood flow. Postoperative cerebral hyperemia increases the risk of disastrous complications, such as intracranial hypertension, cerebral edema, and intracerebral hemorrhage (Townsend and Knowles 1999; Noorani et al. 2010).

Cerebral hyperemia is a complication which may occur after many medical procedures and situations, such as carotid endarterectomy (Noorani et al. 2010), removal of intracranial arteriovenous malformation (Chyatte 1997), or in the recovery period of cardiac arrest (Aaslid et al. 2003) and traumatic brain injury (Macmillan and Andrews 2000; Khatri et al. 2012).

It is known from peri-operative studies that cerebral hemodynamic change is a common phenomenon following complex intracranial surgery (Jabre et al. 1985; Asmussen et al. 1989; Engberg et al. 1989; Felding et al. 1994). But most of these early studies do not link cerebral blood flow (CBF) with brain metabolism. Jabre et al. found the majority of the patients (75%) developed a significant rise in CBF in the immediate postoperative period (the first day after surgery), averaging 18% of the pre-operative value (Jabre et al. 1985). The limitations of Jabre et al.'s research are the comparatively small study population (19 patients with brain tumor) and the absence of brain metabolism status so that the cerebral hyperemia cannot be confirmed. On the other hand, Asmussen et al. detected a decrease of the arterio-venous oxygen content difference (AVDO₂) immediately after extubation, and the decrease sustained during the next 3 h (Asmussen et al. 1989). In their study, Asmussen et al. did not measure CBF at the same time. Neither did they measure the AVDO₂ before anesthesia.

Recently, several researches have revealed that there is about 50% increase in the mean velocity of middle cerebral artery (Vmca) above the awake value at extubation, and the increase of Vmca is significant at least for 2 h after extubation in patients receiving craniotomy under general anesthesia. The simultaneous rise of the

oxygen saturation of jugular venous bulb (SjvO₂) confirms the occurrence of postoperative cerebral hyperemia (Bruder et al. 2002; Grillo et al. 2003; Yang et al. 2013). Moreover, Yang et al. detected a persistent difference of Vmca between the tumor side and the normal side before and after craniotomy (Yang et al. 2013). Their findings indicate that the postoperative cerebral hyperemia is much more pronounced on the side of the tumor.

Pathophysiology

At present, the mechanism of the postoperative cerebral hyperemic response is unclear. It is probably due to multiple factors involving brain tumors, brain tumor resection surgery, and different functional and structural mechanisms of cerebral blood vessels.

Cerebral Autoregulation and Brain Tumor

Cerebral autoregulation is the protective mechanism aiming to protect the brain from excessive changes in arterial blood pressure (Folino 2007). Autoregulation is a feature of many vascular beds, but is especially well developed in the brain, since it requires a high degree of homeostasis with regard to a balance of tissue nutrients and fluids (Townsend and Knowles 1999). This mechanism depends on the constriction or dilatation of arterioles in response to an increase or a decrease in arterial blood pressure (Rose and Mayer 2004; Kulic et al. 2008). Cerebral autoregulation has both myogenic and neurogenic components, the latter consisting predominantly of the sympathetic autonomic innervation (Townsend and Knowles 1999). Within the high and low autoregulatory limits of mean arterial pressure (MAP), normally ranging from about 50–150 mm Hg, CBF remains at a constant level of about 50 ml/min/100 g when at rest (Ravussin and Wilder-Smith 2001).

Conventionally, it is considered that cerebral autoregulation comprised two main components: static and dynamic. Static autoregulation refers to CBF adjustments in response to steady changes

in perfusion pressure. Dynamic autoregulation refers to the rapid regulation of CBF in response to changes in arterial blood pressure that occurs in a few seconds (Folino 2007; Ogoh 2008).

These physiological mechanisms may be impaired by a variety of pathological entities, such as head injury, subarachnoid hemorrhage or hypertension, causing an increase in sensitivity to blood pressure fluctuation (Rose and Mayer 2004).

The brain tumor vessels often lack the tight junctions that are normally present in cerebral microvessels, which results in dysfunction of the blood brain barrier (BBB). The low-grade astrocytomas may contain vessels with close-to-normal barrier characteristics. In contrast, malignant primary tumors (i.e., high-grade astrocytoma, glioblastoma) and brain metastases of systemic cancers contain vessels that are excessively leaky and lack differentiated transport properties of normal BBB vessels (Lee et al. 2006). Up to two-thirds of intracranial meningiomas are accompanied by peritumoral brain edema due to disruption of BBB with leakage of intravascular fluid and proteins out of vessels and into the brain's extracellular space (Lobato et al. 1996; Nakano et al. 2002; Vignes et al. 2008).

As a result, the presence of intracranial tumor can adversely affect cerebral autoregulation. Earlier researches reported the absence of cerebral autoregulation in patients with intracranial tumors (Palvolgyi 1969; Paulson et al. 1972). These researches give rise to the concept of impaired cerebral autoregulation in patients with brain tumors. When autoregulation is abolished, CBF becomes pressure passive, that is, it varies passively to fluctuations of systemic arterial pressure proportionally (Townsend and Knowles 1999). Thus the concept of impaired cerebral autoregulation further highlights it as an important role in the pathophysiology of postoperative cerebral hyperemia.

However, in the last decade, impairment of cerebral autoregulation with brain tumor has been challenged repeatedly. Growing evidence indicates that cerebral autoregulation is intact before and after tumor resection in most patients with brain tumors. Sharma et al. examined cerebral autoregulation and CO₂ reactivity in patients with supratentorial tumors undergoing elective tumor resection surgeries before and after surgery. They found that cerebral

autoregulation was intact before and after tumor resection in 80% patients. Larger tumor size, and midline shift more than 5 mm were associated with impaired cerebral autoregulation (Sharma et al. 2010). Schmieder and colleagues tested dynamic cerebral autoregulation using transcranial Doppler (TCD) in patients with supratentorial tumors and found that autoregulation was preserved before and after tumor resection surgery regardless of tumor location, size or histological classification of the lesion. The patients with accompanying diseases, such as diabetes mellitus and/or hypertension, however, had significantly lower autoregulation values (Schmieder et al. 2000).

The discrepancies in studies concerning cerebral autoregulation status could result from differences in characteristics of patients, brain tumors, or differences in methods of autoregulation testing. One thing to notice is that regional dysfunction of cerebral autoregulation may not be detected by devices which measures global CBF.

Postoperative Hypertension

Postoperative hypertension is a significant contributor to postoperative cerebral hyperemia. Systemic hypertension is a common consequence of stress during neurosurgical recovery, due to sympathetic overdrive, with increased blood catecholamine concentrations leading to systemic hypertension and tachycardia (Bruder et al. 1999). The incidence of systemic hypertension after neurosurgery is more than 50% (Basali et al. 2000; Bilotta et al. 2008). Within the high and low autoregulatory limits of MAP and normal intracranial pressure (ICP), CBF remains constant. Exceed the upper limit of MAP, cerebral vessel constriction is insufficient to maintain cerebral perfusion pressure (CPP) at a constant level and the relationship of CPP and CBF with MAP becomes linear, thereby cerebral hyperemia is inevitable (Nemani and Manley 2004).

In the presence of an impaired cerebral autoregulation, CBF fluctuates even within normal range of MAP. Consequently, patients with dysfunction of cerebral autoregulation are especially vulnerable to systemic MAP changes (Basali

et al. 2000). These observations have resulted in a great emphasis on aggressive control of blood pressure in the patient with brain tumor surgery (Fàbregas and Bruder 2007; Bilotta et al. 2008).

Carbon Dioxide (CO₂)

CO₂ is a potent cerebral vasodilator, resulting in a decrease in arteriole resistance and a corresponding increase in CBF (Curley et al. 2010). The effect of CO₂ on CBF depends on the individual patient, baseline blood flow, and the central nervous system (Curley et al. 2010). Overall, for every 1 mmHg change in arterial blood pressure of CO₂ (PaCO₂), CBF changes by 1–2 ml/min per 100 g of tissue, when it is between 20 and 60 mmHg (Nemani and Manley 2004). Hypercarbia can cause severe cerebral hyperemia postoperatively. Increasing PaCO₂ to 79.5 mmHg will lead to an increase in CBF of greater than twice normal (Townsend and Knowles 1999).

Impaired responsiveness to changes in PaCO₂ occurs in areas of vasoparalysis, which are associated with extensive intracranial disease such as tumor, ischemia, trauma, stroke and infection (Townsend and Knowles 1999). At present, there are still much debates on whether vascular reactivity to CO₂ remains intact in brain areas with tumor. Although Palvolgyi reported the loss of CO₂ reactivity in patients with brain tumors, Paulson et al. observed only a “dissociated vasoparalysis”, that is, loss of autoregulation with preserved CO₂ reactivity (Palvolgyi 1969; Paulson et al. 1972). Dissociated vasoparalysis was described as a sign of tissue acidosis and was thought of as a stage in gradual transition from normal vasomotor function to complete vasomotor paralysis (Paulson et al. 1972). Examined by TCD, Rao et al. found cerebrovascular response to hypocapnia was similar between the normal and the abnormal cerebral hemispheres both under intravenous and inhalational anesthesia (Rao and Pillai 2006). Sharma et al.’s data indicate preservation of CO₂ reactivity in all patients with supratentorial tumors, and dissociated vasoparalysis occurring in 20% patients (Sharma et al. 2010). These studies suggest that vascular response to CO₂ is mainly preserved in patients with brain tumor. But we should realize

that all these clinical researches do not assess the cerebrovascular response to increased CO₂ tensions because of the risk of cerebral hyperemia and increasing intracranial pressure.

Return to normocapnia after several hours of hyperventilation during anesthesia may also play a role in postoperative hyperemia. Contemporary use of hyperventilation has long been the mainstay of intraoperative management of intracranial hypertension during neuroanesthesia to facilitate access or to acutely reduce brain bulk (Gelb et al. 2008; Curley et al. 2010). It is an effective means of reducing elevated intracranial pressure by inducing hypocapnic vasoconstriction of the cerebral arterioles, causing a reduction in CBF and a corresponding decrease in intracranial pressure (Curley et al. 2010). The effect of sustained hypocapnia on CBF decreases progressively because of buffering, which can normalize CBF as quickly as 6 h; subsequent normocapnia can cause rebound cerebral hyperemia and increase intracranial pressure (Albrecht et al. 1987; Curley et al. 2010). However, several hours of hyperventilation does not cause postoperative cerebral hyperemia in patients with normal CO₂ activity undergoing abdominal surgeries, indicating that adaption to hypocapnia may not be a key factor among the mechanisms of postoperative cerebral hyperemia (Yang et al. 2013).

Hemodilution

Most of the O₂ transported is reversibly bound to hemoglobin because the solubility of O₂ into blood is quite low. The amount of O₂ dissolved in the blood is thought to be negligible. Therefore, a decrease in the concentration of hemoglobin in the blood, as can occur through dilution after acute hemorrhage or acute fluid replacement therapy, results in a corresponding reduction in the O₂-carrying capacity of the blood (Nemani and Manley 2004). With hemodilution, cerebral arterioles dilate and CBF increases to compensate for the reduced O₂-carrying capacity (Rebel et al. 2003; Nemani and Manley 2004). There is 2–3% increase in CBF for each percentage decrease in hematocrit (Bruder et al. 1998; Kulic et al. 2008). Whereas the balance between oxygen supply and demand is unchanged (Daif et al.

2012). Therefore, hemodilution does not cause absolute cerebral hyperemia.

Anesthetic Agents

It is generally accepted that some volatile agents can cause cerebral vasodilatation and contribute to cerebral hyperperfusion in a dose-dependent manner (Townsend and Knowles 1999; Randell and Niskanen 2006). But debates still exist in their effects on CBF at different concentrations. Autoregulation is not impaired with 1 minimal alveolar concentration (MAC) isoflurane, but partially impaired with 1.5 MAC (Girard 2004). Sevoflurane has been shown to be a less potent vasodilator than isoflurane (Townsend and Knowles 1999). It is beneficial in preserving cerebral autoregulation. At least under clinical concentration (especially up to 1.5 MAC), it does not affect CBF (Bundgaard et al. 1998; Kaisti et al. 2003; Randell and Niskanen 2006). But some researches find that CBF changes are similar between isoflurane and sevoflurane anesthesia at different MAC level in patients with brain tumor (Petersen et al. 2003; Fukui et al. 2010). Nitrous oxide (N_2O), when addition of other anesthetics, has been reported to increase CBF, cerebral metabolic rate of O_2 ($CMRO_2$), and reduce cerebral autoregulation, but their magnitudes are still being debated (Strebel et al. 1995; Iacopino et al. 2003).

In contrast, decrease in CBF and $CMRO_2$ has been found during propofol anesthesia (Rasmussen et al. 2010; Klein et al. 2011). In a study of patients with brain tumor, the $SjvO_2$ was significantly higher and the $AVDO_2$ significantly lower during anesthesia with isoflurane and sevoflurane compared with propofol (Petersen et al. 2003). Cerebral autoregulation is preserved with propofol anesthesia (Harrison et al. 1999).

In general, opioids have little to no effect on cerebral autoregulation or CO_2 reactivity. Coupling between CBF and metabolism is also maintained. Fentanyl does not affect CBF (Townsend and Knowles 1999). High doses of remifentanyl decrease cerebral blood flow velocity despite perfusion pressure being constant in healthy male volunteers (Klimscha et al. 2003).

Nevertheless, these doses are much higher than in the clinical setting. Likewise, modern nondepolarising myorelaxants have minimal effects on intracerebral haemodynamics (Ravussin and Wilder-Smith 2001).

It is also possible that prolonged anesthesia and residual effect of anesthetics might have affected postoperative CBF. However, most commonly used anesthetics are short-acting and the blood concentration of them is quite low in the period of emergency from anesthesia. Recent studies of patients undergoing craniotomy make this assumption less plausible. In one study, Bruder et al. find no difference of postoperative cerebral hyperemia between propofol and isoflurane anesthesia groups (Bruder et al. 2002). In another study from Yang et al., cerebral hyperemia does not occur in patients after abdominal surgery under isoflurane anesthesia (Yang et al. 2013).

Several studies find that postoperative cerebral hyperemia does not change proportionally with blood pressure or $PaCO_2$. Furthermore, the CBF changes are more severely than those of blood pressure and $PaCO_2$, indicating that some other factors may also contribute to its occurrence (Jabre et al. 1985; Asmussen et al. 1989; Bruder et al. 2002; Grillo et al. 2003; Yang et al. 2013).

Sympathetic Overactivity

It has been well established that many stressful events increase CBF and $CMRO_2$ (Fàbregas and Bruder 2007). Jabre et al. find that the postoperative changes of CBF was not due to a difference in either $PaCO_2$ or BP. They believe that it reflects a normal reaction of the cerebral vasculature to the inevitable disturbance of surgery (Jabre et al. 1985). Grillo et al. further confirm the mechanism of the stress, which is associated with emergence and resultant sympathetic stimulation, since esmolol, a short-acting β -blocking drug, can blunt postoperative cerebral hyperemia (Grillo et al. 2003). However, esmolol decreases cardiac output, which may contribute to the decrease of CBF (Ogoh 2008). This mechanism has an assumption that cerebral BBB is disrupted and cerebral autoregulation is abolished. Because intravenous infusions of norepinephrine has no

affected to CBF in both human (von Knobelsdorff et al. 1996) and animal studies (Myburgh et al. 1998). Besides, there are strong evidences indicating that increases in brain sympathetic activity could mediate vasoconstriction (Mitsis et al. 2009), which make this mechanism of postoperative sympathetic overactivity unsubstantiated.

Surgery Related Factors

The effect of neurosurgery on CBF remains sketchy and controversial. During tumor resection, coagulate an artery or vein could possibly cause cerebral hyperemia. It is known that ischaemia lasting as little as 30 s is sufficient to provoke a three- to four-fold increase in CBF. Such cerebral hyperperfusion has been documented in experimental animals during recovery from ischaemia induced by temporary ligation of arteries. This hyperaemia affects the previously ischaemic tissue only, and persists from minutes to hours (Pulsineli et al. 1982). Obstruction of venous drainage may lead to “occlusive hyperemia”. In an animal experiment, occlusion of two cortical veins impaired cerebral autoregulation, that the cerebral autoregulation-curve lost its plateau-like appearance, with the propensity for hyperperfusion injury at a lower level of MAP (Schaller et al. 2002).

Resection of brain tumor and the application of cerebral retractors may cause postoperative reactive cerebral hyperemia, which is similar as cerebral ischemia/reperfusion injury (Hai et al. 2004). Ischemia occurs under the surgical brain retractors in 5–10% of patients (Ravussin and Wilder-Smith 2001). In the study of intracranial meningiomas and their surrounding tissue, Sergides et al. found that perilesional brain areas were ischemic (Sergides et al. 2009). Pennings et al. detected a decrease of oxygen pressure in peritumoral brain tissue, especially in patients with brain swelling, which indicating ischemia in peritumoral tissue due to increased local pressure. Furthermore, the oxygen pressure in peritumoral tissue increases after dura opening and tumor resection (Pennings et al. 2002). To protect the brain against ischemia, the

microcirculation reacts to hypoxia through induction of vasodilatation (Kulic et al. 2008; Ainslie and Ogoh 2010). After resection of brain tumor, the chronic ischemic brain area restores normal perfusion pressure. It is possible that autoregulatory vessels around the tumor might have disrupted BBB and impaired cerebral autoregulation due to hypoxia (Ainslie and Ogoh 2010), therefore have been chronically dilated and become less responsive, their return to normal caliber might occur gradually after tumor resection. In animal studies, autoregulation was found not to recover acutely after deflation of a balloon inserted into the brain to simulate a localized mass lesion (Miller et al. 1973). It can be regard as a kind of reperfusion process. Interactions between reperfusion and the ischemic tissue can lead to further tissue injury and disruption of BBB, which has been termed reperfusion injury (Khatri et al. 2012). Reperfusion injury occurring in chronic cerebral hypoperfusion area is similar to acute cerebral ischemia/reperfusion injury in rat model (Hai et al. 2004). This mechanism could involve in the development of postoperative cerebral hyperemia even in the presence of normal systemic arterial pressure.

Clinical Complications

Severe cerebral hyperemia, called ‘cerebral hyperperfusion syndrome’, can cause extended cerebral edema, intracranial pressure increase, intracerebral hemorrhage and death (Noorani et al. 2010). There is, however, limited reports of patients’ adverse events and outcome with less severe but acute, extensive, cerebral hemispheric postoperative hyperemia which is short-lasting and self-limiting. Postoperative cerebral hyperemia may lead to devastating postoperative complications such as cerebral edema, intracranial hypertension, or cerebral hemorrhage (Townsend and Knowles 1999). Or maybe, it could potentially cause more insidious and subtle problems, such as memory loss, difficulties with concentration, and modification of thought processes (Falero et al. 2000).

In a retrospective case control study, researchers do find a link between perioperative hypertension and intracranial hemorrhage after craniotomy. Of particular interest was the very strong correlation between hypertension with intracranial hemorrhage when blood pressure remained in the normal range intraoperatively but became increased postoperatively. This suggests that loose surgical hemostasis performed at a low cerebral perfusion pressure may bleed when at a higher perfusion pressure (Basali et al. 2000). Moreover, the structurally weaker capillaries in brain tumor and peritumor region are more likely to rupture and bleed due to the distension during cerebral hyperemia (Sekhon et al. 1997). But a causal relationship between postoperative cerebral hemodynamic changes and the risk of cerebral hemorrhage could not be confirmed yet.

To present, we do not clearly know to what extent early postoperative neurosurgical complications, such as cerebral hemorrhage or edema, may be related to postoperative cerebral hyperemia. There is no definite evidence from randomized controlled trials to demonstrate its effect on patient's outcome, either.

Postoperative Brain Tissue Oxygenation/Metabolism Monitor

Transcranial Doppler

Transcranial Doppler (TCD) is a simple, reliable, non-invasive, bedside technique for beat-to-beat measurement of the CBF velocity in the basal cerebral arteries. TCD measures CBF velocity in the middle cerebral artery (MCA) through a bony window in the temporal fossa (Aaslid et al. 1982). As the diameter of MCA is not affected by autoregulation, TCD is a reliable index of assessing changes in CBF (Serrador et al. 2000). Many studies have confirmed a linear relationship between the changes of CBF velocity and that of CBF in most circumstances (Serrador et al. 2000; Rijbroek et al. 2009; Sorond et al. 2010). These researches validate TCD as a valuable tool to monitor CBF changes postoperatively.

Measurement of CBF alone is insufficient because this parameter does not account for brain metabolism. However, several researches has shown a good correlation between V_{mca} and $SjvO_2$ in monitoring postoperative cerebral hyperemia (Bruder et al. 2002; Yang et al. 2013). Therefore TCD is a well-suited technique in clinical setting for identifying postoperative cerebral hyperemia. Monitoring the CBF velocity of tumor side is recommended, since the postoperative cerebral hyperemia is much more pronounced on the side of the tumor (Yang et al. 2013).

Jugular Bulb Venous Oximetry

Continuous or intermittent estimation of the global balance between cerebral oxygen demand and supply can be measured by jugular bulb venous oximetry. This is done by measuring the $SjvO_2$ through percutaneous retrograde cannulation of the internal jugular vein with a catheter with embedded optical fibers (Nemani and Manley 2004). Normal $SjvO_2$ is assumed to be 54–75% (Macmillan and Andrews 2000). In the absence of anemia and changes in hemoglobin O_2 saturation, increase in $SjvO_2$ above 75% is indicative of absolute or relative hyperemia (i.e., supply in excess of metabolic requirement). A value less than 50% reflects increased oxygen extraction and indicates a potential risk of ischemic injury (Macmillan and Andrews 2000). The major limitation of this technique is that it is unable to detect discrete regions of hyperemia or ischemia unless these are of sufficient magnitude to influence the saturation of brain effluent blood (Nemani and Manley 2004). Besides, its invasive nature also limits its use in postoperative settings.

Near-Infrared Spectroscopy

Near-infrared spectroscopy (NIRS) is a noninvasive optical method for monitoring regional cerebral oxygenation by analyzing light absorption of the brain. NIRS monitors mixed vascular

O₂ saturation in capillaries, arteries, and veins within a small area beneath the sensor. NIRS may penetrate the cerebral cortex by only 2–3 mm, and therefore the region of cerebral tissues sampled probably extends no further than the gray matter (Folino 2007). Various studies have demonstrated that intraoperative transcranial regional cerebral oxygen saturation monitoring using NIRS correlates closely with CBF, as confirmed by animal studies which show oxyhemoglobin concentration is a reliable marker of perfusion (Hoshi et al. 2001).

The major limitations of NIRS include inter-patient variability, variable optical path length, potential contamination from extracranial blood, and lack of definable threshold (Folino 2007). Currently, it is considered a trend monitor, with each patient serving as control. In situations of potential regional hyperemia, bilateral monitoring should be used.

Management of Postoperative Cerebral Hyperemia

Blood Pressure Control

Intensive systemic blood pressure monitoring and control has been recommended after craniotomy (Hatashita et al. 1986; Mayhan 1990; Fàbregas and Bruder 2007). Postoperative poorly controlled hypertension coupled with impaired cerebral autoregulation, may increase the possibility of intracranial hemorrhage, most of which occurs within the first 6 h after surgery (Basali et al. 2000; Wong et al. 2012). The consensus is to control blood pressure at normal values, while the optimal blood pressure after craniotomy is still unknown (Fàbregas and Bruder 2007). Thus treatment of arterial hypertension should be based on physiological data of brain in the individual patient. Postoperatively monitoring techniques and equipment are advocated which allow the direct non-invasive monitor of brain, such as TCD and NIRS.

Many drugs with different mechanisms can be used to control postoperative blood pressure after elective craniotomy. Esmolol, labetalol and

metoprolol have been administered to blunt postoperative hemodynamic changes safely (Felding et al. 1994; Bilotta et al. 2008). Labetalol is given in small incremental doses of 0.25–1 mg/kg to a maximum dose of 2.5 mg/kg (Fàbregas and Bruder 2007). The short half-life of esmolol (9 min) allows a rapid titration of its effects in this context. Thus, esmolol seems particularly suitable to prevent the systemic and cerebral hemodynamic changes after neurosurgery. It is administered with a loading dose of 500 µg/kg followed by a continuous infusion rate of 100–300 µg/kg/min (Bilotta et al. 2008). Administration of dexmedetomidine also has been reported to reduce the hypertensive response to emergence (Tanskanen et al. 2006). Calcium channel antagonists, nitroglycerin, and sodium nitroprusside are cerebral vasodilators, which may increase cerebral blood volume and ICP (Morimoto et al. 2000; Fàbregas and Bruder 2007). Moreover, nicardipine attenuates both static and dynamic cerebral autoregulation and increases CBF (Morimoto et al. 2000; Endoh et al. 2000, 2002). Thus special cautious should be drawn when using these kinds of medicine to control postoperative hypertension after craniotomy.

Nevertheless, blood pressure control alone is insufficient to avoid cerebral hyperemia after neurosurgery, due to the complexity of mechanisms which are involved in postoperative hyperemia.

Maintain Normocapnia

In the neurosurgical context, respiratory extubation criteria need to be strictly observed, such as adequate respiratory drive with acceptable oxygenation and normocapnia, state of consciousness, adequate muscle strength, and etc. (Ravussin and Wilder-Smith 2001). Low PaCO₂ during surgery should be gradually normalized as soon as is feasible (e.g. after adequate tumor debulking), which can decrease the possibility of rebound cerebral hyperemia and ICP rise, in sudden CO₂ normalization (Gelb et al. 2008; Curley et al. 2010).

Esmolol

Esmolol is an ultra-short acting, highly cardio-selective β_1 -adrenergic receptor antagonist which is effective in treating arterial hypertension and tachycardia in the perioperative setting (Bilotta et al. 2008). In a clinical study, esmolol alleviated postoperative cerebral hyperemia after craniotomy (Grillo et al. 2003). Cautious should be made as esmolol can also decrease cardiac output and blood pressure as well as heart rate (Grillo et al. 2003). Interestingly, esmolol does not affect CBF, cerebrovascular reactivity or cognitive performance in healthy volunteers or patients under anesthesia (Heinke et al. 2005; Saito et al. 2005; van der Starre et al. 2008). It is contraindicated in patients who are prone to bronchial asthma, have chronic obstructive pulmonary disease, or a history of congestive heart failure.

Indomethacin

Indomethacin, a cyclooxygenase inhibitor, can act as a cerebral vasoconstrictor and decrease CBF. Animal experiments as well as clinical studies suggest the effectiveness of indomethacin in the management of intracranial hypertension and attenuates the cerebrovascular reaction to hypercapnia (Rasmussen 2005). Intravenous administration of indomethacin rapidly reduces CBF and ICP with no effect on CMRO₂ during craniotomy for supratentorial cerebral tumors (Bundgaard et al. 1996). The mechanism of CBF reduction by indomethacin is not fully understood. The clinical use of indomethacin in managing elevated ICP has been limited due to the risk of causing cerebral ischemia (Rasmussen 2005). However, the clinical reports of indomethacin are confined to patients with severe head injury or during brain tumor resection. There has not been any clinical studies regarding its effect in management of cerebral hyperemia in the postoperative circumstances. Further studies are expected to elucidate the effect of postoperative use of indomethacin and its optimal dosage.

Conclusion and Perspectives

Clinical studies in patients with cerebral tumors provide evidence that postoperative cerebral hyperemia occurs frequently after brain tumor resection surgeries. The mechanism of postoperative cerebral hyperemia is probably multiple-factorial which includes disruption of cerebral autoregulation, hypertension and hypercapnia. Sympathetic overactivity under surgical stress and cerebral ischemia-reperfusion injuries may also involve. Postoperative cerebral hyperemia may lead to postoperative complications such as cerebral edema, intracranial hypertension, or cerebral hemorrhage. Blood pressure control and maintain normocapnia can help to prevent postoperative cerebral hyperemia. However, the reported clinical studies were performed mostly on inconclusive small patient materials using uncontrolled designs. Large-scale randomized studies are urgently needed to reveal detailed information and about cerebral hemodynamics in patients after brain tumor surgeries. Further understanding of its pathophysiology will allow the rational development of prophylactic and therapeutical interventions of postoperative cerebral hyperemia, and eventually improve patient safety as well as outcome.

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Evidence-Based Approach for the Management of Cerebral Metastases from Lung Adenocarcinoma

7

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Contents

Abstract.....	67
Introduction.....	67
Pathology.....	68
Neuroimaging.....	69
Computed Tomography.....	69
Magnetic Resonance Imaging.....	69
Nuclear Medicine Techniques.....	71
Management.....	72
Surgery.....	72
Radiotherapy.....	73
Chemotherapy.....	73
Conclusion.....	74
References.....	75

Abstract

Adenocarcinoma is the most common histologic subtype of lung cancer in most countries, accounting for almost half of all lung cancers. A considerable proportion of non-small-cell lung cancer [NSCLC] patients will develop central nervous system [CNS] metastases throughout the course of their disease and these manifestations cause significant morbidity and mortality. Thus, aggressive management of cerebral metastases is vital to preventing death from neurological causes and prolonging functional independence. This chapter focuses on current knowledge of pathology, imaging and management of brain metastases from lung adenocarcinoma and outlines a practical approach for the management of cerebral metastases.

Introduction

Cancer is the leading cause of death in economically developed countries and the second leading cause of death in developing countries. About 12.7 million cancer cases and 7.6 million cancer deaths are estimated to have occurred in 2008. Lung cancer is the leading cancer site in males, comprising 17% of the total new cancer cases and 23% of the total cancer deaths. Among females, it was the fourth most commonly

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diagnosed cancer and the second leading cause of cancer death. Lung cancer accounts for 13% [1.6 million] of the total cases and 18% [1.4 million] of the deaths in 2008 (Jemal et al. 2011). Adenocarcinoma is the most common histologic subtype of lung cancer in most countries, accounting for almost half of all lung cancers.

Brain metastases are the most common complication of systemic cancer and hospital and autopsy-based studies estimate that these tumors develop in 20–40% of all adult cancer patients (Posner and Chernik 1978). Empiric screening of patients with newly diagnosed NSCLC identify brain metastases in 3–10% of cases. Brain metastases, particularly those from NSCLC, constitute an increasing proportion of this global cancer burden. It is estimated that 40% of cancer patients will develop brain metastases and that 50% of all patients with brain metastases will have a primary lung malignancy (Yokoi et al. 1999).

Taking everything into account, the presence of brain metastases is associated with high morbidity and mortality; in general, more than 50% of these patients will die within 6 months of diagnosis. In the study of Nussbaum et al. (1996), the median duration from diagnosis of cancer to brain involvement was 3–6 months for lung cancer and the duration from brain metastases to death was 4 months.

The actual management of cerebral metastases from lung adenocarcinoma, as with other metastatic tumors, is complex. This chapter summarizes the crucial actions for the treatment of cerebral metastases from lung adenocarcinoma, also along with some common advices for other cerebral metastases.

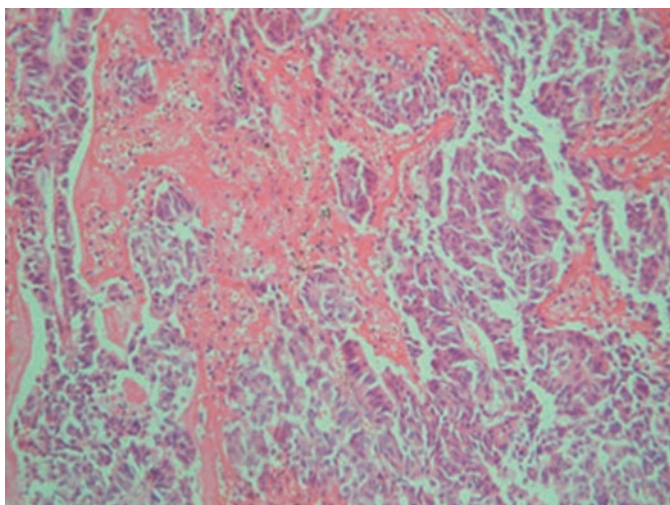
Pathology

Brain metastasis formation in lung adenocarcinoma involves multi-step, highly discriminating molecular events with complex interactions between tumor and host cells. Following complex genetic alterations affecting tumor cell adhesion, motility, protease activity and internal signaling pathways, systemic tumor cells with a more motile and aggressive phenotype usually

travel to the brain by hematogenous spread through the arterial circulation (Nathoo et al. 2005). All those complex steps have to be successfully completed for tumor cells to invade through blood–brain barrier [BBB], a dynamic interface that separates the brain from the circulatory system, and survive. Seeding of tumor cells are more likely to be promoted with narrowing of cerebral vasculature and slowing of blood flow and so brain metastases are commonly located beneath the gray-white matter junction and superficial distal arterial fields. The distribution of brain lesions generally reflects the blood flow. The most common locations for metastases are the frontal and temporal lobes, other lobes of the cerebrum, the cerebellum and diencephalic region. Unusual presentations were also seen (Samanci et al. 2011). After overcoming the BBB, tumor cells face up with components of the local microenvironment including the extracellular matrix, resident brain parenchymal cells, and paracrine signaling molecules including cytokines and growth factors (Preusser et al. 2012). Once the metastatic cells have traveled to the nervous system, the brain provides adequate blood supply via angiogenesis, making brain metastases grow and proliferate better. Although new knowledge is achieved over time, these mechanisms still remain unclear.

Brain metastases usually form sharply demarcated, spherical masses in the brain parenchyma. They often do not infiltrate the surrounding brain parenchyma, but cause edema due to mass effect. Larger tumors tend to have necrotic centers and the tumor is usually surrounded by an extensive amount of vasogenic edema. However, this edema is sometimes disproportional to the size of the lesions. Occasionally, metastases from lung carcinomas appear cystic. The lesions can be single or multiple. However, patients with lung adenocarcinoma are more likely to have multiple metastatic lesions. Nussbaum et al. (1996) examined the records of 729 patients with brain metastases and 52% of the patients with lung cancer had multiple cranial lesions. The histological features of brain metastases from lung adenocarcinoma are usually similar to those of the primary neoplasm on microscopic examination (Fig. 7.1).

Fig. 7.1 Photomicrograph of sheets of highly pleomorphic malignant cells with epithelial characteristics. Note glandular formation. H&E @ 200× (Reproduced from Samanci et al. 2011)



Neuroimaging

Increased sensitivity and specificity of cerebral metastasis detection is possible with modern neuroimaging techniques.

Computed Tomography

As Computed Tomography [CT] is a readily available and fast imaging modality, non-contrast CT scans are often the initial diagnostic test for symptomatic patients presenting to emergency department. On non-contrast CT scans, metastases are typically hypodense to isodense, well-circumscribed solid lesions. Most metastases are surrounded by low attenuation vasogenic edema. Grey-white matter differentiation is maintained and the edema involves mainly white matter, extending in finger-like fashion. The edema may be quite extensive in relation to the size of the lesion (Fig. 7.2).

Asymptomatic patients with known systemic NSCLC are screened for brain metastases with a contrast-enhanced study. Iodinated contrast enhancement is vital to the detection of metastases on CT, and brain metastases may demonstrate ring, nodular, or solid enhancement. More metastases are visible on delayed imaging with high-dose contrast and the size of a given metastasis

may appear to increase on delayed imaging (Sidhu et al. 2004).

Computed Tomography may fail to identify lesions with small size and low contrast. Lesions missed on CT are almost always less than 10 mm in size. Although not as sensitive as magnetic resonance imaging [MRI], there is no clinical difference between these imaging methods in terms of survival time or rate in NSCLC. Alternatively, if contrast-enhanced CT shows multiple brain metastases, there may be little added value of obtaining brain MRI (Soffietti et al. 2006).

Magnetic Resonance Imaging

Magnetic resonance imaging is a sensitive screening test for brain metastasis and is suggested in cases where surgery or stereotactic radiosurgery [SRS] is planned, in order to detect additional lesions; in cases where CT is negative but there is a strong clinical suspicion for metastases in patients with known malignancy; and in patients for whom CT is not conclusive in determining whether a lesion is neoplastic or non-neoplastic (Soffietti et al. 2006).

On MRI, most metastases are usually hypointense to iso-intense on T1-WI and iso-intense to hyper-intense on T2-WI. Surrounding edema is hypo-intense on T1-WI and more hyper-intense on T2-WI. Hypointensity of metastatic lung

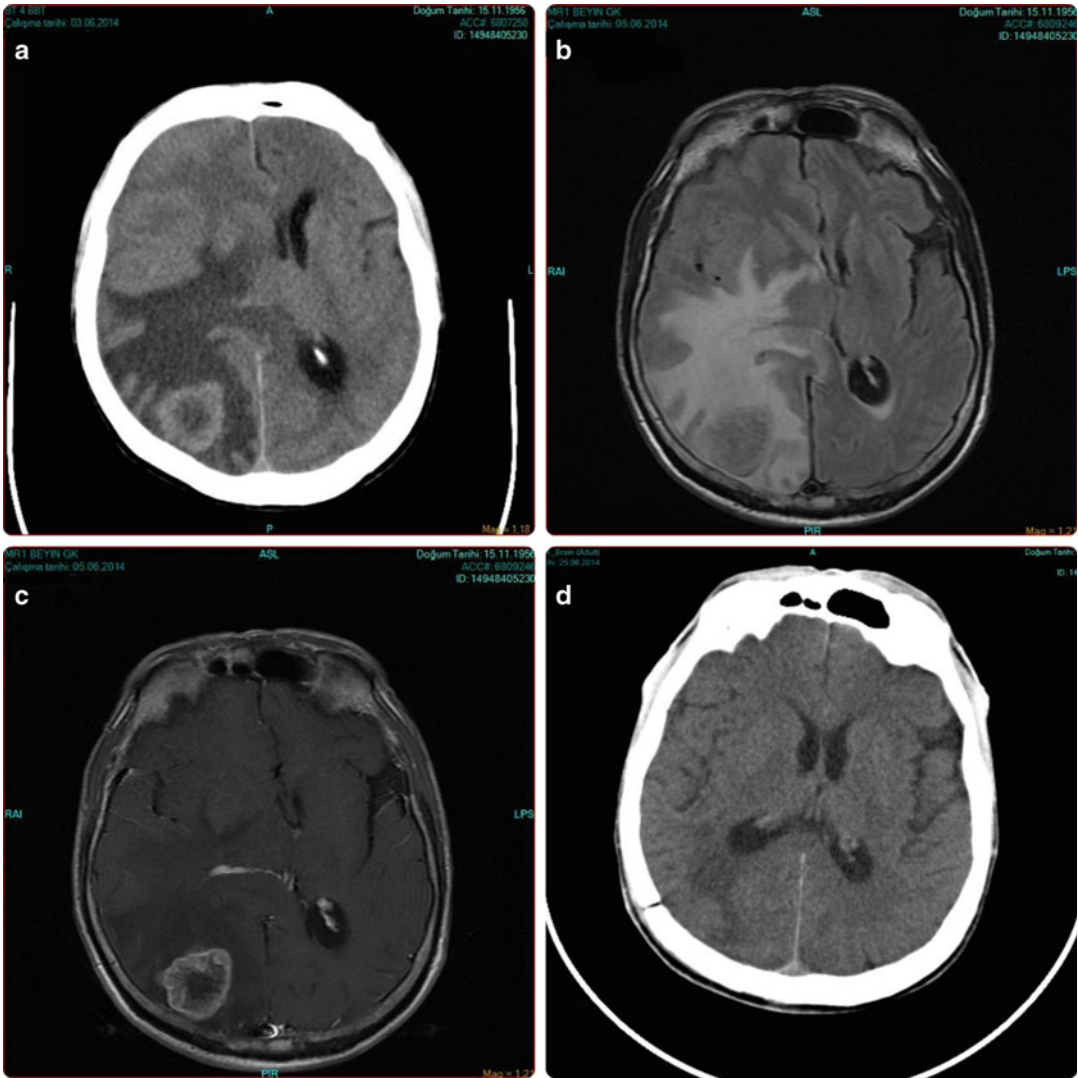


Fig. 7.2 A 58-year-old man with lung adenocarcinoma metastasis. (a) Axial non-contrast CT image showing a heterogeneous hyperdense mass lesion in *right* occipitoparietal area with extensive edema around and observed hypodense areas in the *middle*. (b) FLAIR image showing

vasogenic edema extending to frontotemporoparietal and occipital *white* matter and marked *midline* shift. (c) Contrast-enhanced axial T1WI image showing thick, circular, non-homogeneous contrast enhancement with irregular contours. (d) Post-operative [20 days] axial CT image

adenocarcinomas on T2-WI has been documented in a few reports but no correlation was found regarding the presence of mucin, blood products, iron or calcium (Carrier et al. 1994).

Small cortically based metastases may not demonstrate any visible edema, and can therefore be missed on unenhanced MRI. Unenhanced MRI can also fail to demonstrate small nodules hidden by edema from other more dominant

lesions. Gadolinium contrast enhancement is vital to detect small metastases and distinguish non-neoplastic white matter disease from metastases. But contrast enhancement may be altered by radiation, chemotherapy or steroid therapy. The standard gadolinium contrast dose is 0.1 mmol/kg of body weight and a useful technique for increasing lesion detection is increasing the contrast dose beyond that. Although

lesions greater than 10 mm will be seen at all dose levels, the utility of higher dose gadolinium mainly lies in the ability to reveal smaller lesions less than 5 mm. Studies utilizing incremental doses of intravenous contrast dose over the range of 0.05–0.3 mmol/kg resulted in more lesions detected (Baleriaux et al. 2002). Apart from increasing the contrast dose, delayed MRI with standard-dose gadolinium was found to be useful and one study concluded that the post-contrast scan should be started with a 2–5 min delay after contrast medium administration (Akeson et al. 1997). Also some studies have suggested the use of standard gadolinium dose with magnetization transfer, a technique that increases the contrast of enhancing and non-enhancing lesions by suppressing background signal, is equivalent to those reported for triple-dose gadolinium-enhanced MRI with conventional spin-echo techniques (Knauth et al. 1996). Higher field magnetic resonance magnets above 1.5 Tesla [T] now have widespread clinical availability and it has been shown that 3.0 T imaging increases subjective assessment of brain metastases (Ba-Ssalamah et al. 2003). Besides MRI techniques, contrast agents with higher T1 relativity have been shown to improve the sensitivity of brain metastasis detection (Colosimo et al. 2001).

Metastatic tumors with higher tumor cellularity or high nuclear-to-cytoplasmic ratio can show restricted diffusion on diffusion-weighted images [DWI]. Diffusion imaging aids in distinguishing metastases from pyogenic abscesses and can play a role in describing cystic from solid components in metastases (Bukte et al. 2005). A small study evaluating DWI signal intensity and apparent diffusion coefficient values of various histologic types of metastases found lower DWI signal intensity in enhancing portions of well-differentiated adenocarcinomas as compared with poorly differentiated carcinomas (Hayashida et al. 2006).

Perfusion-Weighted Imaging can be used as an adjunct to conventional imaging to assess the degree of tumor microvascularity and to evaluate grading and malignancy. Perfusion imaging may help differentiate between brain metastases and

cerebral abscesses, which can appear identical on anatomic imaging (Hakyemez et al. 2006).

Magnetic Resonance Spectroscopy provides semi-quantitative evaluation of metabolite levels to characterize brain tumors. Commonly measured metabolites include choline, N-acetyl aspartate, lactate, lipids and creatine and creatine phosphate. N-acetyl aspartate is low or absent in metastases, consistent with the lack of neuroglial elements (Al-Okaili et al. 2006).

Nuclear Medicine Techniques

Single-photon Emission Computed Tomography [SPECT] and Positron Emission Tomography [PET] provide a means of examining regional cerebral blood flow, metabolism, and pharmacology in vivo. These molecular imaging techniques rely on radiolabeled molecules and PET utilizes short-lived positron emitting isotopes, whereas SPECT uses lower energy γ -emitting isotopes.

Positron Emission Tomography has the advantage that it is more sensitive, has higher intrinsic resolution and also allows for the evaluation of suspected metastatic lesions and whole-body staging in a convenient single examination. Single-photon Emission Computed Tomography is, however, cheaper and more widely available (Brooks 2005).

Thallium-201 is a potassium analogue tracer for SPECT and has been used chiefly in the evaluation of primary tumors, but also shows accumulation in metastatic tumors. One study reported a sensitivity of 70% for both thallium-201 and Technetium-99 m-MIBI SPECT in the detection of cerebral metastases (Kojima et al. 1994).

FDG-Positron Emission Tomography and PET are increasingly utilized tools particularly in the staging of lung cancer. FDG-Positron Emission Tomography has not been found to be as sensitive as MRI in the evaluation of brain metastases from lung cancer. Positron Emission Tomography was only 27% sensitive compared with contrast-enhanced MRI for brain metastases (Kruger et al. 2011).

Management

Surgery

There is now an important role for surgical resection of metastatic brain tumors but a number of factors should be carefully considered for a patient with metastatic brain tumor in developing a treatment plan. The benefit of surgery in the treatment of brain metastases has been validated by data from randomized trials (Patchell et al. 1990; Vecht et al. 1993; Vogelbaum and Suh 2006). The study by Patchell et al. (1990) showed that surgical resection plus irradiation improved patient outcome in comparison to the irradiation alone. Median survival and functional independence was significantly longer in the surgical group. Vecht et al. (1993) randomized a group of 63 patients with solitary brain metastases and median survival was significantly longer in patients treated with surgery plus whole brain radiotherapy [WBRT] versus patients treated with WBRT alone.

The Radiation Therapy Oncology Group [RTOG] used a recursive partitioning analysis [RPA] of patients with brain metastases to define three classes according to pre-treatment characteristics with prognostic significance to select homogeneously candidates for local interventions (Gaspar et al. 1997). Recursive partitioning analysis class I patients have a Karnofsky performance score [KPS] ≥ 70 and age < 65 years, with the primary tumor controlled and without extra cranial metastases. Recursive partitioning analysis class III is defined by a KPS of < 70 , and RPA class II covers the remaining patients. The median survival times for the 3 RPA classes are 7.1, 2.3, and 4.2 months, respectively. Patients in Class 1 are the best candidates for aggressive treatment. Patients in Class 3 are unlikely to derive benefits from invasive therapy. Patients in RPA Class 2 comprise a heterogeneous subset with intermediate survival rates, and for those who have a better prognostic index, such as controlled primary disease or good performance status, aggressive control of intracranial disease may still be justified (Pollock et al. 2003).

In addition to considering these RPA class prognostic factors, it is also important to consider tumor histology. Guo et al. (2012) assessed the impact of histologic subtypes on survival stratified by the graded prognostic assessment [GPA]. The authors analyzed 780 patients with NSCLC brain metastases treated from 1982 to 2004. Graded prognostic assessment classification variables included age, KPS, number of brain metastases, and presence of extra cranial disease. Mean survival time, in months, by histology was: adenocarcinoma 6.2 [n=464], large cell 4.1 [n=98] and squamous 4.2 [n=108] [p=0.0549]. For GPA 3.5–4.0, mean survival times did not differ significantly by histology. Differences in mean survival times by histology were noted for GPA 3.0 [p=0.04], GPA 1.5–2.5 [p=0.01], and GPA 0–1.0 [p=0.02]. For all patients with brain metastases, brain metastases from NSCLC, mean survival times by GPA score were: GPA 3.5–4.0, 12.6; GPA 3.0, 10.2; GPA 1.5–2.5, 5.8; and GPA 0–1.0, 2.7. Adenocarcinoma showed a statistically significant higher mean survival time than other histologies of NSCLC for patients with GPA 0–3.0.

Patients with solitary brain metastases are the best candidates for surgical removal. But with better procedures, surgery is also an option for patients with multiple brain metastases. Bindal et al. (1993) reported similar survival rates for patients undergoing surgery for multiple metastases to that of solitary metastases. The outcome of surgical resection for multiple metastases appears similar to that of patients having only single metastases for patients with high performance status and controlled systemic disease (Pollock et al. 2003).

Surgery has been shown to improve survival and quality of life in patients with recurrent disease. It provides an opportunity to confirm tumor histology; remove necrotic tissue; and use localized adjunctive therapies. Both radio-necrotic and recurrent tumors will benefit from surgical debulking and intraoperative histological confirmation (Sills 2005).

Recursive partitioning analysis provides a general framework for patient selection, but in some occasions, surgical treatment may become an option for those who do not meet standard

criteria, and vice versa. Surgery is the preferred treatment if there is a significant mass effect, if the therapeutic delay associated with radiation treatment is not acceptable and if SRS results in an unacceptably high radiation dose to the surrounding brain parenchyma. Although a small single lesion might be considered for SRS, extensive peritumoral edema, refractory seizures and resistance to radiation therapy lead one to surgery. Surgery might also be indicated for obtaining histology and sufficient tissue samples for providing optimal treatment (Sills 2005). Although with modern techniques, there are few areas of the brain that are considered surgically inaccessible, lesions that are deep and within eloquent areas are inevitably associated with higher surgical morbidity. In metastases that impinge on eloquent areas or the motor strip, subtotal resection may be the only option, especially if SRS is not appropriate (Tangjitgamol et al. 2004).

Various advanced techniques and surgical adjuncts [stereotactic intra operative guidance, functional MRI, diffusion tractography, intraoperative MRI, awake craniotomy, intraoperative confocal microscopy] are in use for decreasing the risk of morbidity and enhancing the benefit of surgery (Sanai et al. 2011; Orringer et al. 2012). Taken together, these technologies enable more insightful operative decision-making and risk assessment.

Radiotherapy

Whole brain radiotherapy has been used as the primary therapeutic modality for the treatment of brain metastasis. Median survival across all studies based on previous RTOG protocols ranged from 2.4 to 4.8 months, thereby proving WBRT alone is inadequate in the long-term control of brain metastasis. After receiving WBRT, most patients experience neurocognitive sequel and declines in quality of life including headache, seizures, impaired mentation, cerebellar dysfunction and motor deficits (Welzel et al. 2008). In patients with a solitary brain metastasis, randomized studies have demonstrated improved overall survival when either surgery (Patchell et al. 1990)

or SRS (Hatiboglu et al. 2013) is combined with WBRT. Combination of WBRT with chemotherapeutics yielded disappointing results. Studies with Temozolomide in patients with brain metastases from NSCLC provided limited or no benefit compared with WBRT alone (Chua et al. 2010). However combining WBRT with targeted drugs resulted in decreased toxicities and increased median survival times. Welsh et al. (2013) reported acceptable toxicity profile and promising median survival time in patients with brain metastases associated with NSCLC. But further studies with various chemotherapies and targeted therapies are needed.

The benefit of prophylactic cranial irradiation [PCI] in small-cell lung cancer is well established and it has also been investigated in patients with NSCLC, but with less compelling evidence of benefit. The authors concluded that PCI may reduce the incidence of brain metastases, but there is no evidence of a survival benefit and that there is insufficient evidence to support the use of PCI in clinical practice (Gore et al. 2011).

To avoid toxicity issues of WBRT, SRS alone has been used in patients with better prognosis and a limited number of metastases. Pooled results from three randomized trials have revealed an apparent survival advantage in younger patients treated with SRS alone compared with WBRT and SRS. Although these studies demonstrated that patients with better prognosis receiving SRS alone had a similar survival to patients who received WBRT and SRS, a meta-analysis showed reduced local and distant control of brain metastasis after treatment with SRS alone compared with WBRT and SRS. But salvage with either repeat-SRS or WBRT was found to be comparable to initial treatment with WBRT and SRS (Tsao et al. 2012). There are also advancing radiotherapy techniques [‘under-dosing’ the bilateral hippocampi, hippocampal-avoidance WBRT] for possible use in the future.

Chemotherapy

Chemotherapy has become a more viable option for the treatment of metastatic brain tumors in

recent years. The upfront association of carboplatin plus pemetrexed was reported to allow simultaneous treatment of cerebral and systemic disease in patients with NSCLC with newly diagnosed brain metastases and appear to be particularly interesting in terms of radiologic response and overall survival (Bailon et al. 2012).

Bearz et al. (2010) confirmed the good tolerability of Pemetrexed and reported a progressive disease in 12 patients [30.8%], a stable disease and partial response in 12 [30.8%] and 15 [38.4%] patients respectively, with an overall clinical benefit in 69% of patients.

Single agent Temozolomide was reported to achieve a 41% disease control in patients with recurrent brain metastases from a variety of primary malignancies, half of them suffering from NSCLC, with minimal toxicity (Abrey et al. 2001). In another study, authors concluded that there is a lack of therapeutic activity of single-agent Temozolomide in patients with stage IV NSCLC (Dziedziszko et al. 2003). A study, which sought to confirm the benefit of adding Temozolomide to WBRT in patients with NSCLC with brain metastases, showed no significant difference between the two treatment arms (Chua et al. 2010). Therefore, the role of Temozolomide in treating brain metastases remains unresolved.

A randomized pilot trial investigated whether primary chemotherapy was feasible in terms of efficacy, survival, toxicity profile, and quality of life compared with WBRT given first in chemotherapy-naïve patients NSCLC with synchronous brain metastasis when neurologic symptoms or signs are absent or controlled by supportive care and it was reported that primary chemotherapy is more feasible and can be an appropriate option for patients with synchronous brain metastasis when neurologic symptoms or signs are absent or controlled (Lee et al. 2008).

A phase III study to determine if the timing of WBRT with respect to chemotherapy with Cisplatin and Vinorelbine would influence survival in patients with NSCLC and concurrent brain metastasis suggests that the timing [early or delayed] of WBRT did not influence survival of NSCLC with brain metastasis treated with con-

current chemotherapy (Robinet et al. 2001). Upfront systemic standard platinum-based combination chemotherapy rather than standard WBRT has been proposed as an alternative (Peters et al. 2012).

The recent expansion of knowledge regarding the molecular biology of neoplasia and the metastatic phenotype has led to intense development of several targets of therapeutic intervention. Routine baseline tissue-based assessment of epidermal growth factor receptor [EGFR] mutations in patients with NSCLC and so treatment of these mutation-positive patients with EGFR tyrosine-kinase inhibitors [TKI] has been shown to result in improved survival (Rosell et al. 2012). Epidermal growth factor receptor TKI therapy might be the treatment of choice for metastatic brain tumors in NSCLC patients harboring an activating EGFR mutation (Park et al. 2012).

There is limited data on efficacy of anaplastic lymphoma kinase [ALK] inhibitor Crizotinib on brain metastases. In a study by Shaw et al. (2013), Crizotinib was found to be superior to standard chemotherapy in patients with previously treated, advanced NSCLC with ALK rearrangement.

A meta-analysis demonstrated that Bevacizumab significantly prolonged overall survival and progression free survival when added to first-line platinum-based chemotherapy in patients with advanced NSCLC (Soria et al. 2013). Data on efficacy of other targeted therapies in lung cancer patients with brain metastases, including BRAF inhibitors, HER2 inhibitors or RET inhibitors are lacking.

Conclusion

The following data could be used for basic management of brain metastases along with some exceptions:

- Computed tomography and MRI remain the primary modalities utilized for the detection of metastatic tumors of the CNS.
- While certain findings on standard and advanced imaging are suggestive of a particular

diagnosis, imaging is not able to reliably predict the histology of a brain metastasis.

- For patients with single brain metastases, the data supports surgical resection for patients with good performance status and controlled primary cancer [RPA Class 1 and subset of Class 2].
- For patients with unfavorable prognosis [RPA Class 3], local aggressive treatment must still be considered for palliation of neurological symptoms or avoidance of death.
- For patients with multiple brain metastases, the potential of a combined surgical/radiosurgical approach holds promise.
- For small [<2 cm], asymptomatic and deep-seated tumors, SRS provides a feasible non-invasive alternative.
- Response rates of systemic therapy vary according to prior brain-directed treatments, number of previous lines of systemic treatments, and molecular features of the tumor guiding appropriate targeted therapies.

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Intramedullary Spinal Cord Tumors: Long-Term Follow-Up After Surgery

8

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Contents

Abstract	79
Introduction	80
Personal Series: Clinical Presentation and Surgery	80
Patients Population.....	80
Personal Series: Operative Procedures	82
Personal Series: Surgical Results	83
Ependymomas.....	83
Low Grade Astrocytomas	83
High-Grade Astrocytomas	84
Other Histological Types	86
Personal Series: Clinical Long-Term Follow-Up	86
Pain	86
Impaired Sensitivity	86
Sphincter and Sexual Dysfunction.....	86
Motor Disorder.....	87
Cord Tethering	87
Discussion	87
Conclusions	92
References	93

Abstract

We review a series of patients who underwent surgical removal of intramedullary spinal cord tumors, focusing on the long-term functional outcome, recurrence rates for the various tumors, and technical problems continually debated in neurosurgical practice. From December 1972 to December 2011, 202 patients underwent removal of intramedullary tumors. Lesions were located in the cervical spinal cord in 61 patients (30%), at a dorsal site in 60 (29%), at a cervicodorsal site in 51 (25%), and in the medullary cone in 30 (15%). The most frequent histological tumor types were astrocytomas (86 patients, 42%) and ependymomas (68 patients, 34%).

Of the 68 ependymomas, 55 (81%) were completely removed and 13 (19%) incompletely removed. In 66% of the patients (42 patients), the presenting signs and symptoms remained unchanged at long-term follow-up; in 25% (16 patients), they improved; and in 9% (6 patients), the clinical status worsened. Of the 27 Grade I astrocytomas, 22 (81%) were completely removed and 5 (19%) incompletely removed. Functional assessment of the 23 patients available at “late” follow-up showed that 26% (6 of 23 patients) had improved, 9% (2 of 23 patients) had worsened, and 66% (15 of 23 patients) remained unchanged from preoperative status. Conversely, of the 41 Grade II astrocytomas,

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only 5 (12%) were completely removed, and 10% had improved. Only 2 of the 22 high-grade astrocytomas (10 Grade III astrocytomas and 12 glioblastomas) could be completely removed. The clinical postoperative status worsened in 14 patients (63.6%), was unchanged in 7 patients (31.8%), and there was 1 case of intraoperative death (4.5%). None of the 22 patients showed improvement in their neurological status postoperatively. In this series, excluding the 1 intraoperative death, all patients died of progression of the malignancy. Determinant predictors of a good outcome after surgery for intramedullary spinal cord tumors are histological type of lesion, complete removal of the lesion, and a satisfactory neurological status before surgery. Concerning the high grade astrocytoma, surgical treatment did not ameliorate the postoperative neurological status; instead, in the majority of cases, it prompted a worsening of the deficit. Radiotherapy and chemotherapy have a little influence on the length of survival. In this series, multimodality treatment of intramedullary high-grade astrocytomas has been shown to increase length of survival without improving the neurological status.

Introduction

Tumors of the intramedullary spinal cord are relatively rare lesions, accounting for only 2–4% of all central nervous system tumors (Stein and McCormick 1992). In adults, they account for approximately 20% of spinal tumors, whereas in children, the percentage reaches 35%. The most frequent histological types of intramedullary tumors are glial tumors (80%), astrocytomas, and ependymomas, whereas oligodendrogliomas and subependymomas and gangliomas and cavernomas are less frequent (McCormick and Stein 1990; Miller 2000; Schwartz and McCormick 2000). Hemangioblastomas account for 3–8% of all intramedullary tumors (Neumann et al. 1989) and in 15–25% are associated with von Hippel-Lindau syndrome, autosomal dominant inherited disease with incomplete penetrance (Parsa et al.

2000). Only 2% of all intramedullary tumors are metastases, usually from primary lesions in the lungs or breast (Raco et al. 1992). Ependymomas account for 15% of spinal tumors and, in adults, 60% of intramedullary spinal cord tumors, being the most frequent intramedullary tumor in adults. In patients older than 60 years, the incidence of astrocytomas and ependymomas is equal (Chang et al. 2002). The incidence of intramedullary tumors correlates closely with age. In children and young adults, astrocytomas, usually of low grade, are two to four times more frequent than ependymomas. Only 10% of astrocytomas in childhood are anaplastic lesions or glioblastomas, whereas in adults, the percent-age reaches 25% (Constantini et al. 2000; Epstein and Farmer 1990; Guidetti et al. 1981; Innocenzi et al. 1996). Astrocytomas in children show an indolent biological behavior only in part explainable by the histological features, age being a favorable prognostic factor. Low-grade fibrillary astrocytomas are biologically more benign in children than in adults. In a study conducted by Sandler, 60% of the patients younger than 21 years at diagnosis were still alive without recurrence 10 years after surgery, whereas for patients older than 21 years at diagnosis, the percentage dropped to 40% at 5 years without recurrence (Sandler et al. 1992).

The series of patients we review span a long time and several surgeons, chairmen at Sapienza University, contributed along the years to these results. Among them, the late professor Beniamino Guidetti, Professor Gianpaolo Cantore, professor Roberto Delfini and myself A.R.

Personal Series: Clinical Presentation and Surgery

Patients Population

A consecutive series of 241 patients underwent surgery for intramedullary tumors at the Neurosurgical Division of the University of Rome between December 1972 and December 2011.

The population of 241 patients comprised 142 male patients (59%) and 99 female patients (41%), whose ages ranged from 8 to 72 years (mean, 42.3 years).

Patients with tumors involving the filum terminale were excluded because their clinical and therapeutic implications differ from those of intramedullary tumors.

Histological features, site and extension were evaluated.

Clinically, the duration of clinical symptoms before diagnosis and the presenting symptoms.

Patients treated early in the series underwent diagnostic imaging with myelography or computed tomographic myelography or both procedures, and those treated from 1986 onward underwent gadolinium-enhanced spinal magnetic resonance imaging (MRI) (111 patients). Five patients with suspected vascular lesions also underwent spinal angiography.

All patients underwent pre- and postoperative neurological evaluation and functional assessment, in which we used a simplified version of the McCormick classification system as follows: Grade I (neurologically normal, mild focal deficit not significantly affecting the function of the involved limb, deficit affecting the function of the involved limb, mild to moderate gait difficulty, and severe pain or dysesthetic syndrome impairing the quality of life, although the patient can still function and walk independently); Grade II (presence of a motor and sensory deficit affecting the function of the involved limb, mild to moderate gait difficulty, and severe pain or dysesthetic syndrome impairing the quality of life, although the patient can still function and walk independently);

Grade III (severe deficit, with the patient requiring a wheelchair, cane, or brace, with bilateral upper-extremity impairment; the patient is not independent).

In all but 1 of 241 patients with intramedullary tumors treated, the tumors were surgically removed through a posterior median approach with a laminectomy, or a laminotomy in children, and a median continuous myelotomy or discontinuous myelotomy (20 patients) (Cantore et al. 2002). One patient harboring a ventrally placed intramedullary cavernous angioma was operated on via a transthoracic anterior approach.

From 1990 onward, all patients underwent postoperative MRI follow-up within 72 h of surgery, at 4–6 months, and thereafter yearly.

Of the 241 patients, 226 (94%) were available for “late” follow-up review.

We analyzed our surgical and clinical results according to the various histological tumor types. Special attention was paid to III and IV grade astrocytoma according to WHO classification, that constitute a different subgroup.

The average follow-up for the entire population of 190 patients available is 8.3 years.

Progression-free survival (PFS), defined as the absence of any clinical sign of recurrence of the disease, and overall survival were estimated by use of the Kaplan-Meier technique.

We tested the PFS in the different histopathological types (Fig. 8.1a) and in the different kinds of resection (gross total or partial) (Fig. 8.1b).

The most frequent histological tumor types were astrocytomas (101 patients, 42%) and ependymomas, including one subependymoma (81 patients, 34%), followed by epidermoid-dermoid tumors (11 patients, 5%), hemangioblastomas (11 patients, 5%), oligodendrogliomas (5 patients, 2%), metastases (5 patients, 2%), neuromas (5 patients, 2%), and meningiomas (2 patients, 1%). The series also included 20 cavernous angiomas (8%), accounting for 22% of the 72 intramedullary neoplasms treated from 1991 to 2003. At histological classification of the astrocytic intramedullary tumors according to the World Health Organization, 29 (29%) were Grade I (pilocytic astrocytomas), 50 (49%) Grade II, 10 (22%) Grade III and 12 grade IV lesions. A specific subgroup formed by 22 Grade III and IV astrocytomas were considered for the evaluation of results.

The duration of clinical symptoms before diagnosis ranged from 2 months to 20 years (mean, 3.1 year), being shorter for group with malignant lesions (ranged from 2 months to 14 months, mean 6 months).

The presenting symptoms and signs were hypesthesia or paresthesia or both complaints in 167 patients (69%), motor disorders in 48 patients (20%), and sphincter dysfunction in 26 patients (11%). High grade astrocytomas were located in the cervical spinal cord in 12 patients, and in the thoracic site in 10, and the presenting symptoms and signs were hypesthesia, paresthesia, or both in 20 patients (90.9%); motor disorders in 19 (86.3%); and sphincter dysfunction in 9 (40.9%).

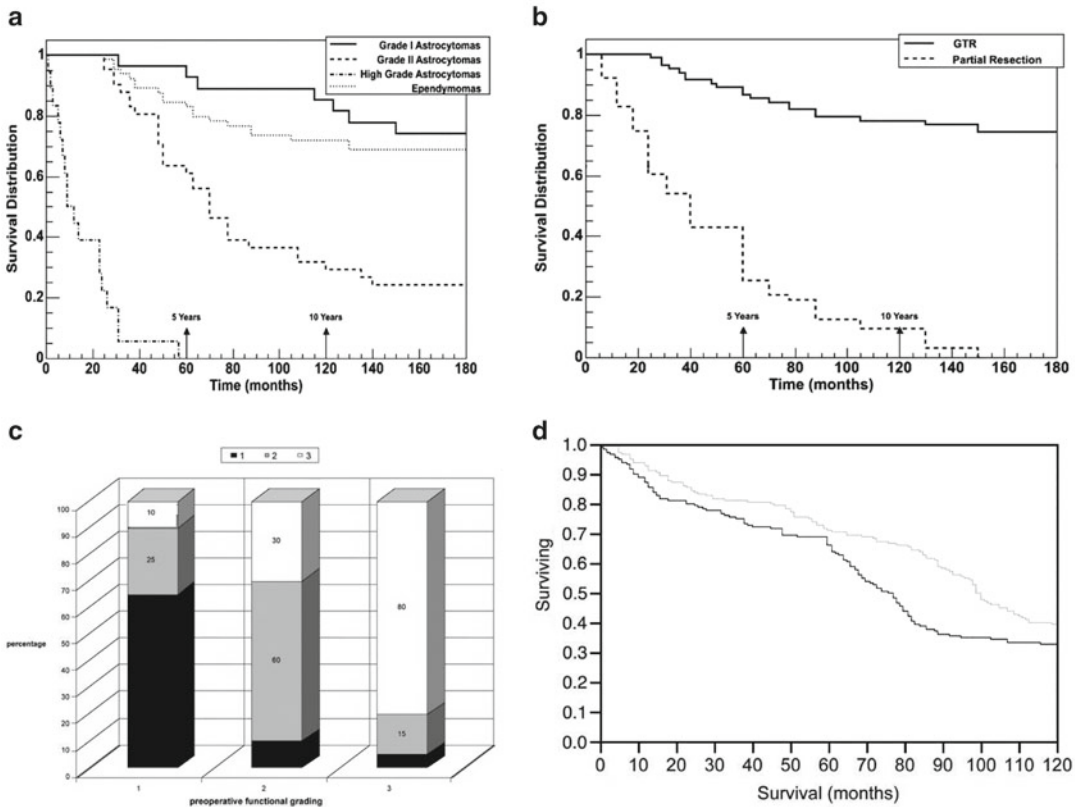


Fig. 8.1 (a) Graph showing Progression-free Survival (PFS) for intramedullary astrocytomas and ependymomas. (b) Graph showing effect of extent of resection on

PFS. (c) Graph showing comparison of preoperative and postoperative functional assessments. (d) Graph showing PFS (upper curve) and survival at 10 years (lower curve)

Lesions were located in the cervical spinal cord in 73 patients (30%), at a dorsal site in 70 (29%), at a cervicodorsal site in 61 (25%), and in the medullary cone in 37 (15%). The extension was, apart from two holocord tumors, from one to five vertebral segments.

The neurological symptoms and signs at surgery included back pain in 72 patients (30%), radicular pain in 38 patients (16%), central pain in 50 patients (21%), paresthesias in 74 patients (31%), sensory deficits in 93 patients (39%), motor deficits in 132 patients (55%), and sphincter disturbance in 36 patients (15%). Interestingly, considering only the subgroup of malignant lesions, the symptoms and signs at surgery were motor deficit in 21 patients (95%), hypoesthesia and paresthesia in 17 patients (88%), cordal or radicular

pain in 8 patients (44%) and sphincter impairment in 8 patients (44%).

Personal Series: Operative Procedures

Patients were placed in the prone position. For cervical and cervicodorsal lesions, a Mayfield skull clamp was used, and the head was flexed and elevated to reduce physiological cervical lordosis.

Surgery began with a midline skin excision, followed by subperiosteal dissection of the muscle fascia. The lesion was then accessed with a laminectomy or, in children, a laminotomy.

The spinal dura was incised at the midline and suspended away from the field to provide maximum exposure.

Under the operating microscope, the arachnoid membrane was incised above the midline: the arachnoidal layer was preserved as much as possible in an effort to limit postoperative scarring (Guidetti et al. 1981). In most patients, we preferred a longitudinal midline incision at the level of the posterior columns, or occasionally, a more lateral incision in the area of greatest cord swelling or bulging of the lesion. Once the tumor poles were identified, pial retraction sutures (5–0) were applied. These served to open the spinal cord wide without retracting it directly and to limit annoying cord pulsation under the operating microscope.

In patients with glial lesions, intraoperative frozen sections were obtained for an intraoperative histological diagnosis to guide the surgical strategy. This was useful primarily in malignant tumors. If a cleavage plane was found, the tumor was ablated with a CUSA (Soma Technology, Inc., Bloomfield, CT) or with Sonopet (Stryker, Inc., Kalamazoo, MI) at low suction. If the neoplasm had infiltrated normal tissues, the Cavitron ultrasonic surgical aspirator or the Sonopet was used to achieve internal decompression in the pulsed mode without suction, and for removing the areas close to the normal spinal cord parenchyma.

In some patients, we elected to use a discontinuous myelotomy (Cantore et al. 2002) so as to avoid damaging the arteriovenous vessels on the dorsal surface of the spinal cord. For hemostasis, we prefer to use Surgicel (Ethicon, Inc., Somerville, NJ) or fibrillar Surgicel or irrigation, reducing as much as possible the use of bipolar coagulation. Once bleeding stopped, we removed the Surgicel to avoid artifacts in the postoperative MRI scan (Miyazawa et al. 2000). The dura mater was closed with a nylon monofilament running suture to prevent cord tethering, and a duraplasty was performed using Tutopatch, a dural substitute. Since the early 1990s, all patients have undergone perioperative neurophysiological monitoring with somatosensory evoked potentials, and in the last 6 years, we used motor evoked potentials.

Intraoperative ultrasound proved useful for localizing nonbulging lesions, especially for small tumors such as cavernomas, and in more extensive masses for identifying the lesion poles.

In the last five cases we used intraoperative MRI coupled with – in the last two cases – 1.5 T fiber tracting, to test the efficacy of such techniques in recognising tumor residual and intramedullary long fibers involvement during surgical procedure.

Personal Series: Surgical Results

Independently of histopathological type, total removal of the tumor was significantly associated with a longer interval free of disease (Fig. 8.1b).

Ependymomas

Of the 81 ependymomas, 65 (81%) were completely removed and 16 (19%) incompletely removed. One patient (1%) died as a result of perioperative respiratory tract problems related to the disease. Mean follow-up was 7.1 years (range, 13 months to 20 year) in the 77 patients available (1 died, and 3 were lost to follow-up). PFS at 5 years was 83% (53 patients).

In 51% of the patients (41 patients), the presenting signs and symptoms remained unchanged at long-term follow-up; in 38% (31 patients), they improved; and in 10% (8 patients), the clinical status worsened. In patients with ependymomas, the outcome depended strictly on the completeness of removal. In 7 of the 16 patients who underwent incomplete removal, tumor regrowth developed, causing neurological and clinical evidence of compression. All 7 patients underwent reoperation, and 3 patients who had been treated early in the series and in whom we were not able to remove tumor at the second operation underwent radiation therapy at a mean dose of 30 Gy.

Low Grade Astrocytomas

Of the 101 astrocytomas, only 33 (31%) were completely removed. One patient who had a Grade I lesion died (1%), again as a result of perioperative respiratory tract problems.

The mean follow-up in the 81 patients available (1 dead and 4 lost to follow-up) was 6.7 years (range, 5 months to 18 years).

Of the 27 Grade I astrocytomas, 22 (78%) were completely removed and 7 (22%) incompletely removed. Functional assessment of the 28 patients available at “late” follow-up showed that 38% (11 of 28 patients) had improved, 8% (3 of 28 patients) had worsened, and 50% (14 of 28 patients) remained unchanged from preoperative status. One patient died and three patients were lost to follow-up. PFS at 5 years was 91% (21 patients) (Fig. 8.1a).

Conversely, of the 41 Grade II astrocytomas, only 9 (18%) were completely removed, and the remaining 41 (82%) were incompletely removed. One patient was lost to follow-up.

Functional assessment at “late” follow-up showed that 14% (7 of 49 patients) had improved, 45% (22 of 49 patients) had worsened clinically as the lesion progressed, and 41% (20 of 49 patients) remained unchanged from preoperative status. PFS at 5 years was 63% (25 of 40 patients) (Fig. 8.1a).

High-Grade Astrocytomas

None of the 10 Grade III and 12 grade IV astrocytomas could be completely removed. Only 2 of the 22 high-grade astrocytomas could be treated with gross-total removal. Seventeen patients had a subtotal excision and 3 patients received only a biopsy with a duroplasty. The postoperative radiological imaging confirmed subtotal removal in 20 of the 22 lesions.

The mean follow-up was 18 months (range, 3–39 months).

The postoperative clinical status worsened in 14 patients (63.6%), was unchanged in 7 patients (31.8%), and there was 1 case of intraoperative death (4.5%). None of the 22 patients showed any improvement in their neurological postoperative status.

In our series, we had three cases of GBM with CNS dissemination; one of these cases with Grade III C3–5 astrocytoma presented in the follow-up as a T7–8 dissemination, as was demonstrated on an MR imaging study performed 3 months after surgery. The patient had an overall

survival of 10 months. Another patient with a Grade III C6–7 astrocytoma revealed a new right occipital lesion on an MR imaging study performed 1 month after surgery. This patient had an overall survival of 18 months.

Another patient with a previously reported 6 C3–7 GBM developed a left occipital GBM seeding 13 months after surgery; it was detected on contrast-enhanced CT scans. The lesion was removed and the patient died 16 months after total removal of the intracranial lesion.

The mean postoperative survival was 18.7 months (range 0–56 months). For patients with GBM, the mean postoperative survival was 18.1 months (range 0–40 months), and for those with Grade III astrocytomas it was 18.2 months (range 9–56 months) ($p=0.8$, logrank test) (Fig. 8.2a).

For the patients with cervical tumors, the mean survival was 16.3 months, whereas for the thoracic ones it was 22 months ($p=0.3$, log-rank test) (Fig. 8.2d).

In our series all patients were followed until they ultimately died: the cause of death was respiratory paralysis in 7 cases (31.8%), pneumonia in 8 (36.3%), and pulmonary embolus in 5 (22.7%), with 1 case of intraoperative death (4.5%) and 1 case unavailable for follow-up.

Five patients (22.7%), 2 of them female and 3 male, with a mean age of 30.8 years (range 10–59 years) underwent only postoperative radiotherapy. The histological finding in these 5 patients was 4 GBMs and 1 Grade III astrocytoma. The mean postoperative survival in this group was 19.4 months (range 3–36 months) ($p=0.6$, log-rank test).

Ten patients (45.4%), 4 of them female and 6 male, with a mean age of 33.6 years (range 12–56 years), underwent radiotherapy and chemotherapy during the postoperative period. The histological finding in these 10 patients was 4 GBMs and 6 Grade III astrocytomas. the mean postoperative survival in this group was 23.3 months (range 9–56 months) (Fig. 8.2b).

We compared patients’ survival according to the treatment administered: surgery alone versus surgery plus radiotherapy and chemotherapy. The “surgery alone” group had a mean survival duration of 10.6 months, whereas the surgery plus radiotherapy

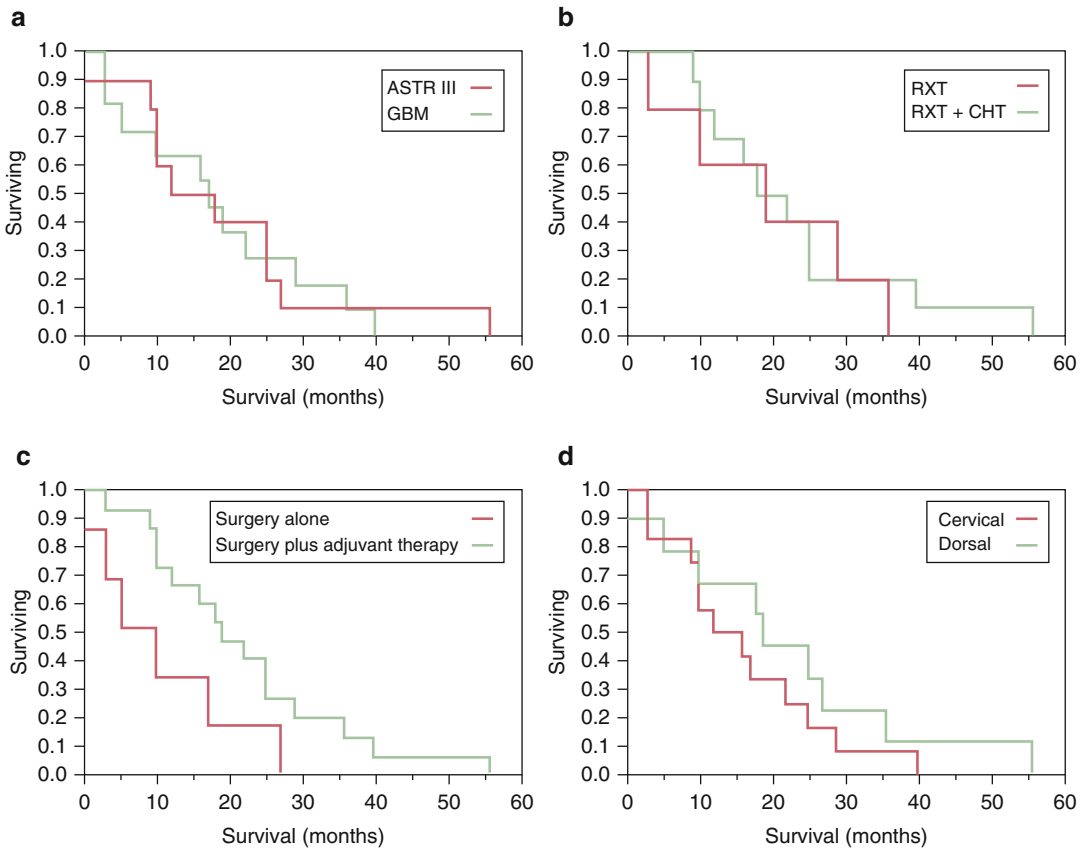


Fig.8.2 (a) Comparison of patient survival times between groups with anaplastic astrocytomas and GBMs. There is no statistically significant difference in survival between the 2 groups, even though patients with anaplastic astrocytomas fared better, with a mean survival of 18.2 months compared with 18.1 months for those with GBMs ($p=0.8$, logrank test). ASTR III=astrocytoma Grade III. (b) Comparison between the group of 5 patients (4 with GBMs and 1 with astrocytoma) who were treated only with 40 Gy radiotherapy over the neuraxis and the group of 10 patients (4 with GBMs and 6 with astrocytomas) treated with radiotherapy and chemotherapy with TMZ. The mean survival duration in the first group was 19.4 months, compared with the mean survival duration in the second group, which was

23.3 months. These data show an increase in mean survival, although not statistically significant, of about 4 months. CHT chemotherapy, RXT radiotherapy. (c) Comparison between length of survival and preoperative neurological status. We found a rather longer median survival duration for patients in McCormick Grade I (22 months) with respect to the other patients (20.4 months for Grade II and 15.5 for Grade III) ($p=0.6$, logrank test). (d) Comparison between patient survival and localization of lesions, not taking into account the histological grade. Patients with thoracic lesions fared much better than those with cervical lesions (22 vs 16.3 months) ($p=0.3$, log-rank test) even though, as expected, the numbers are too small to show a statistically significant increase of survival

and chemotherapy group had a mean 22-month survival ($p=0.05$, log-rank test) (Fig. 8.2c).

Cavernous Angiomas of 20 cavernomas, 19 were completely removed, while only in one case a second intervention was required to remove a residual malformation.

In 69% (14 of 20 patients), the clinical and neurological symptoms improved, and in only

1 patient did the preoperative deficit persistently worsen. In this case CUSA was used to remove the cavernoma. Five patients (22%) remained unchanged.

In 16 patients a standard laminectomy was used as surgical approach, while in 3 patients a arthro-pedunclectomy was performed. In another patient a trans-thoracic approach was utilized.

Other Histological Types

The 11 epidermoid-dermoid tumors were completely removed in 45% of the patients (5 patients). The clinical symptoms remained substantially unchanged from the preoperative status in seven of the eight patients who were available for long-term follow-up. In one patient, the clinical symptoms and signs progressively worsened, especially the motor symptoms, primarily owing to spinal arachnoiditis that developed in the long term. Two patients were lost to follow-up.

Of the 11 hemangiomas, 10 (91%) were completely removed. The patient who underwent incomplete removal was lost to follow-up. Of the remaining ten patients, seven are currently alive and three died as a result of causes unrelated to the intramedullary lesions.

In patients with oligodendrogliomas, complete removal was possible in 60% (three of five patients); two patients died 3 years after surgery, one died at 5 years, and the other two are still alive respectively 8 and 3 years after surgery.

In all 5 patients with metastases, the lesions were radically removed. One patient was lost to follow-up; one died 3.5 years after surgery, and despite meticulous screening, the site of the primary lesion remained undisclosed; The three remaining patients died, one at 6 months, one at 16 months and the other at 18 months after surgery.

The five patients who had schwannomas and the two patients with meningiomas all underwent complete removal of their tumors. In all seven patients, the clinical status improved substantially after surgery. Six of the seven patients are alive, and one died as a result of causes unrelated to the intramedullary tumor.

Personal Series: Clinical Long-Term Follow-Up

Pain

Most patients (153 of 190 patients, 80%) experienced pain or painful paresthesias during the postoperative course. In most patients, these disturbances became gradually more bearable

until in 30% of the patients (45 patients), they disappeared. In the 107 patients (56%) in whom pain persisted at follow-up, pain was equally distributed among the three characteristic types: spinal cord pain in 35% (37 patients), root pain in 40% (43 patients), and sublesional pain in 25% (27 patients). Spinal cord pain is a burning, biting, unbearable sensation, a deep ache at the site of the surgical procedure. Sublesional pain is primarily a burning sensation often accompanied by paresthesia below the level of the lesion. In most of these patients, medical therapy satisfactorily relieved the symptoms (pain relief 50%). Gabapentin (Neurontin; Parke-Davis, Morris Plains, NJ) was used in an escalating manner, starting at 600 mg/day up to 1200 mg/day for at least 4 months. Buprenorphine (Temgesic; Schering Plough, Kenilworth, NJ) was used as a second-line treatment (0.2 mg twice a day) in 15 patients. Of the 107 patients who had persistent postoperative pain, only 3 patients (2%) had chronic untreatable pain. All three patients had burning cord pain that was resistant even to intrathecal pump infusion of opioids (pain relief 50%).

A common radiological finding in these three patients was MRI evidence of dorsal cord tethering at the surgical site. In the one patient who agreed to undergo reoperation for this complication, persistent postoperative pain remained unchanged. Dorsal tethering complicated 17% of the surgical interventions for intramedullary spinal cord tumors in this series.

Impaired Sensitivity

In 45% of the patients (85 of 190 patients), disturbance of sensitivity remained unchanged after surgery; in 40% (76 of 190), it worsened; and in 15% (29 of 190), it improved.

Sphincter and Sexual Dysfunction

Sphincter dysfunction complicated the postoperative course in 25% of the patients (47 of 190). Of these 47 patients, 5% (2 of 47 patients)

required a urinary catheter and 10% (5 of 47) complained of urinary incontinence or urgent micturition. A total of 5% of the male patients in the series (9 of 190) had sexual problems (erectile dysfunction, difficulty in ejaculation).

Motor Disorder

Comparison of preoperative and postoperative functional assessments by use of a simplified McCormick classification in the 83 patients who underwent gross total removal of the intramedullary tumor (27 astrocytomas and 55 ependymomas) gave the following results. Of the patients classified preoperatively as McCormick Grade I, at 6 months after surgery, 65% remained functionally unchanged, 25% worsened to Grade II, and 10% worsened to Grade III. Of patients classified preoperatively as Grade II, 60% remained in this functional class at 6 months, 30% worsened, and only 10% improved. Of the patients classified preoperatively as Grade III, 80% remained unchanged at 6 months after surgery, 15% improved their functional status to Grade II, and only 5% improved to Grade I (Fig. 8.1c).

According to these results, the McCormick functional assessment, which was made during the follow-up neurological evaluation, shows 1 patient with McCormick Grade I, 5 with Grade II, and 15 with Grade III (there was 1 case of intraoperative death).

Cord Tethering

In our experience, cord tethering complicated 17% of the cases (32 of 190 patients). In 8% (15 of 32 patients), tethering caused painful symptoms. Reoperation for posterior cord tethering proved necessary in 5% of our series (9 of the 190 patients); in 4 of these 9 patients, the symptoms subsequently regressed, and another 2 patients had a partial relief. The remaining 3 patients continued to complain of unbearable posterior column pain. A pump implanted for intrathecal opioid infusion failed to provide satisfactory pain relief. In the last 30 patients in the

series, in whom we used a continuous dural monofilament nylon suture, only 1 patient had asymptomatic cord tethering.

Discussion

When our institute reported the long-term results for the treatment of intramedullary tumors in 1981, the operative technique was still controversial (Guidetti et al. 1981). Today, the surgical treatment of intramedullary tumors is well codified (Brotchi 2002; Jallo et al. 2001; Stein 1983).

In the series of 202 patients we report here, who were treated in the microsurgical era, adjuvant treatment (chemotherapy and radiation therapy) had only a role in the management of high grade lesions, as it did in other reported series (Balmaceda 2000; Isaacson 2000; Minehan et al. 1995).

Although the surgical management of intramedullary lesions is now standardized, some technical refinements we routinely use make certain problematic steps easier and avoid major pitfalls; these technical refinements are of little use in malignant lesions, because the complete removal is often impossible.

Pial retraction is an extremely helpful tactic that allows the neurosurgeon to divaricate the posterior columns in an anatomic and atraumatic manner; it also eliminates the annoying cerebrospinal fluid pulsation that hinders surgical maneuvers under the microscope. In our previously published series of intramedullary tumors, tumors had to be removed without the CUSA. The CUSA, now an indispensable tool in intramedullary surgery, especially the last-generation device with a microtip (Valleylab Division, Tyco Group LP, Boulder, CO), is used differently according to whether it is needed for working within the lesion (pulsed mode, low suction) or for peeling the tumor bed (pulsed mode, no suction) (Guidetti et al. 1981). Coagulation must be reduced to a minimum, preferring hemostatic materials instead. Once bleeding has stopped, the operative field should be irrigated with physiological solution to remove as much Surgicel as possible so as to avoid artifacts that may be misinterpreted at

intraoperative or postoperative neuroradiological imaging.

Another useful technical point is that except for laterally bulging lesions with paramedian extension documented on the preoperative MRI scan and seen even more clearly at surgery, we use a longitudinal midline medullary incision centered on the lesion. As a first step, we usually debulk the tumor, then we remove its lower and upper poles by angling the handle and the micro-tip of the CUSA. Conversely, if vessels of importance cross the midline on the dorsal surface of the cord, we prefer to preserve them by means of a discontinuous myelotomy comprising two separate incisions at midline level, one above and the other below the vessel or vessels involved (Cantore et al. 2002). We then remove the tumor with the CUSA at the level of the two incisions and later beneath the bridging area. The distance between the two incisions must be as small as possible to avoid excessive traction on the bridging area. In the patients with intramedullary tumors in this series, these technical refinements proved especially useful in preserving spinal cord function.

Even though Epstein and Farmer consider “extensive debulking using the CUSA critical before any effort is made to identify the interface between the tumor and adjacent spinal cord” (Epstein and Farmer 1990), we find that in removing ependymomas, a cleavage plane must be sought immediately, preferably reducing the use of the CUSA to the minimum.

Since the early 1990s, all patients have undergone perioperative neurophysiological monitoring with somatosensory evoked potentials.

In the last 6 years, we routinely used motor evoked potentials.

We currently use standard combined monitoring (SEPs and MEPs) in averaging modality with D-wave recording. We usually test patients before surgery because of the possibility that D-wave is not detected. Primarily SEPs are very useful in the beginning of surgery to perform posterior myelotomy when it is difficult to identify the midline and anatomy is distorted by the tumor. Usually SEPs are lost during myelotomy and often they do not recover, but this is not a criteria to stop surgery

because the SEPs loss do not correlate with a significant postoperative sensory or motor deficit. Secondly, in debulking of the tumor, especially when margins are not so clear, MEPs and D-wave are very useful to predict a transient or a definitive motor deficit. In fact, when MEPs are lost uni or bilaterally but D-wave is still unchanged or above 50%, we adopt some corrective measures: irrigation with warm physiological solution, cotton patties soaked in papaverin, increase of the average arterial pressure, modifying the site of removal or modify the removal technique (CUSA vs piecemeal removal using microscissors). Usually MEPs recover. If not, surgery can proceed and patient will suffer a transient motor deficit, which they usually recover in a short time. When MEPs are lost uni or bilaterally and D-wave decrease below 50%, surgery must be interrupted immediately. All the corrective procedure must be taken and if D-wave do not recover, surgery should be abandoned, otherwise patient will suffer a permanent motor deficit (Sala et al. 2006). Patient with deficit are monitored repeatedly in the postoperative course in the following months to check any modifications of the neurological status.

In my personal experience the D wave is the single most important factor in choosing whether to go or to stop the surgical procedure.

At the same time I do believe that increase of the blood pressure is the single most important factor in trying to balance a possible neurological deficit.

Our experience shows that the surgical technique must be planned according to the radiological appearance of the lesion. Neuroimaging techniques have had the same progress than surgical technique.

All patients underwent neuroimaging using myelography and CT myelography before 1986, and from 1986 until the present, Gd-enhanced spinal MR imaging has been used. In the pre-MR imaging era, these kinds of lesions were always surgically treated, because myelography or CT myelography did not give any clue regarding the histological type of the lesion.

Magnetic resonance imaging supplies a much clearer image, showing the site, size, and some-

times even the nature of lesions (Balmaceda 2000; Goto et al. 2003; Guidetti et al. 1981). Currently, the introduction of MR imaging spectroscopy for this type of disease allows us to acquire some data on the glial nature of the lesion and its mitotic activity, which aids our preoperative planning, that is really important especially for malignant lesions.

Before the introduction of MR imaging spectroscopy, our therapeutic protocol consisted of surgery, followed by adjuvant radiotherapy (Isaacson 2000) and chemotherapy (Balmaceda 2000) with TMZ. Nowadays, due to advancements in preoperative neuroimaging, our therapeutic protocol consists of the following: (1) diagnostic suspicion based on clinical and radiological features; (2) biopsy; (3) histological confirmation; (4) duraplasty; and (5) radiotherapy plus chemotherapy.

The clinical pattern and the results of diagnostic neuroimaging frequently establish the diagnosis, which should be confirmed by surgery or biopsy. This therapeutic strategy is mandatory in Italy, because the administration of medullary radiotherapy without histological confirmation is a medicolegal issue. Duraplasty is a fundamental step in the treatment of these lesions to prevent the complication of possible swelling of the spinal cord due to radiotherapy.

Thus advanced neuroimaging techniques are fundamentals in diagnosis, in conditioning surgical strategy and in early postoperative evaluation of surgical results.

Although nowadays, the extent of surgical resection is based primarily on the early postoperative MRI scan and secondarily on the neurosurgeon's impression, in the past, the only reliable means to judge the extent of removal was the surgeon's impression.

Interestingly in the last few cases we used intraoperative MRI and fiber tracking. The utility of such techniques, differently from intracranial surgery, must be still evaluated. To date, we have never changed our surgical strategy according to the informations collected with intraoperative imaging techniques.

Interestingly, diffusion tensor imaging tractography has been already used preoperatively to

predict resectability of spinal cord tumors (Cantore et al. 2002; Setzer et al. 2010), but these reported preliminary data should be further confirmed.

Further studies are needed to understand the usefulness of such imaging techniques, especially in predicting short and long term clinical results.

Our review underlines that the most important prognostic factor in determining postoperative neurological status in the long term is the patient's functional status at surgery (Fig. 8.1c), hence the need for an early diagnosis. An early diagnosis markedly influences patients' functional prognosis and quality of life. It may be interesting to note that, because of the progress of neuroimaging, whereas in the first 10 years of our series, the average diagnosis time was 3.7 years, in the last 10 years, the duration of clinical symptoms before surgery was 2.1 years.

In our institute all patients underwent pre- and postoperative functional assessment in which a simplified version of the McCormick classification (Minehan et al. 1995) was used. Some patients were actually retrospectively classified according to the McCormick scale; this clinical evaluation seems to be very important for prediction of the functional postoperative status.

This scale demonstrated an high prognostic value.

In our series, even patients with large-sized tumors extending over several metameres that had not caused major functional motor deficits had a good chance of maintaining a stable neurological status after surgical resection. In this series, two determinant predictors of long-term functional outcome and survival were the histological type of tumor and the extent of surgical removal (Fig. 8.1a, b). These two interrelated factors cannot be analyzed separately. Accordingly, as others have reported, more than 80% of the Grade I ependymomas and astrocytomas in our series but only 10–15% of Grade II astrocytomas and none of the more aggressive glial tumors (anaplastic astrocytomas, oligodendrogliomas, and glioblastomas) could be microscopically completely removed (Chang et al. 2002; Sandler et al. 1992; Schwartz and McCormick 2000).

Patients with low-grade ependymomas and astrocytomas that were totally removed had a greater chance of leading a normal life after surgery and a higher rate of PFS than those whose tumors were incompletely removed (PFS, 26% at 5 years). Some of the incomplete resection, especially in ependymomas, were attributable to the fact that some remnants were visible only at postoperative MRI and were not detected at surgery; in four patients (high cervical lesion), resection could not be completed because of intraoperative vital function disturbance. Even in the patients with gross total resection, 20% of late deterioration is expected, and it is a result of tumor recurrence.

Clearly, the poor prognosis of these highly aggressive tumors (Grade III astrocytomas and glioblastomas) in our series depended on both the histological grade and the incompleteness of surgical removal.

The results of survival and PFS for ependymomas and astrocytomas in our series are deeply influenced by the higher presence of astrocytomas rather than ependymomas (86 patients versus 68). The comparison with other important studies shows remarkable differences. Constantini et al. 2000, in their study of 164 pediatric patients with intramedullary tumors, reported a 3-, 5-, and 10-year PFS and survival, respectively, of 80, 71, and 54% and 80, 76, and 70% (Fig. 8.1d). The 3-, 5-, and 10-year survival statistics in our series were 83, 73, and 42%; the 3-, 5-, and 10-year PFS statistics were 77, 70, and 34%. The main difference between our series and that of Constantini regards the PFS and the survival at 10 years (PFS, 34 versus 54%; survival, 42 versus 70%). This difference may be explained by the fact that Constantini's series is entirely a pediatric one, whereas our series is composed primarily of adult patients.

Intramedullary astrocytomas account for ~1% of all CNS tumors and 6–8% of spinal cord tumors (Raco et al. 2005). The vast majority of intramedullary astrocytomas are slowgrowing lesions. Glioblastomas account for ~7.5% of all intramedullary gliomas and barely 1.5% of all spinal tumors (Miller 2000; Raco et al. 2005; Schwartz and McCormick 2000). Unlike the

intracranial variety, intramedullary high-grade astrocytomas have received little attention in the literature. This histotype shows a slight predilection for patients in the earlier decades of life, (Constantini et al. 2000; Townsend et al. 2004) with the mean age ranging from 28.5 to 18.1 years according to the review of the literature by Fortuna and Giuffr  in 1971. In our series of 22 cases, the mean age (34.1 years) was higher; only 2 patients were younger than 15 years.

According to the literature, the most affected spinal regions were the cervical and thoracic ones, especially the upper cervical region, (Guidetti et al. 1981; Houten and Cooper 2000; Miller 2000; Raco et al. 2005; Schwartz and McCormick 2000; Townsend et al. 2004) as in our series. As a rule, the clinical history is marked by initial symptoms that may be nonspecific and are often underestimated (hypesthesia, paresthesia, and sensory deficits) (Ciappetta et al. 1991; Houten and Cooper 2000; Raco et al. 2005; Santi et al. 2003; Schwartz and McCormick 2000), whereas in the second stage, the neurological pattern worsens swiftly (pain, sensory and motor disorders, paraplegia, tetraplegia, and sphincter disturbances).

In our series we found 3 cases of CNS dissemination of the malignancy: 1 intramedullary and 2 intracerebral. In the literature several cases of CNS dissemination of intramedullary high-grade astrocytomas are described (Ng et al. 2001), but the mechanism of dissemination is still unclear; surgical manipulation may facilitate tumor cell seeding to the subarachnoid space, resulting with an intracranial dissemination.

Immunosuppression is another mechanism that leads to CNS dissemination, but no patient in our series revealed this pathological mechanism in the routine serum examinations.

In case of intramedullary astrocytomas, the most important prognostic factors in determining postoperative survival are as follows: histological grade of the lesion, application of adjuvant radiotherapy and chemotherapy, localization, and preoperative neurological status (Innocenzi et al. 1996; Raco et al. 2005).

We compared survival duration according to the histological grade in patients with anaplastic

astrocytomas and GBMs (Fig. 8.2a). There is no statistically significant patients with anaplastic astrocytomas fared better, with a median survival of 18.2 months compared with 18.1 months for those with GBM ($p=0.8$, log-rank test).

As shown in Fig. 8.2c, the adjuvant treatment compared with surgery alone yielded an important improvement in survival (22 months compared with 10.6 months; $p=0.05$, log-rank test). In our review we compared a group of 5 patients (4 with GBMs and 1 with astrocytoma) who were treated only with 40 Gy radiotherapy over the neuraxis with the group of 10 patients (4 with GBMs and 6 with astrocytomas) who were treated with radiotherapy and chemotherapy with TMZ. The mean survival in the first group was 19.4 months, and in the second group it was 23.3 months. These data show an increase in mean survival time, although not a statistically significant one, of ~4 months (Fig. 8.2c).

We also compared survival with respect to the localization of lesions (Fig. 8.2d), not taking into account the histological grade. Patients with thoracic lesions fared much better than those with cervical lesions (22 months compared with 16.3 months; $p=0.3$, log-rank test) even though, as expected, the numbers are too small to show a statistically significant increase in survival.

Next we compared length of survival with respect to the preoperative neurological status: we found again a rather longer survival for patients categorized in McCormick Grade I (survival 22 months) with respect to the other patients (20.4 months for Grade II and 15.5 months for Grade III; $p=0.6$, log-rank test), but this probably reflects the fact that this first group of patients had probably been diagnosed at an early stage of the disease.

None of the 22 patients showed any improvement in postoperative neurological status. The cause of postoperative worsening could be attributed to increased spinal cord swelling due to edema or to surgical manipulation of the cord for an uncertain tumor margin.

The prognosis for high-grade astrocytomas, like their intracranial counterparts, is extremely poor, and virtually all patients die as a consequence of progressive disease.

As in patients with malignant intracranial astrocytomas, widespread leptomeningeal metastasis and hydrocephalus are common. The terminal event in many patients with to the cervicomedullary area, with respiratory paralysis. Pulmonary embolus and pneumonia may prove fatal to patients bedridden by progressive disease.

Our experience in managing the cavernous angiomas in this series suggests that the first option for symptomatic lesions is total surgical excision to avoid recurrence and the risk of further hemorrhage. Especially in patients who present with mild paraparesis of relatively recent onset, the surgical outcome combines low mortality with a high probability of functional recovery. When removing cavernous angiomas, we rarely find the CUSA or Sonopet helpful.

For the surgical removal of hemangioblastomas, we use primarily bipolar coagulation to diminish blood flow and shrink the lesion; this maneuver makes it easier to dissect the lesion away from the normal parenchyma, thus allowing en bloc removal. In excising dermoid and epidermoid lesions, the lesion must be meticulously dissected away from the spinal cord parenchyma with the help of microforceps and the dissector, removing the tumor piecemeal. In some cases, no attempt should be made to remove the tumor pseudocapsule completely, because it often adheres tightly to the cord and its removal might provoke microhemorrhage and medullary petechiae.

The surgical technique for removing intramedullary metastases resembles that for glial lesions, even though the prognosis depends closely on the primary lesion. The small number of oligodendrogliomas in this series makes it difficult to comment on the outcome of treatment. An unusual finding in our series was the four intramedullary mesenchymal lesions (two meningiomas and two neuromas, one malignant), all of which were totally removed; three of these lesions were easily cleaved, but the malignant neuroma resembled a high-grade glial intramedullary tumor.

Some state that the extent of surgery has no significant influence on the prognosis (Sandler et al.

1992). In the series we review here, patients whose tumors were macroscopically completely removed survived longer. This finding was also confirmed by the outcome in patients with cerebral ependymal lesions. From a technical viewpoint, removing an ependymoma is often an arduous task because these tumors tend more than other astrocytomas of similar grade to bleed intraoperatively.

Nevertheless, because ependymomas usually have a better cleavage plane than low-grade astrocyte lesions, most of them can be completely removed.

A special consideration should be made for our result according to the patient age.

Our series of 202 patients with intramedullary spinal cord tumors spanned a wide age range (8–72 years); 25 patients were children. Most tumors exhibited a more benign biological behavior in children than in adults. Despite the higher incidence of astrocytomas than ependymomas in children (18 astrocytomas versus 7 ependymomas), we found no significant difference in the prognostic criteria in the outcome in terms of quality of life and functional status between children and adults (Isaacson 2000).

Tethering: postoperative spinal cord tethering is a problem frequently overlooked in most of the series regarding treatment of intramedullary tumors.

In our experience, although 17% of the patients (32 of 190) had radiologically documented cord tethering, only 8% (15 of 32 patients) were symptomatic.

Drug-resistant postoperative pain is almost invariably associated with tethering of the spinal cord at the operation site. Dorsal tethering, in our experience, is not associated with deterioration of neurological function.

Despite surgical revision and pain therapy, some patients with cord tethering remain untreatable. Actually, as previously reported in the Results section, three of our patients who nonetheless had a pump infusion device implanted had no relief. They should be considered as having a failure of the surgical therapy. Hence the importance of searching for the possible onset of cord tethering in the postoperative MRI scan so that debridement can be planned. Even though

the literature documents complex techniques for reducing the incidence of cord tethering, we find that closing the dura mater with a running suture in monofilament nylon (5-0) instead of using silk interrupted stitches will suffice (Goto et al. 2003).

In managing large-sized tumors with solid and cystic components, we center the laminectomy or laminotomy on the solid component. If the solid component is removed, the coexisting syringomyelic cavity disappears from the postoperative MRI scan within 2 months. The same surgical tactic provides a good outcome in cervical masses extending into the bulbo-cervical area.

Conclusions

From our experience in treating this large series of patients with intramedullary spinal cord tumors, we conclude that the determinant predictors of a good outcome after surgery for intramedullary spinal cord tumors are histological type of lesion, complete removal of the lesion, and a satisfactory neurological status before surgery.

Surgery for intramedullary spinal cord tumors should aim at complete removal of low-grade ependymomas and Grade I astrocytomas; “smart” removal of Grade II astrocytomas; and complete removal of cavernous angiomas, hemangioblastomas, and rare lesions such as schwannomas and intramedullary meningiomas.

Radical surgery is neither advisable nor feasible for epidermoid and other benign lesions in cases of dense adhesion to the spinal cord, making removal a hard task that puts patients at risk of severe neurological deficits.

Total surgical removal of the high-grade lesions is rarely possible because of the anaplastic aspects; that is, infiltrative behavior. Surgical treatment did not ameliorate the postoperative prognosis prompting a worsening of the deficit. Radiotherapy and chemotherapy have a moderate impact on survival. Multimodality treatment of high-grade intramedullary astrocytomas has been shown, in our series, to increase survival without improving the patient’s neurological status.

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Index

A

- AG. *See* Angiocentric glioma (AG)
- 5-aminolevulinic acid (5-ALA), 37
- Anesthetic agents, 59
- Angiocentric glioma (AG)
 - clinical presentation and imaging, 2–3
 - differential diagnosis, 4–5
 - genetic abnormalities, 4
 - histopathology, 3–4
 - historical context, 2
 - treatment and prognosis, 5
- Angiogenesis
 - importance of, 26–28
 - K1-5, 30
 - treatment of malignant gliomas, 27, 29
 - tumor vasculature, 27
- Asmussen, J., 56

B

- Bailey, P., 45
- Bearz, A., 74
- Belachew, A., 43–51
- Bevacicumab, 27
- Bindal, R.K., 72
- Blood pressure control, 62
- Brat, D.J., 8

C

- Cantore, G., 80
- Cao, R., 26, 29
- Cao, Y., 29
- Carbon dioxide (CO₂), 58
- Cerebellopontine angle (CPA) meningiomas
 - anatomy, 44
 - chemotherapy, 51
 - history, 45, 48
 - neuroimaging, 48–49
 - pathology, 45
 - radiation therapy, 50–51
 - risk factors, 47
 - symptoms, 48

treatment options

- observation, 50
- surgery, 49–50
- trigemino-cardiac reflex, 49
- types and classification, 45–46
 - premeatal tumor location, 46–47
 - retromental tumor location, 47
- Chowdhury, T., 43–51
- Cimatti, M., 79–92
- Computed tomography (CT), 69–70
- Constantini, S., 90
- Cord tethering, 87
- Cottingham, S.L., 5
- Crizotinib, 74
- Cushing, H., 45

D

- Delfini, R., 80

E

- EGFR tyrosine-kinase inhibitors (TKI), 74
- Ependymomas, 83
- Epidermal growth factor receptor (EGFR), 29
- Epithelial membrane antigen (EMA), 4
- Epstein, F.J., 88
- Esmolol, 63

F

- Farmer, J.P., 88
- Fisher, M.J., 13–22
- Forbes, J.A., 8
- Fortuna, A., 90

G

- Giuffré, R., 90
- Grillo, P., 59
- Guidetti, B., 80
- Guo, S., 72
- Guzowski, 46

H

Hefti, M., 35–40
 Hemodilution, 58–59
 High-grade astrocytomas, 84–85
 Huang, M.-C., 25–32

I

Impaired sensitivity, 86
 Indomethacin, 63
 Intermediate pilomyxoid tumor (IPT), 8
 Intramedullary spinal cord tumors
 clinical long-term follow-up
 cord tethering, 87
 impaired sensitivity, 86
 motor disorder, 87
 pain, 86
 sphincter and sexual dysfunction, 86–87
 clinical presentation and surgery, 80–82
 diffusion tensor imaging tractography, 89
 D-wave, 88
 immunosuppression, 90
 intramedullary astrocytomas, 90
 low-grade ependymomas and astrocytomas, 90
 magnetic resonance imaging, 88–89
 McCormick classification, 89
 McCormick Grade I, 91
 operative procedures, 82–83
 pial retraction, 87
 pulmonary embolus and pneumonia, 91
 surgical results
 ependymomas, 83
 high-grade astrocytomas, 84–85
 histological types, 86
 low grade astrocytomas, 82–84
 tethering, 92
 Islamian, A.P., 10

J

Jabre, A., 56, 59
 Jeon, Y.-K., 7
 Jugular bulb venous oximetry, 61

K

Komotar, R.J., 7
 Kringle 1-5 (K1-5)
 animals, 31
 decreases hypoxic area, 28, 30
 effects, 32
 inhibition effect, 28, 31
 inhibits tum/angiogenesis, 28, 30
 lower VEGF expression, 30–31
 structure and antiangiogenic effect, 29–30

L

Lin, Y.-L., 25–32
 Louis, A., 45

Low grade astrocytomas, 82–84
 Lum, D.L., 4
 Lung adenocarcinoma
 cerebral metastases management, 68
 management
 chemotherapy, 73–74
 radiotherapy, 73
 surgery, 72–73
 neuroimaging
 computed tomography, 69–70
 magnetic resonance imaging, 69–71
 nuclear medicine techniques, 71
 pathology, 68–69

M

Magnetic resonance imaging (MRI), 69–71
 Malignant gliomas
 angiogenesis
 antiangiogenesis in treatment, 27–29
 importance of, 26–28
 tumor vasculature, 27
 Kringle 1-5
 animals, 31
 decreases hypoxic area, 28, 30
 effects, 32
 inhibition effect, 28, 31
 inhibits tum/angiogenesis, 28, 30
 lower VEGF expression, 30–31
 structure and antiangiogenic effect, 29–30
 vascular endothelial growth factor, 26
 Masica, D.N., 46, 49
 Meningiomas
 5-aminolevulinic acid, 37
 intratumoral heterogeneity and fluorescence intensity, 37–38
 non-radical resections, 36
 photodynamic diagnosis, 36–37
 photodynamic therapy, 38–40
 Simpson grading system, 36
 Miller, D.C., 1–10
 Minturn, J.E., 13–22
 Miscusi, M., 79–92
 Motor disorder, 87

N

Nagaishi, M., 7
 Nager, G.T., 46, 49
 Near-infrared spectroscopy (NIRS), 61–62
 Normocapnia, 62
 Nuclear medicine techniques, 71
 Nussbaum, E.S., 68

O

Odem, J.L., 1–10
 O'Reilly, M.S., 29
 Oya, S., 48

P

- Pain, 86
- Palvolgyi, R., 58
- Paraskevopoulos, D., 8
- Patchell, R.A., 72
- Paulson, O.B., 58
- Pediatric low-grade gliomas
 - classification, 14
 - cytotoxic chemotherapy, 20
 - diagnosis, 15–16
 - dissemination, 14
 - fibroblast growth factor receptor, 20
 - neurofibromatosis type 1, 21–22
 - oncogenic KIAA1549-BRAF fusion gene, 20
 - Ras-mediated signal transduction pathways, 21
 - stereotactic fractionated radiotherapy, 20
 - symptoms, 14
 - treatment and outcomes
 - adjuvant therapy, 19
 - biopsy, 19
 - carboplatin and vincristine, 17
 - complete surgical resection, 16
 - favorable prognostic factors, 19
 - focal radiation therapy, 17
 - multi-agent chemotherapy, 17–18
 - positron emission tomography, 17
 - post-operative management, 16
 - temozolomide, 18
 - TPCDV, 18
 - vinblastine, 18
 - tuberous sclerosis, 22
- Pennings, F.A., 60
- Perfusion-weighted imaging, 71
- Perri, S.R., 31
- Phillips, J.J., 9
- Photodynamic diagnosis (PDD), 36–37
- Photodynamic therapy (PDT), 38–40
- Pilomyxoid astrocytoma (PMA)
 - clinical presentation/imaging, 5–6
 - differential diagnosis, 7
 - genetic abnormalities, 7
 - histopathology, 6–7
 - historical context, 5
 - prognosis and treatment, 7–8
- Pituicytoma
 - clinical presentation/imaging, 8–9
 - differential diagnosis, 9–10
 - genetic abnormalities, 9
 - histopathology, 9
 - historical context, 8
 - prognosis and treatment, 10
- Platelet-derived growth factor receptor (PDGFR), 29
- PMA. *See* Pilomyxoid astrocytoma (PMA)
- Positron emission tomography (PET), 71
- Postoperative cerebral hyperemia
 - clinical complications, 60–61
 - management
 - blood pressure control, 62
 - esmolol, 63

- indomethacin, 63
 - normocapnia, 62
- pathophysiology
 - anesthetic agents, 59
 - carbon dioxide, 58
 - cerebral autoregulation and brain tumor, 56–57
 - hemodilution, 58–59
 - postoperative hypertension, 57–58
 - surgery related factors, 60
 - sympathetic overactivity, 59–60
- postoperative brain tissue oxygenation/metabolism monitor
 - jugular bulb venous oximetry, 61
 - near-infrared spectroscopy, 61–62
 - transcranial Doppler, 61
 - velocity of middle cerebral artery, 56
- Preusser, M., 4
- Protoporphyrin IX (PpIX), 36–37

R

- Raco, A., 79–92
- Rao, G.S., 58
- Ricci-Vitiani, L., 27

S

- Samanci, M.Y., 67–75
- Sandler, H.M., 80
- Sandu, N., 43–51
- Schaller, B., 43–51
- Schmieder, K., 57
- Secci, F., 10
- Sergides, I., 60
- Sharma, D., 58
- Shaw, A.T., 74
- Sievert, A.J., 13–22
- Single-photon emission computed tomography (SPECT), 71
- Sphincter and sexual dysfunction, 86–87
- Spiriev, T., 43–51

T

- Tatevossian, R.G., 4
- Temozolomide, 74
- Terasaki, M., 8
- Thallium-201, 71
- Thetering, 92
- Tihan, T., 5, 7
- Transcranial Doppler (TCD), 61
- Trigemino-cardiac reflex, 49
- Tsugu, H., 8

V

- Vascular endothelial growth factor (VEGF), 26
- Vecht, C.J., 72
- von Baumgarten, L., 29
- von Campe, G., 35–40

W

Wang, M., 2

Wang, R., 27

Welsh, J.W., 73

X

Xu, H.-Z., 55–63

Y

Yang, X.-Y., 55–63

Z

Zhang, W., 31

Zhou, S.-J., 55–63